

# DRAW — User Manual

**A pixel art editor written in QB64-PE — that exports your artwork as QB64 source code.**

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*Inspired by Dazzle Draw, Deluxe Paint, Grafx2, Inkscape, The GIMP, and yes... Photoshop*

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# About this Manual

This manual is the companion reference to the DRAW pixel art editor. It mirrors the structure of the upcoming 55-episode video tutorial series and walks you from "I just installed DRAW" through real-world pixel-art workflows, advanced layer techniques, and the quirky touches that only DRAW offers — like exporting your artwork as runnable QB64 source code.

You can read it cover-to-cover, jump to any chapter from the Table of Contents, or use it as a quick-reference once you know your way around. Every chapter ends with practical exercises and every keyboard shortcut you'll see in the manual is also listed in the appendix and the live, registry-generated `SHORTCUTS.md` (keyboard **and** mouse).

## Who this is for

- 🎨 **Pixel artists** moving from DPaint / Aseprite / ProMotion who want a free open-source alternative.
- 🎮 **Game developers** who need fast sprite, tile, and palette workflows with deep export options.
- 💻 **QB64-PE / QBasic enthusiasts** who appreciate that the whole editor and its native export format is QB64 code.
- 🆕 **Newcomers to pixel art** who want a guided walkthrough rather than a feature dump.

## Conventions used in this manual

| Convention | Meaning |
|---|---|
| `Ctrl+S` | Press the keys together. On macOS, `Ctrl` is `⌘ Cmd` unless the manual says otherwise. |
| `B` | Press a single key (no modifier). |
| `[ ]` | Brackets around a key reference (often used for brush size keys). |
| **Click** / **Right-click** / **Middle-click** | Mouse buttons L / R / M. |
| **Drag** | Press and hold the left mouse button while moving. |
| 🎯 *Goal* | What you'll be able to do after the section. |
| 🎨 *Try it* | A short, practical exercise. |
| 💡 *Tip* | Something useful but optional. |
| ⚠️ *Gotcha* | Something that commonly trips people up. |

## How this manual is organized

```
docs/
├── MANUAL.md            ← this cover + master TOC
└── MANUAL/
    ├── 01-introduction.md       … 22-workspaces-capture.md
    ├── SCREENSHOTS.md   ← capture checklist for missing visuals
    └── images/          ← captured screenshots + placeholder.svg
```

If you are reading this on GitHub, every chapter link above will jump straight to the rendered chapter file. If you are reading offline in VS Code, open the Markdown preview ( `Ctrl+Shift+V` ) on this file and the entire manual is one click away.

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## Contributing & feedback

DRAW and this manual are open source. If you find a mistake, or want to add a workflow, please open an issue or pull request on the GitHub repository. Thank you for reading — now go make something pixelated.

— *grymmjack*

# Ch. 01 🎬 Introduction & Setup

> **What you'll learn:** What DRAW is, why it exists, how to install or build it on Windows / Linux / macOS, and how to find your way around the interface in five minutes.

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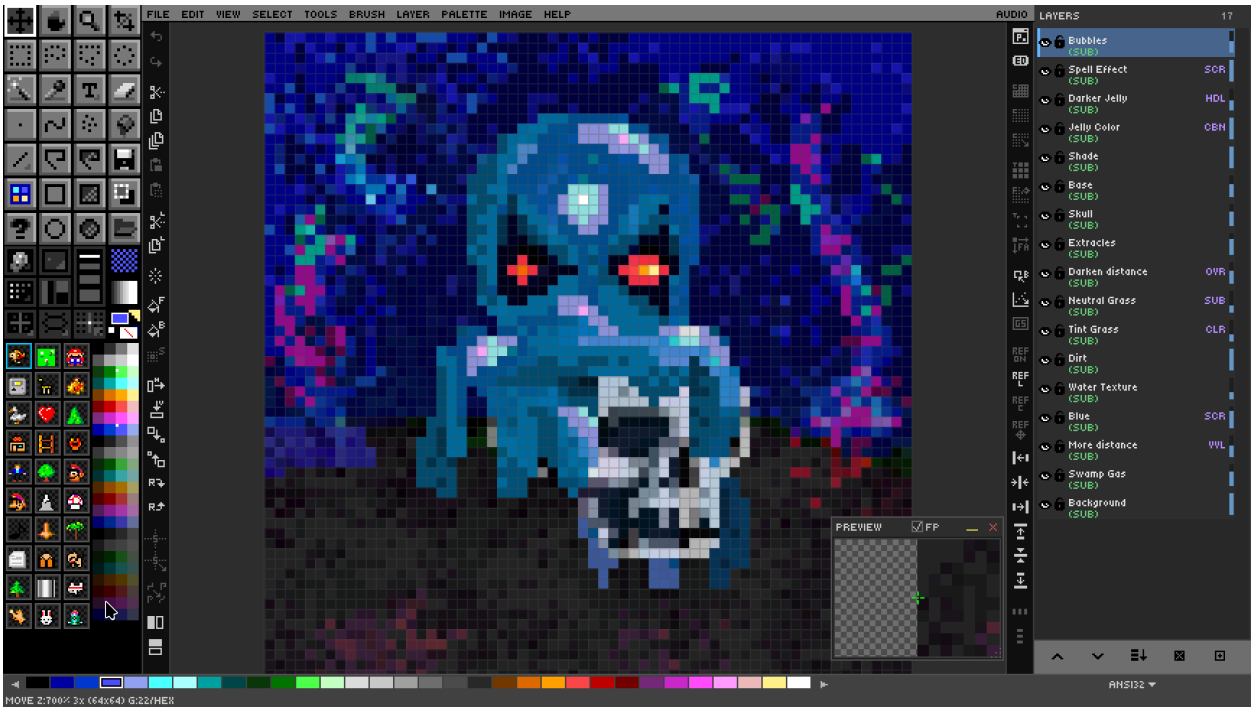
## What Is DRAW?

> 🎯 **Goal:** Understand what DRAW is and why it exists.

DRAW is a pixel art editor written entirely in QB64-PE — a modern reimagining of Microsoft's QuickBASIC compiler. That alone would make it a curiosity. What makes it genuinely different is its signature party trick: **DRAW can export your pixel art as runnable QB64 source code**. Open the resulting `.bas` file in any text editor, hit compile, and your sprite is reborn as a self-contained BASIC program that draws itself pixel by pixel.

The project takes its inspiration from the legends of the genre — BR0DERBUND's **DazzleDraw** (on the Apple ][), Electronic Arts' **Deluxe Paint** (DPaint DP2 enhanced for DOS), **Grafx2** by Sunset Design, Cosmigo's **ProMotion**, a little **Inkscape**, a little **GIMP**, and of course Adobe **Photoshop**, as well as the other DP spiritual heirs that followed including **DPaint.js**, etc. If you grew up with Amiga art, DOS demoscene tools, or the modern Aseprite, or LibreSprite, DRAW will feel familiar within seconds.

DRAW is **cross-platform** (Windows, Linux, macOS), **open source** (MIT licensed), and **dependency-free** at runtime — once it's built, you have a single executable plus its `ASSETS/` folder. It ships with **64 layers**, **19 blend modes**, a complete **text system** with rich formatting, **theming**, **audio**, **custom brushes**, and **symmetry drawing**. Its native `.draw` file format is a regular PNG file with a custom embedded chunk, so any image viewer can preview your art even though no other application can re-edit it. You don't need to save your file as .draw.png or .draw, as it recognizes draw files automatically.



# Getting DRAW — Download, Build & Install

> 🎯 **Goal:** Get DRAW running on your machine.

There are two paths to a working install: **download a pre-built release** (recommended for almost everyone) or **build from source** (recommended if you want to hack on DRAW itself).

## Option A — Pre-built release

Visit the DRAW Releases page and grab the archive that matches your platform. Each release ships as a self-contained zip containing the `DRAW` executable, the entire `ASSETS/` tree, and a default `DRAW.cfg`. There are no separate library installs, no runtime dependencies, and no internet connection required for normal operation.

## Option B — Build from source

You will need the QB64-PE compiler (from github.com/QB64-Phoenix-Edition/QB64pe). You need v4.4+. Once it is installed and on your `PATH`:

```
git clone https://github.com/grymmjack/DRAW.git
cd DRAW
qb64pe -w -x -o DRAW.run DRAW.BAS
```

The repository also ships a `Makefile` with shortcuts (`make`, `make clean`, `make run`).

## Platform-specific setup

- **Linux** — the `INSTALL/` folder contains a `.desktop` file and MIME-type registration for the `.draw` format so you can double-click `.draw` files in your file manager and have them open in DRAW.
- **Windows** — an installer registers the application in the Start Menu and associates `.draw` with the executable in the Registry.
- **macOS** — run `INSTALL/install-mac.command` once; it copies the app bundle into `/Applications` and sets up Launch Services associations.

## First launch

On first run, DRAW probes your monitor with `_DESKTOPWIDTH` / `_DESKTOPHEIGHT` and chooses a sensible **display scale** (1× through 8×) and **toolbar scale** (1× through 4×). It writes a fresh `DRAW.cfg` next to the executable. That file is plain text — you can hand-edit it any time, and you can override the defaults entirely with `--config /path/to/your.cfg`.

# Your First 5 Minutes — UI Tour

> 🎯 **Goal:** Navigate the interface confidently.

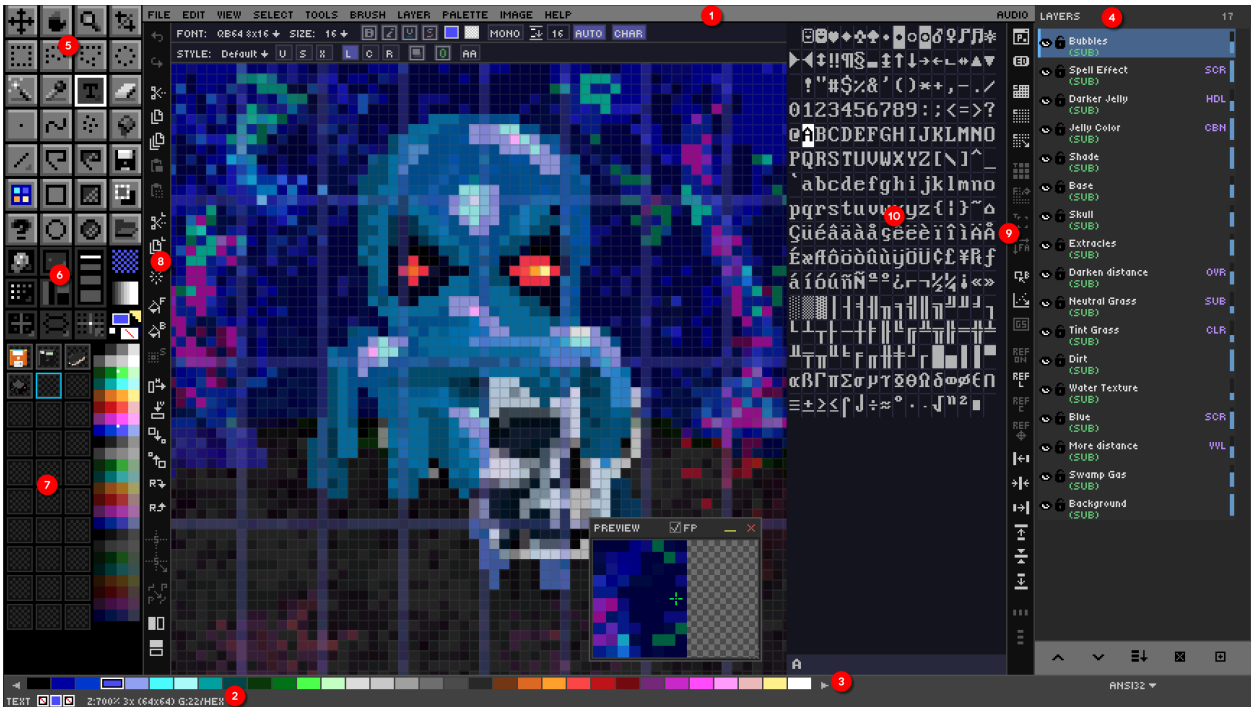
When DRAW launches you'll see six regions. From top to bottom and outside in:

| Region | Purpose |
|---|---|
| **Menu bar** | 11 menus from File through Audio. |
| **Toolbar** | A 4×7 grid of 28 tool icons on the left. |
| **Canvas** | The drawing surface, centered. |
| **Layer panel** | Stack of layers on the right. |
| **Palette strip** | The current palette across the bottom. |
| **Status bar** | Coordinates, zoom, current tool, grid state. |

A second tier of **hidden panels** can be summoned on demand:

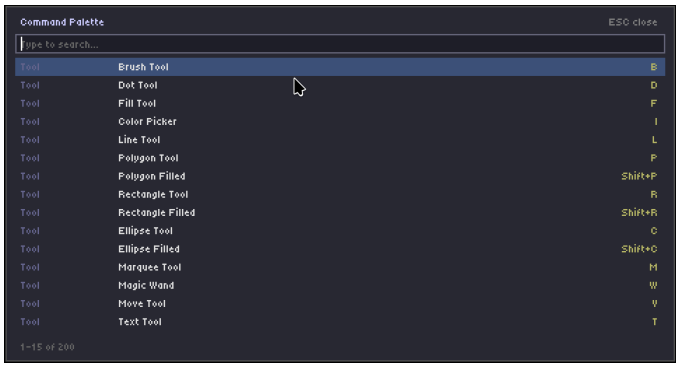
- **Organizer Widget** — brush presets and toggles.
- **Drawer Panel** — 30 reusable brush / pattern / gradient slots.
- **Edit Bar** ( `F5` ) — quick edit actions.
- **Advanced Bar** ( `Shift+F5` ) — 26+ toggles.
- **Preview Window** ( `F4` ) — magnifier or floating image.
- **Character Map** ( `Ctrl+M` ) — 16×16 glyph grid for text-mode work.

Every dockable panel can be flipped to the opposite side with **Ctrl+Shift+Click** on its title or icon area. Panels can also be dragged by their title into stacks, tabs and floating windows, and the floating windows dock beside them (see Chapter 22). `F11` toggles *all* UI for distraction-free drawing; `Ctrl+F11` keeps only the menu bar; `Tab` toggles the toolbar alone.




1. Menu Bar
2. Status Bar
3. Palette Strip
4. Layer Panel
5. Tool Box
6. Organizer
7. Drawer Bins
8. Edit Bar
9. Advanced Bar
10. Character Map

The single most important keystroke to remember on day one is `?` — that opens the **Command Palette**, a fuzzy-searchable list of every command in the app. If you can't remember a shortcut, type a few letters and hit Enter.




➡️ Next: Chapter 2 — Core Drawing Fundamentals

# Ch. 02 🖌️ Core Drawing Fundamentals

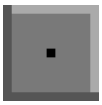
> **What you'll learn:** Free-hand brushwork, geometric shapes, polygons, the flood-fill tool, the spray can, and the eraser — the seven tools that account for 90% of pixel art.

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## Brush, Dot & Freehand Drawing

> 🎯 **Goal:** Master basic freehand pixel drawing.

The two beginner tools you'll meet first are the **Brush** ( `B` ) for freehand strokes and the **Dot** ( `D` ) for single-pixel placement. Both share the same brush size and shape state.

| Tool | Icon | Key | Behaviour |
|---|---|---|---|
| Brush |  | `B` | Drag to paint a continuous stroke. |
| Dot |  | `D` | Click to stamp one pixel (or one brush footprint). |

**Brush size** runs from 1 to 50 pixels and is adjusted with `[` (smaller) and `]` (larger). The **brush shape** toggles between circle and square with `\`. To preview the current footprint and color before committing, hold the backtick key ( `` ` ``).

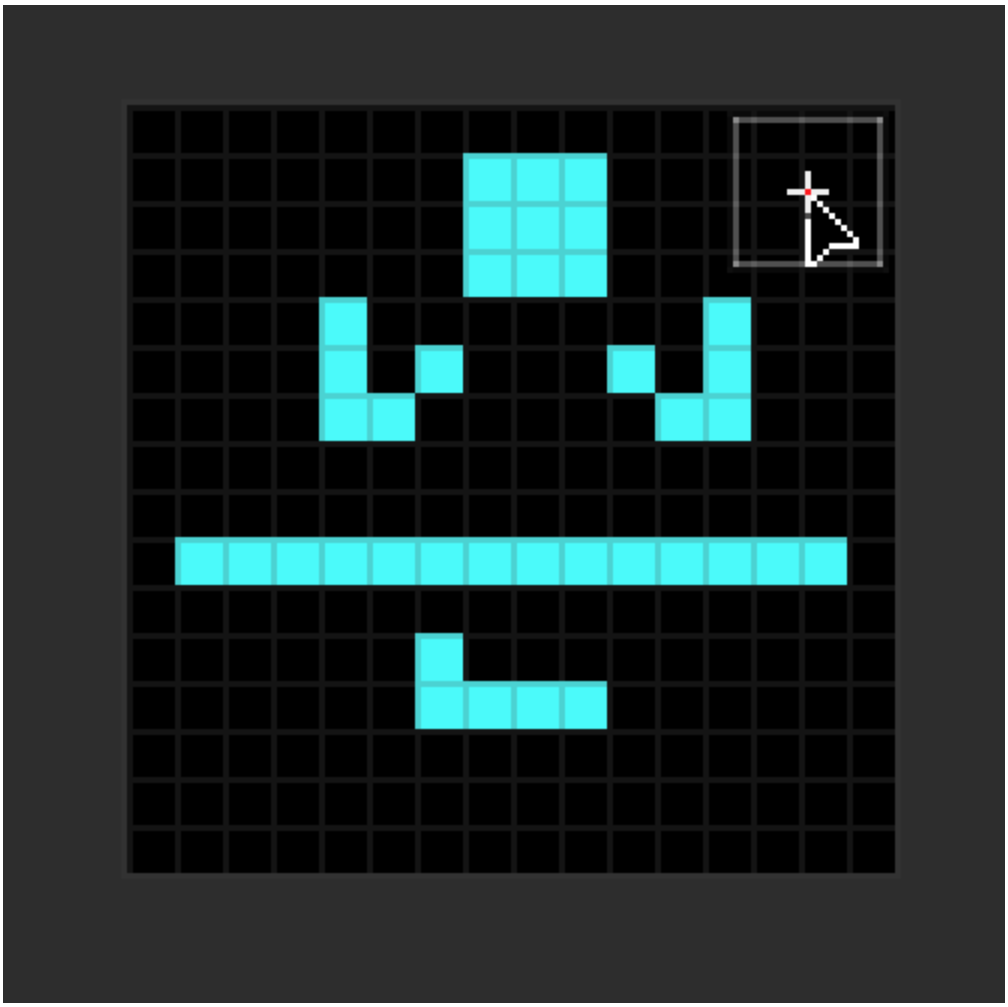
DRAW follows the universal "left = foreground, right = background" convention. **Left-click paints with the FG color**, **right-click paints with the BG color**. Holding `Shift` while drawing constrains the stroke to a perfectly horizontal or vertical axis. Holding `Shift` and right-clicking draws a **connecting line** from the previous click to the current position — invaluable for stitching together long straight lines without ever touching the Line tool.

If you have a steady hand but find that single-pixel stairsteps creep into your strokes, enable **Pixel Perfect mode** with `F6`. DRAW will retroactively remove L-shaped corners as you draw, leaving cleaner outlines.

The Organizer widget on the left contains **four brush size presets** (1, 3, 5, 9 pixels for sprite work).

**🎨 Try it — your first 16×16 sprite**

1. `Ctrl+N` for a new canvas at 16×16 pixels.
2. Press `D` for the Dot tool, set size to 1 with `[`.
3. Sketch out the silhouette of a simple character.
4. Switch to `B` (Brush), increase to size 2, and fill the larger regions.
5. Save as `.draw` so you can come back later.



### Pen Tablets & Pressure

DRAW reads **pen pressure** from a drawing tablet: Wacom, Huion, XP-Pen and Surface Pen are all supported, plus a Force Touch trackpad on a Mac. Pressure can change the Brush and Eraser **size** and **opacity**, and how much paint the mix brush lays down. It changes smoothly within a single stroke. A mouse is never affected.

**View → Pen Pressure** opens a small panel with everything in one place:

- **Live meter:** shows how hard you're pressing.
- **Size, Opacity, Mix Paint:** each has a checkbox and a **LIGHT** slider, which is the value at the lightest touch. Full pressure always gives 100%.
- **Curve slider:** shapes the response; 1.00 is linear.
- **TIP slider:** the tip threshold. The pen draws only once it presses at least this hard (default 3%). It's shown as a red mark on the meter. This stops stray dots when the pen grazes the tablet and registers a click. A graze that never passes the threshold draws nothing and adds no undo step.
- **Response graph:** plots the curve, with a dot that follows your pen.

Opacity by pressure shows live as you draw. Going over the same spot again never builds the paint up past the strongest pressure you used there.

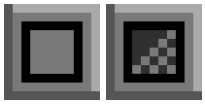
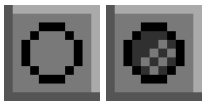
- **Setup.** Install the tablet maker's driver.
  - **Windows:** leave **Windows Ink** turned on in the driver's pen settings.
  - **macOS:** keep the driver's tablet app running in the menu bar.
  - **Linux:** most tablets work without any setup, and the maker's driver works too.
- **Settings → General → Pen Tablet:**
  - **Pen Pressure** turns the feature on or off.
  - **Pressure Sets Brush Size** controls whether pressure changes the size. It applies to brushes bigger than 1px; a 1px pencil stays 1px.
  - **Lightest Touch Size %** is the size at the lightest touch (default 20% of the brush).
  - **Pressure Curve** sets how pressure maps to size, opacity and paint.
  - **Tip Threshold %** is the pressure the pen needs before it draws (default 3%). 100 is linear, lower makes light touches count more, and higher means you press harder.
  - **Pressure Sets Opacity** and **Lightest Touch Opacity %** (default 10%): a light touch paints faintly. Off by default.
  - **Pressure Sets Mix Paint** and **Lightest Touch Paint %**: with the mix brush, a light touch smears and full pressure lays paint.
- **Export.** Pressure-varied strokes export to BAS/QB64 as pixels.

## Lines, Rectangles & Ellipses

> 🎯 **Goal:** Draw clean geometric shapes.

When freehand stops being precise enough you reach for the geometric trio. All three respect the current brush size, color, and symmetry settings.

| Tool | Icon | Outlined | Filled |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Line |  | `L` | — |
| Rectangle |  | `R` | `Shift+R` |
| Ellipse / Circle |  | `C` | `Shift+C` |

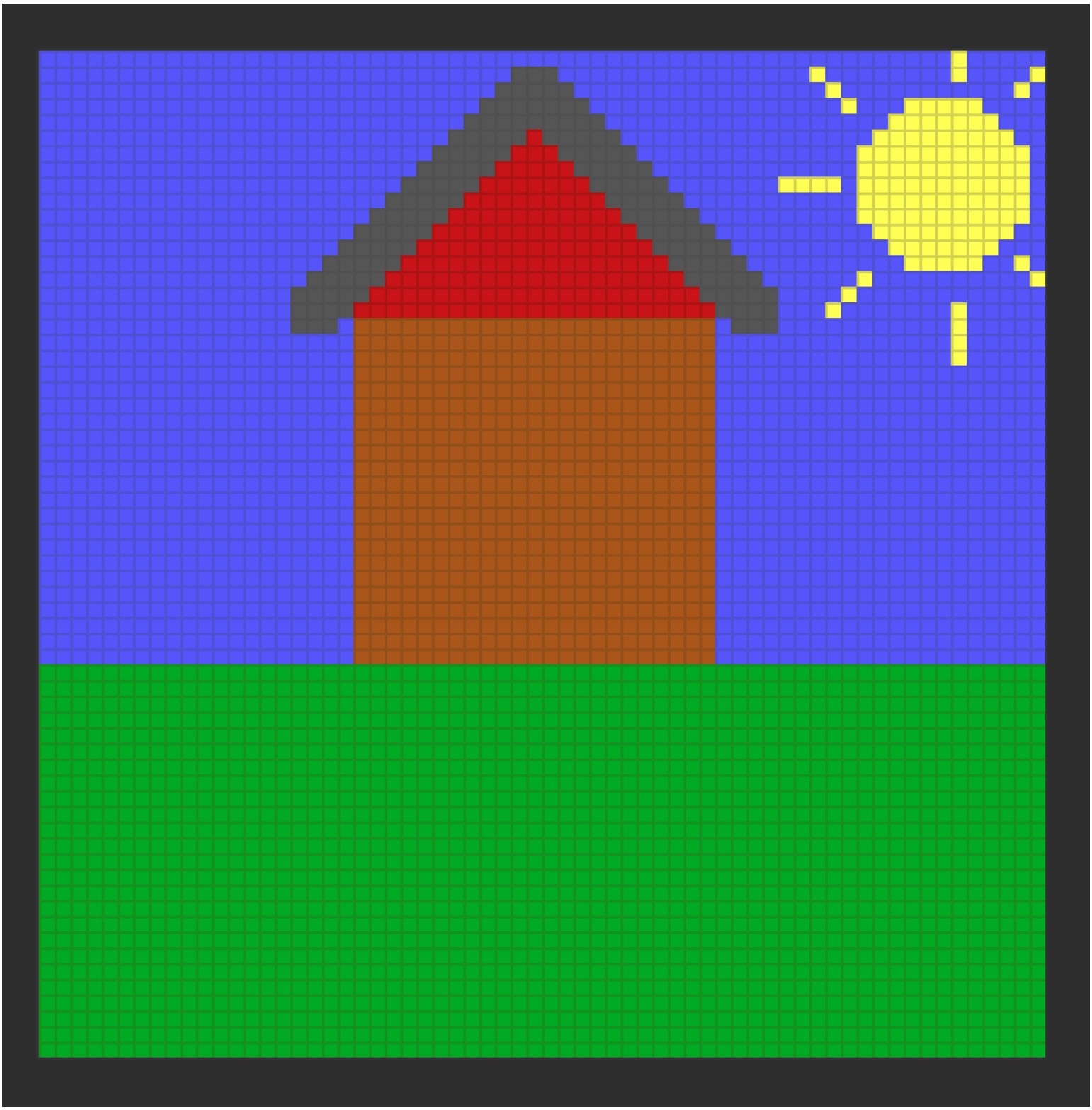
**Tip:** While dragging any shape, hold SPACE to move the entire shape with the mouse instead of resizing it. Release SPACE to resume resizing from the new anchor. This works for all geometric and Smart Shape tools, and honors grid snap.

The shared modifier vocabulary across all three:

- **Shift** — constrain (line: H/V; rect: square; ellipse: circle).
- **Ctrl** — perfect aspect (forces square / circle even after you start dragging).
- **Ctrl+Shift while drawing the rect or ellipse** — anchor from the *center* instead of the corner.
- **Ctrl+Shift while drawing a line** — angle snap to 15°/30°/45°/90° increments.

🎨 **Try it — house in 30 seconds**

1. Filled rectangle for the body. `Shift+R`
2. Two lines forming the roof triangle. `L`
3. Filled circle (the sun). `Shift+C`
4. A few short lines for the sun's rays. `L`
5. Filled rectangle for the ground. `Shift+R`



# Polygons & the Fill Tool

> 🎯 **Goal:** Draw complex shapes and fill regions.

The **Polygon** tool comes in two flavours: outlined ( `P` ) and filled ( `Shift+P` ). Each click adds a vertex to the in-progress polygon; press `Enter` to close and commit. As with the line tool, **Ctrl+Shift** snaps the segment angle.

The **Flood Fill** tool ( `F` ,  ) pours the FG color into every pixel contiguous with the click point that shares its starting color. By default it samples only the active layer; hold `Shift` to sample from the **merged visible composite** (useful when you have outlines and fills on separate layers).

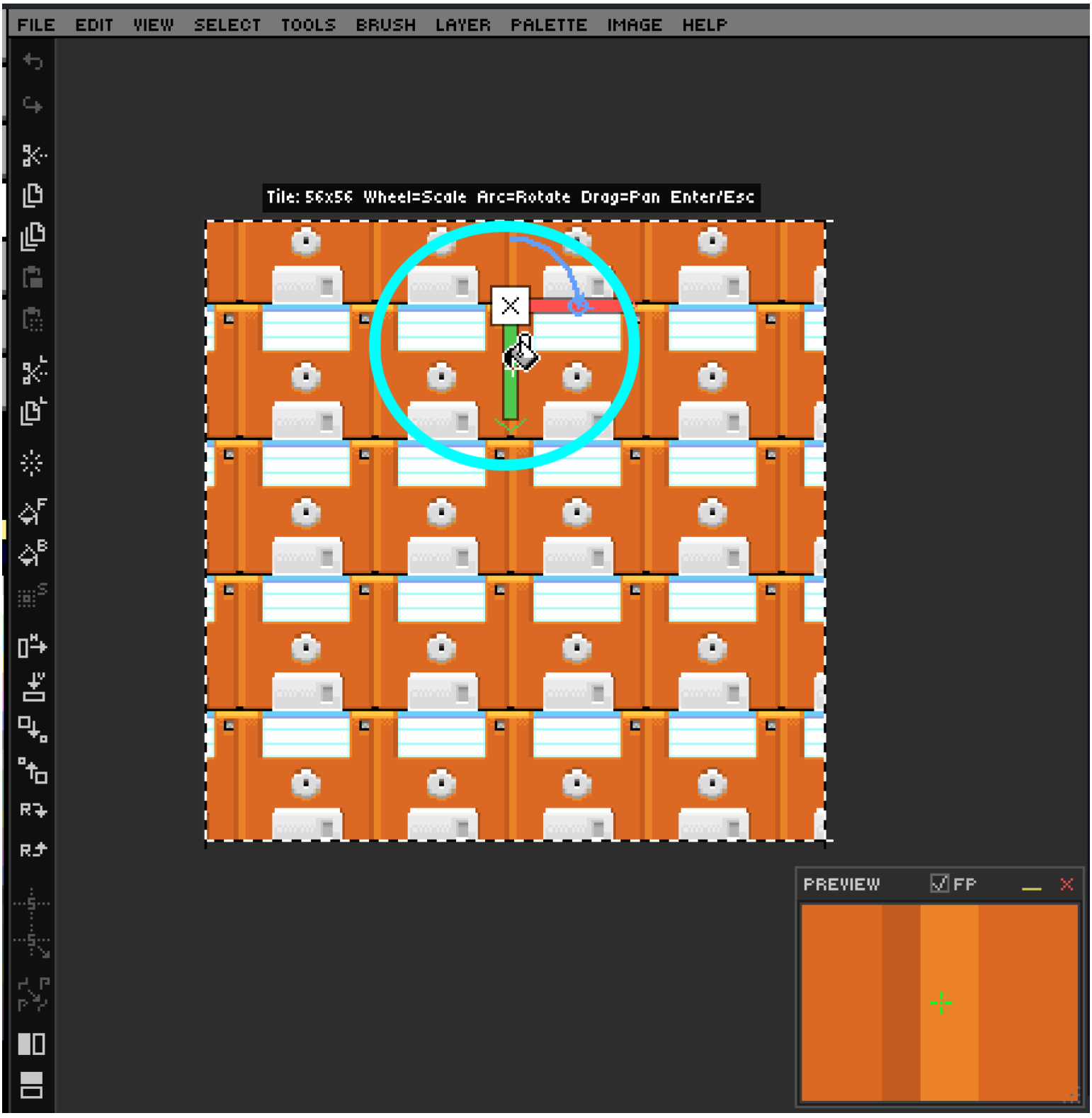
DRAW's flood fill is unusual in two ways:

1. It supports **custom brushes as a tiled fill** — if you have a brush captured (Chapter 9), the fill is rendered as a seamless tiling of that brush rather than a flat color.
2. It also honours the **paint mode** — set Pattern or Gradient mode (Chapter 9) and your fill becomes a dithered gradient or tiled pattern instead of a solid color.

Press `F8` to enable **Fill Adjustment overlay**. Now when you fill using pattern mode or gradient mode, or with custom brushes in color mode (loaded from the bins), drag the canvas to reposition the tile origin, mouse-wheel for uniform scale, drag the L-handle for independent X/Y scaling, and drag the rotation handle (small arc) to rotate the tile. `Enter` applies, `Esc` cancels.

**🎨 Try it — tileable pattern fill**

1. Draw a small 8×8 motif.
2. Capture it as a custom brush (see Chapter 9).
3. On a fresh layer, flood-fill a large rectangle with the brush active.
4. Press `F8` and experiment with scale, rotation, and offset.



## Spray Tool & Eraser

> 🎯 **Goal:** Use spray-can effects and clean up mistakes.

The **Spray** tool ( `K` ,  ) emits randomized dots within a circular nozzle. The nozzle radius **doubles for every brush-size step**, and density scales with radius — small brush = pinpoint mist, large brush = thick coverage. Spray respects custom brushes too: every "drop" stamps the brush instead of a single pixel, which is wonderful for foliage, dust, and confetti.

The **Eraser** ( `E` ,  ) paints fully transparent pixels — it does not paint with the BG color, it removes alpha. Two non-obvious tricks make it indispensable:

- **Hold `E` to temporarily eraser anything** while a different tool is active. Release `E` to return to your previous tool. This is the fastest way to nudge an outline or clean a stray pixel.
- **Smart Erase** ( `Shift` while erasing) operates on **all visible layers at once**, with per-layer history tracking so each layer's undo step is independent.

The eraser uses your current brush size, shape, and even custom-brush stamp. The status bar shows `FG:TRN` to remind you transparent painting is active.

> 💡 **Tip:** Combine the eraser with a layer's **Opacity Lock** (Chapter 4) to clean up overpaint while preserving the layer's silhouette.

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## Smart Shapes & Advanced Tools

> 🎯 **Goal:** Draw parametric shapes, 3D dice, and advanced curves.

### Smart Shapes Tool Group

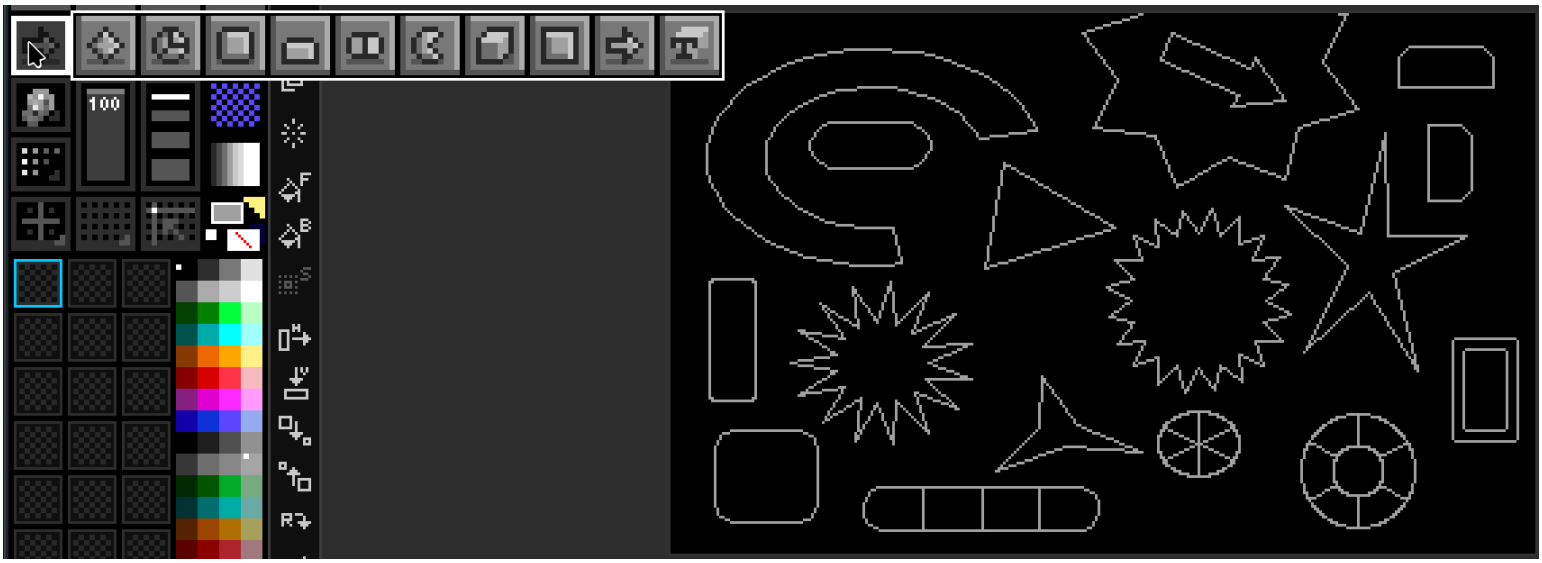
The **Smart Shapes** tool group is accessed from the toolbar. **Long-press or right-click** the Smart Shapes button to open a flyout menu and pick a sub-tool. Each shape has live drag-time arrow-key tweaks and unique modifiers.

**Tip:** While dragging any Smart Shape, hold SPACE to move the entire shape with the mouse instead of resizing it. Release SPACE to resume resizing from the new anchor. Grid snap is honored.

**Sub-tools:**

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### Smart Shapes — Detailed Controls & Behaviors



The idea of smart shapes is that you begin drawing, then use keys to modify the shape as you create it. All smart shapes snap to the grid and honor grid/symmetry settings, just like regular rectangles and ellipses.

Below are the controls and behaviors for each smart shape:

### Polygon

> Minimum sides: 3, Maximum: 30

Draws a regular polygon. While dragging, use:

- **Up arrow**: Add edges
- **Down arrow**: Remove edges
- **Left arrow**: Decrease point/star depth (makes a star/burst)
- **Right arrow**: Increase point/star depth

### Pie / Donut

> Create pie charts or donut shapes with equidistant or custom slices. Left/right arrows add a hole (donut mode). Minimum slices: 0, Maximum: 30. Max hole radius: 90% of shape size.

- **Up arrow**: Add segments
- **Down arrow**: Remove segments
- **Left arrow**: Decrease hole size
- **Right arrow**: Increase hole size

### Rounded Rectangle

> Rectangle with rounded corners. Corner radius can be 1 (almost square) up to 30 (fully rounded), but never so large that corners intersect.

- **Up arrow**: Increase corner radius
- **Down arrow**: Decrease corner radius

### Tab

> Rounded rectangle with one flat edge. Flat edge can be cycled to any side (top, bottom, left, right).

- **Up arrow**: Round top corners more
- **Down arrow**: Round top corners less
- **Left arrow**: Cycle flat edge left
- **Right arrow**: Cycle flat edge right

### Pill

> Capsule/oval with dividers for options (like a segmented control). Horizontal by default.

- **Up arrow**: Increase roundness
- **Down arrow**: Decrease roundness
- **Left arrow**: Decrease segments
- **Right arrow**: Increase segments

### Pac-Man

> Pie chart with a single slice cut out (the mouth). Mouth can be opened/closed. Inner hole for dial/knob effects.

- **Up arrow**: Increase mouth size
- **Down arrow**: Decrease mouth size
- **Left arrow**: Decrease inner hole
- **Right arrow**: Increase inner hole

### 3D Cube / 3D Polygon (Dice)

> Draws a 3D cube or polyhedral dice wireframe. Press number keys while dragging for dice types: 4 = D4, 6 = D6, 8 = D8, 0 = D10, 1 = D12, 2 = D20, 3 = D30.

- **Up arrow**: Increase Z-depth
- **Down arrow**: Decrease Z-depth
- **Left arrow**: Rotate left
- **Right arrow**: Rotate right
- **Mouse wheel up**: Rotate up
- **Mouse wheel down**: Decrease Z-depth (angle)

### Bevel Rect

> Rectangle with beveled (angled) edges. Bevel and border size adjustable. Can switch between inner/outer bevel.

- **Up arrow**: Increase bevel size
- **Down arrow**: Decrease bevel size
- **Left arrow**: Decrease border size
- **Right arrow**: Increase border size
- **I**: Switch to inner bevel
- **O**: Switch to outer bevel
- **Mouse wheel up**: Increase Z-depth (angle)
- **Mouse wheel down**: Decrease Z-depth (angle)

### Arrow

> Arrow with adjustable stem and head. Head can be made concave/curved.

- **Up arrow**: Make stem fatter
- **Down arrow**: Make stem skinnier
- **Left arrow**: Make arrow head shorter
- **Right arrow**: Make arrow head longer
- **Mouse wheel up**: Increase head concavity
- **Mouse wheel down**: Decrease head concavity

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- **Right-click drag** — solid mode (BG color fill, FG color wireframe)
- Dice type: `4` / `6` / `8` / `0` / `1` / `2` for D4/D6/D8/D10/D12/D20

## Bezier Curve Tool ( `Q` )

- `Escape` — cancel

## Line Tool — End Caps

## While dragging the Line tool, press `s` / `e` to cycle the start / end cap shape (none → arrow → diamond → circle → square → ...). End caps render in the same FG color as the line.

➡️ Next: Chapter 3 — Color & Palette Mastery

# Ch. 03 🎨 Color & Palette Mastery

> **What you'll learn:** How DRAW thinks about color, the FG/BG/swap dance, the Color Picker and live Color Mixer, the 56 bundled palettes, and the palette-ops mode that lets you remap entire compositions in seconds.

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## Color Basics — FG, BG & Palette Strip

> 🎯 **Goal:** Select, swap, and manage colors.

The palette strip across the bottom of the screen is the fastest way to choose colors. **Left-click a swatch** to set the foreground (FG) color; **right-click a swatch** to set the background (BG) color. The active FG and BG swatches are echoed in the status bar.

| Action | Key / mouse |
|---|---|
| Swap FG and BG | `X` |
| Reset to white FG / black BG | *Palette menu → Default Colors* |
| Set BG to transparent | `Shift+Delete` |
| Scroll the palette strip | Mouse wheel over strip |
| Fast-scroll 32 colors at a time | `Shift` + Mouse wheel |
| Temporary FG eyedropper (any tool) | `Alt` +Left-click on canvas |
| Temporary BG eyedropper (any tool) | `Alt` +Right-click on canvas |

**Paint opacity** — the keys `1` through `9` set strokes to 10–90% opacity, and `0` returns to fully opaque 100%. DRAW uses **per-stroke compositing**, which means a 50% stroke that overlaps itself does not get *more* opaque the way Photoshop's brush would; the stroke is rendered once at the end as a single 50% pass. This makes for predictable, pixel-art-friendly results.

### The Opacity Slider

Use the opacity slider in the organizer of the toolbox to set whatever opacity level you wish from 1% to 100%.




| Action | Key / mouse |
|---|---|
| Set Opacity from 10% to 100% | `1-9` for 10% to 90%, or `0` for 100% opacity |
| Mouse Wheel Up | Increase opacity by 1% |
| Mouse Wheel Down | Decrease opacity by 1% |
| `Shift` + Mousewheel | Increase or decrease opacity by 10% |

> Opacity slider will be outlined in a bright color if you set opacity to `⩽` the percentage according to `DRAW.cfg` setting: `OPACITY_WARN_THRESHOLD`

# Color Picker, Color Mixer & Custom RGB

> 🎯 **Goal:** Use the full RGB color picker and the live Color Mixer.

Click either the FG or BG swatch in the status bar to open the **Color Picker**. It supports true 24-bit RGB selection plus a hex input field. The Picker tool ( `I` ) on the canvas pairs an eyedropper with a **loupe overlay** that magnifies the area under the cursor and prints the RGB and hex values for the pixel you'd sample.

The **Color Mixer panel** is a floating, persistent alternative to the modal picker. Open it from `View → Color Mixer` (or via the Command Palette). It exposes:

- RGB sliders — adjust Red, Green, Blue independently.
- HSV sliders — adjust Hue, Saturation, Value.
- A hex input field — paste any hex value directly.
- FG and BG swatches — click to apply the mixed color.

Because the mixer is non-modal, you can keep it open while drawing and tweak colors live. Its visibility is persisted in `DRAW.cfg` .

## 3D Color Space — OKLab, CIELAB, XYZ & RGB

`View → 3D Color Space` opens a floating panel that shows every color your screen can display (the sRGB gamut) as a 3D solid. Different color spaces arrange those same colors differently:

- **OKLAB** (default): a perceptual space (Björn Ottosson, 2020). Equal distances look like equal differences to your eye. It looks like a tilted droplet with black at the bottom and white at the top.
- **LAB**: CIELAB (1976), the classic perceptual space.
- **XYZ**: the CIE 1931 space that all the others are defined from.
- **RGB**: the familiar RGB cube, standing on its black-white diagonal.

How to use it:

- **Left-click or left-drag on the solid** to pick the color under the cursor. It becomes your FG color, and what you see is exactly what you get.
- **Right-drag or middle-drag**, or left-drag on empty space, to spin the view. The **wheel** zooms.
- **SOLID / CLOUD** switches between the gamut's surface and a grid of color dots through the whole volume.
- **SLICE** cuts the solid at a lightness you set with the slider. The cut face shows every color at that lightness, so interior colors become pickable.
- The readout below the view shows the hovered (or current) color in hex/RGB, OKLab, OKLCh, CIELAB and XYZ.

The panel auto-hides while you draw over it and is included in `F11` (hide/show all UI). The space, view, slice, rotation and zoom are remembered in `DRAW.cfg` .

## Perceptual Gradients & OKLCh Ramps

Gradient fills blend in **OKLab** by default ( `Settings → Panels → Color Blending + Ramps → Gradient Blend` ). Plain RGB blends sag in the middle: red→green passes through a muddy olive. OKLab passes through a clean yellow-orange instead. Choose *sRGB (classic)* for the old behavior, *Linear light* for physically mixed light, or *Pigment (paint)* to blend like mixed paint (see below).

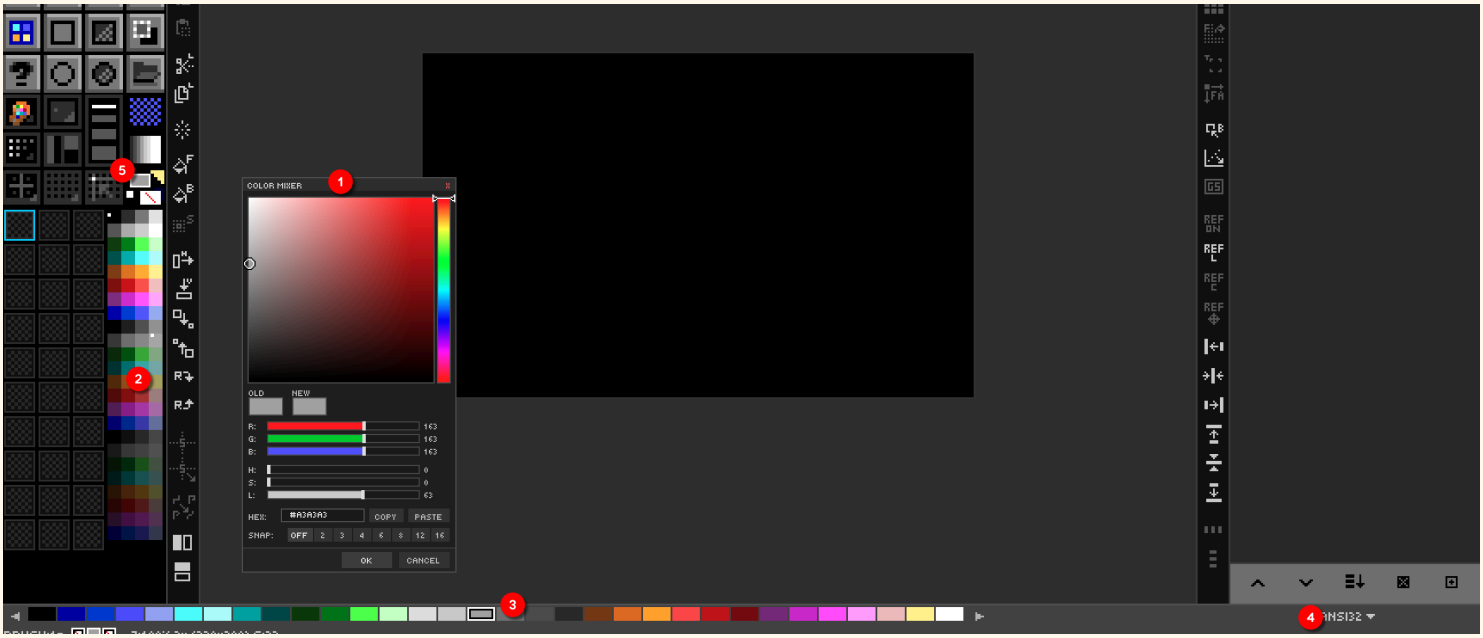
`Palette → Generate OKLCh Ramp from FG` turns the FG color into a new palette: an evenly spaced dark-to-light ramp of the same hue. It's saved under your *Created* palettes as `Ramp RRGGBB` and selected straight away. The same Settings section controls the number of colors, the darkest and lightest lightness, a hue shift at the ends (warmer lights and cooler darks), and how much color the ends keep.

### Pigment Mixing — Mix Brush & Mix Palettes

Light mixes additively, so on a screen blue + yellow makes gray. Paint mixes *subtractively*: each pigment absorbs part of the spectrum, so blue + yellow paint makes green. DRAW models paint with **Kubelka–Munk** pigment mixing, using a port of Spectral.js.

- **`Brush → Mix Colors (Pigment)`** turns on the mix brush. The brush picks up the colors it crosses and smears them along, mixing them with your paint like wet paint (a smudge model after MyPaint's).
  - Blue dragged through yellow paints green, and through red a deep brown.
  - The brush stays wet: carry it off the paint and it keeps laying down the mixed color for a while.
  - **Smudge** (default 50%) is how much of each dab is the picked-up color. 0% means plain paint; 100% is a pure smudge tool that only pushes existing paint around.
  - **Smudge Length** (default 50%) is how long the brush holds a color. Low values follow the canvas closely; high values drag colors far; 100% keeps the first color picked up.
  - **Smudge Radius** (default 100%) is how big an area under the brush is sampled.
  - The brush samples the layer as it was when you started the stroke.
  - All three are in `Settings → Panels → Color Blending + Ramps`.
  - The eraser and custom brushes don't mix.
  - Pixel-perfect mode and undo work as usual.
- **`Palette → Generate Pigment Mix FG > BG`** makes a palette that mixes the FG color into the BG color like paint. It's saved under *Created* palettes as `Mix RRGGBB-RRGGBB`. The number of colors comes from the *Ramp Colors* setting. The BG can't be transparent.
- The *Pigment (paint)* gradient blend (above) applies the same mixing to gradient fills.

DRAW's various color widgets and doo-dads




1. Color Mixer — `View → Color Mixer`
2. Mini Palette — Always visible as part of Tool Box

3. Color Strip — Can be hidden, shows (4) Palette
4. Palette Picker — Choose a palette for editing
5. Active Foreground (FG) and Background (BG) colors

# Palette Management — 56 Built-in Palettes

> 🎯 **Goal:** Switch, browse, and manage palettes.

DRAW ships with **56 palettes** in GIMP's `.gpl` format. They live under `ASSETS/PALETTES/` and include classics — **NES**, **PICO-8**, **Commodore 64**, **Game Boy**, **Endesga 32/64**, **DawnBringer 16/32**, **AAP-64**, **Sweetie 16**, **Resurrect 64**, **CGA**, **EGA**, **VGA**, **Amiga**, **MSX** — plus 40 more.

Click the palette name above the strip to open the dropdown. **Pressing a letter** while the dropdown is open jumps to the next palette starting with that letter — handy when you know the name but not the position.

Palette workflows DRAW supports natively:

- **Download from Lospec** — DRAW can fetch palettes from Lospec's online database directly.
- **Create palette from existing image** — distill the unique colors of any open image into a new palette.
- **Import / Export** `.gpl` — interchange with GIMP, Aseprite, Krita.
- **Remap existing artwork** — recolor a finished piece into a different palette while preserving structure.



# Swatches Panel — Your Swatch Sets, Laid Out Your Way

> 🎯 **Goal:** Keep your own swatch sets — ramps, gaps and all — one click away, beside the palette strip.

**View → Swatches** opens a panel with a grid of color swatches. It does not replace the palette strip: the strip stays the document's palette, and the Swatches panel shows any swatch set you like next to it.

- **Left-click** a swatch → it becomes the **FG** color. **Right-click** → **BG**.
- The FG swatch gets a double frame and the BG swatch a single one, so you can see where you are in the set.
- Hover a swatch to see its hex value and name on the panel's bottom line.
- **Wheel** scrolls the grid; drag the thin scroll bar on the right for big sets.

**Choosing a set.** Click the set's name at the top of the panel. The list shows:

- **Document Palette** — the palette the strip shows, 16 to a row.
- **My Swatches** — sets you loaded (see below).
- **Swatches** — the sets that ship with DRAW.
- **Palettes** — the 56 built-in palettes, and your own saved ones under **My Palettes**.

**Loading your own swatches.** Gear → **Load Swatches...** (or ☰ → **Load...**) opens Photoshop `.aco` swatch files, Adobe `.ase` swatch exchange files (Illustrator, InDesign, Photoshop), GIMP / Aseprite / Inkscape / Krita `.gpl` palettes, Photoshop `.act` color tables, JASC / RIFF `.pal`, Lospec `.hex`, Paint.NET `.txt` and `.png` images. DRAW never changes your original: it saves a `.gpl` copy in **My Swatches** (Gear → **Open My Swatches Folder** shows where) and uses that.

- An `.ase` file's groups each start a new row, and the first swatch of a group carries the group's name. CMYK, Lab and gray swatches come in as their RGB colors.
- A `.png` gives one swatch per distinct color, in reading order (left to right, top to bottom); transparent pixels are skipped. The number of colors on its first row sets the columns, so a palette strip or a grid of color cells comes back laid out the same way. An image with more than 1024 colors is refused.

**Saving a set in another format.** ☰ → **Save As...** writes the set as `.gpl` (the default, and the only one that keeps empty cells, row breaks and locks), `.aco`, `.ase`, `.act`, `.pal`, `.hex`, `.txt`, or a `.png` of the grid as you see it (one cell per swatch, row breaks start a new row, empty cells are transparent). Type a name without an extension and you get a `.gpl`. The panel keeps showing the set you were on.

**Laid out the way you made it.** A `.gpl` remembers its column count. A Photoshop `.aco` does not, so DRAW works it out from the ramps. Many Photoshop swatch sets pad their rows with black to line the ramps up; with **Black filler = empty cells** on (the default), that padding becomes empty cells and only the first black stays as a color.

**The gear** (the cog at the top right) changes the layout:

| Option | What it does |
|---|---|
| **View** | **GRID** = swatches side by side · **LIST** = one row per swatch: the color, its hex and its name. A row break shows as a divider. Everything else works the same; in a list a drag drops above or below a row |
| **Columns** | **SET** = the set's own column count · **FIT** = as many as fit · or a fixed 8 / 12 / 16 / 24 / 32 |
| **Cell** | swatch size, 6 – 24 px |

| | |
|---|---|
| **Gap** | space between swatches, 0 – 3 px |
| **Stretch to width** | swatches grow (or shrink) to fill the panel's width |
| **Square cells** | keep swatches square when stretching |
| **Cell borders** | a thin border around every swatch |
| **Recent colors row** | a row above the swatches with the FG colors you used last, newest at the left (each once). **Click** one = FG, **right-click** = BG. DRAW remembers them between sessions |

### Swatches and the palette strip

The **palette strip** is the document's own palette; a swatch set is a separate collection you keep across documents. To move colors between them:

- **Drag** selected swatches **onto the strip**: they are inserted after the chip under the pointer (drop past the last chip to add them at the end). The panel's bottom line reads **ADD n TO THE STRIP** while you are over it. The swatches stay in the set.
- ☰ → **Selected to Strip: Replace / Add** does the same from the menu.
- ☰ → **New Set from Strip** goes the other way.

Each change to the strip is one undo step: `Ctrl+Z` with the pointer over the canvas takes it back.

### Selecting swatches

The edits below act on the **selected** swatches. A selected swatch gets a thin frame inside it, in a color that stands out against it.

| Do this | To |
|---|---|
| **Click** a swatch | select it alone (it also becomes the FG) |
| `Shift` **+click** | select the range from the last clicked swatch (row breaks are skipped) |
| `Ctrl` **+click** | add it to the selection, or take it out |
| **Drag a box** from an empty place in the grid | select every swatch it touches; drag below or above the grid and it scrolls |
| `Ctrl` **+drag** (or `Shift`) a box | add to the selection |
| **Click an empty place** | select nothing |
| `Ctrl+A` with the pointer on the panel | select every swatch (over the canvas, `Ctrl+A` still selects the canvas) |
| `Esc` with the pointer on the panel | select nothing |

Right-click still sets the BG and leaves the selection alone. Switching to another set clears the selection. With more than one swatch selected, the panel's bottom line counts them.

### Editing swatches — your copy, saved as you go

DRAW never changes a swatch set it did not make. The first time you edit a shipped set, one of the palettes, the Document Palette set or a file you pointed DRAW at, it saves a copy of it in **My Swatches** under the set's name ( `Name`

`(1)` if that name is taken) and edits the copy from then on. The status bar says so. After that, every edit is saved straight away, so there is no Save step.

**The edit menu.** Click the ☰ button (left of the gear), or `Shift` **+right-click** in the grid to open it at the pointer. Shift+right-clicking a swatch that isn't selected selects it alone first; on a selected swatch, or an empty place, the selection stays. The menu's top line counts the selected swatches.

| Menu item | Key (pointer on the panel) | What it does |
|---|---|---|
| **Add FG Color** | `Ins` | adds the FG color after the last selected swatch (nothing selected: at the end) and selects it |
| **Delete** | `Del` | deletes the selected swatches; locked swatches stay |
| **Lock / Unlock** | `L` | locks the selected swatches (a small corner mark); if they are all locked already, unlocks them |
| **Duplicate** | | copies the selected swatches right after the last one; the copies become the selection |
| **Edit Color...** | double-click the swatch | opens the color dialog for it |
| **Rename...** | | names the swatch (the name shows when you hover it, and is saved in the `.gpl`) |
| **Insert Empty Cell** | | an empty cell before the first selected swatch (nothing selected: at the end) |
| **Insert Row Break** | | the swatches from the first selected one start a new row |
| **Remove Duplicates** | | keeps the first of each color; a locked swatch is never removed |
| **Ramp First > Last** | | the colors between the first and last selected swatch blend from one to the other (in the *Gradient Blend* space, like the palette strip's Shift+click ramp); empty cells are skipped, locked swatches keep their color |
| **Insert Ramp FG > BG** | | inserts **Steps** swatches from the FG to the BG after the selection (nothing selected: at the end) |
| **Insert OKLCh Ramp from FG** | | inserts an even dark-to-light ramp of the FG, using the *Generate OKLCh Ramp* settings (Settings → Panels → Color Blending + Ramps) |
| **Steps** chips | | how many swatches an inserted ramp has (the same setting as Palette → Generate OKLCh Ramp) |
| **Sort** chips: Hue · Light · Sat · Luma · Rev | | sort the selection (two or more selected), else the whole set, by hue, OKLab lightness, chroma or luminance — or reverse it. Only colors move: empty cells, row breaks and locked swatches stay put |
| **Selected to Strip: Replace** | | the selected swatches become the palette strip, in order (nothing selected: **All to Strip** — the whole set). The FG and BG keep their colors |
| **Selected to Strip: Add** | | adds them to the end of the strip; colors the strip already has are skipped, and it holds 256 at most |
| **Load...** / **Save As...** | | open a swatch file as a new set · save this set in another format (see above) |

| | | |
|---|---|---|
| **New Set from Strip** | | a new set in My Swatches with the strip's colors |
| **New Set from Image / from Selection** | | a new set with every color the image uses (all visible layers, as you see them), in reading order — only the selection's colors when there is a selection. More than 1024 colors is refused: select a smaller area |
| **Lospec...** | | opens the Lospec browser (Palette → Load from Lospec); each palette you download also becomes a set |
| **Find FG Color** | `F` | selects the swatch that is the FG color and scrolls to it — or, when the set doesn't have it, the nearest one (by how close they look, OKLab). The FG stays as it is |
| **Select All / Select None** | `Ctrl+A` / `Ctrl+D` or `Esc` | |
| **Undo / Redo** | `Ctrl+Z` / `Ctrl+Y` | the last swatch edit |

**Changing a swatch's color.** Double-click a swatch for the color dialog. Or keep one swatch selected and use the **Color Mixer**, **Advanced Color Picker** or **3D Color Space**: the swatch follows them live, and each drag is one undo step. The panel's bottom line reads `COLOR PANELS EDIT #RRGGBB` while that is on. Only those panels change it — picking a color from the palette strip, the canvas or another swatch just sets the FG. Turn it off with gear → **Color panels edit the selected**.

**Drag to rearrange.** Drag a selected swatch and every selected swatch goes with it; scattered ones land side by side. A bright bar shows where they will land: the left half of a swatch drops them before it, the right half after it, and past the end of a row drops them at the end of that row. Hold `Ctrl` when you let go to **copy** them instead (the copies become the selection). Drag below or above the grid and it scrolls. Let go off the panel, or right where they already are, and nothing changes. A plain click on a selected swatch still selects it alone — it just happens when you let go, so a drag can start first.

A locked swatch stays where it is: it survives Delete and Remove Duplicates, and a drag moves the swatches around it (a locked one in the selection stays put). Each set keeps its own undo steps while DRAW is running. With the pointer anywhere else these keys keep their usual jobs — `L` is still the Line tool, `F` the Fill tool, `Del`, `Ins`, `Ctrl+Z`, `Ctrl+Y`, `Ctrl+A` and `Ctrl+D` still work on the canvas.

**Docking.** Like the Preview and the color pickers, the Swatches panel floats in its own window (drag the corner grip to resize it) or docks: drag its title bar onto a dock target. Docked, it takes the slot's size and scrolls. Right-click its title bar → **Scale** to give it its own size.

# Palette Ops — Edit Colors Directly

> 🎯 **Goal:** Modify palette colors and remap on canvas.

**Palette Ops mode** is one of DRAW's signature features. Toggle it from the Organizer panel. Once it's on, the palette strip becomes editable, and the canvas follows every change you make to the palette.

A swatch works like an entry in an indexed palette: its pixels belong to it. Change the swatch and every pixel of that color on the canvas changes with it.

| Gesture on the palette strip | Effect |
|---|---|
| Left-click a swatch | **Select** it (yellow frame). It becomes the foreground color. While it's selected, **any** color change edits the swatch and its pixels live: the Color Mixer, Advanced Color Picker, 3D Color Space, picker tool or hex input. One `Ctrl` + `Z` undoes the whole edit. Half a second after the click, the matching pixels are also wand-selected. |
| Left-click the selected swatch | Deselect it. |
| Left-click an empty part of the strip | Deselect the swatch and clear every marker. |
| `Shift` +Left-click another swatch | **Ramp**: the swatches between the selected one and this one become a smooth blend between the two colors, and their pixels follow. The blend uses the *Gradient Blend* setting. |
| Double-click a swatch | Open the color picker for that swatch. |
| Drag a swatch | Move it to a new position. Dragging a **marked** swatch moves *all* marked swatches together. |
| Right-click, or right-drag across swatches | Mark or unmark them. A drag gives every swatch it passes the same state as the first one. |
| Middle-click a swatch | Delete it and remap its pixels to the nearest remaining color. On a **marked** swatch, this deletes every marked swatch. |
| Middle-drag across swatches | Delete the whole range. It's outlined in red until you release. |
| `Shift` +Middle-click a swatch | Insert the **current foreground color** after it, so you can build a palette straight from the color panels. |

> 💡 **Help card.** In Palette Ops, hover the palette strip for a moment and a **help card** pops up just above the strip, with every gesture in this table drawn as a mouse diagram plus its modifier keys. While color cycling is on, it shows the cycling gestures instead, along with the hovered swatch's range. You can turn the card off or move it to another edge of the canvas area (bottom, top, left or right) in **Settings → General → Help Card**.

Undo and redo, and opening the color panels or the Preview, keep Palette Ops on. Other commands, switching tools, `Esc`, and right-clicking the Organizer button turn it off.

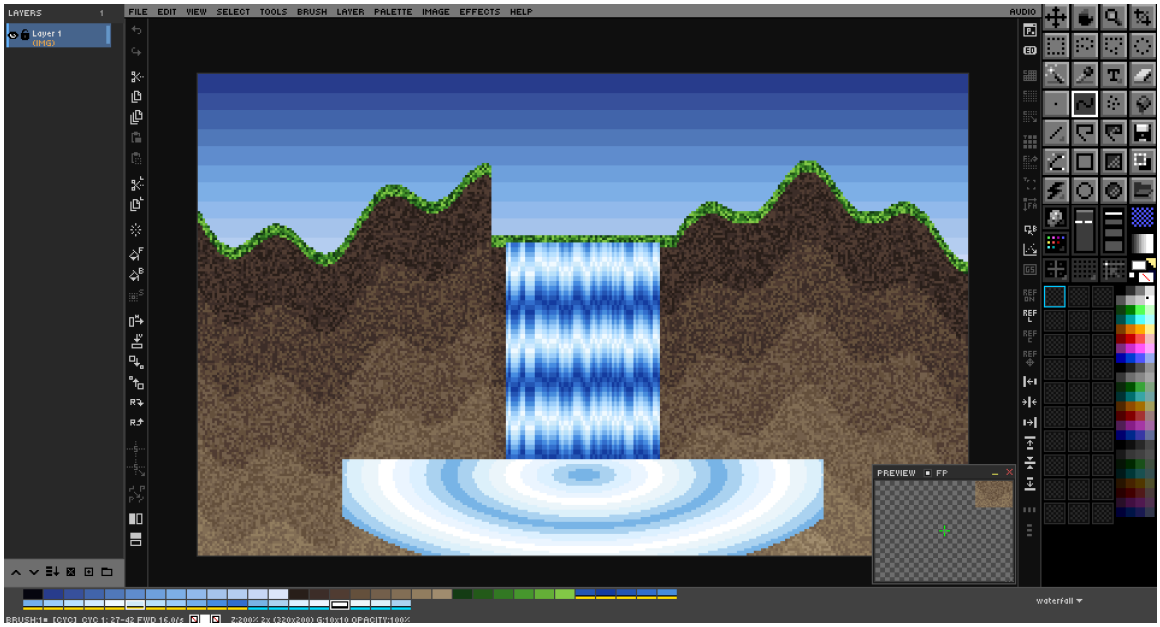
When you first enter Palette Ops, DRAW automatically creates a `[DOCUMENT]` **palette** that snapshots the current state. This means experimentation is safe — you can hop back to the original palette at any time without losing your remapping.

> 🎨 **Try it — colorway exploration**
>
> 1. Open a finished sprite.
> 2. Toggle Palette Ops.
> 3. Click a swatch, then drag in the Color Mixer to shift its hue. The sprite updates as you drag.
> 4. Compare colorways. When one feels right, exit Palette Ops to bake it in.

# Color Cycling — Animate the Palette

> 🎯 **Goal:** Make art move by rotating palette colors, the way DeluxePaint and GrafX2 do, without redrawing a pixel.

Color cycling turns a run of palette swatches into a loop. While it runs, the colors in that run rotate one step at a time, so every pixel painted with them appears to move. Waterfalls flow, fire rises, chase lights run, and a spiral turns. The pixels never change, only the colors they show. That is why one still picture can carry a whole animation.




## Turn it on and off

Press `Shift` + `Tab`, or choose **Palette** → **Cycle Colors**. The status bar shows `[CYC]` while cycling runs. The canvas, the Preview window and the palette strip animate together. Press `Shift` + `Tab` again to see the true colors.

Cycling is a **view**: it never changes your layers. Saving, exporting a PNG and flattening all use the real colors. The cycle ranges themselves are saved in the `.draw` file.

## Make a cycle range

A **range** is a run of neighboring swatches, such as chips 16–31. You can have up to 16 ranges, and they can't overlap. A new range replaces any range it overlaps. There are two ways to make one, both in **Palette Ops** mode:

1. **Mark and convert.** Right-click or right-drag the swatches to mark them, then choose **Palette** → **Color Cycling** → **Range From Marked Chips**. The range runs from the first marked swatch to the last.
2. `Ctrl` **+drag.** Hold `Ctrl` and left-drag across the swatches. A white band previews the span, and the range is created when you release.

Each range shows as a **band along the bottom of its swatches**:

| Band | Meaning |
|---|---|
| Solid | **Forward**: colors move toward higher swatch numbers. |
| Dash-dot | **Reverse**: colors move toward lower swatch numbers. |

| | |
|---|---|
| Dashed | **Ping-pong**: colors move forward to the end of the range, then back. |
| Dimmed | The range is **paused** and holds its current step. |
| Red checker | This swatch's color also appears **on an earlier swatch**, so its pixels belong to that swatch and can't cycle (see below). |

Hover any swatch in a range and the status bar reads it out, for example `CYC 2: 32-47 REV 8.0/s`. In Palette Ops with cycling on, the help card above the strip shows the same readout, the gestures below, and this band legend.

### Ranges in Palette Ops ( `Ctrl` gestures)

| Gesture on a swatch inside a range | Effect |
|---|---|
| `Ctrl` +Left-drag across swatches | Create a new range over the span. |
| `Ctrl` +Left-click | Change direction: Forward → Reverse → Ping-pong. |
| `Ctrl` +Right-click | Pause or resume the range. |
| `Ctrl` +Middle-click | Delete the range. |
| `Ctrl` +Wheel | Change speed through a table from 0.5 to 60 steps/s. Wheel up is faster. |
| `Ctrl` + `Shift` +Wheel | Fine speed control, 5% per notch. |

The same commands are in **Palette → Color Cycling**: *Delete Range*, *Clear All Ranges*, *Change Direction*, *Pause / Resume Range*, *Faster*, *Slower*, *Restart All* and *Check Duplicate Colors*. A menu command acts on the range under the hovered swatch, then the selected swatch, then the foreground color. Every range edit can be undone with `Ctrl` + `Z`. A burst of wheel steps counts as a single undo step.

### What cycles, and what doesn't

- **A pixel cycles when its color exactly matches a swatch inside a range.** This is the Palette Ops rule: a swatch *is* its pixels.
- **Duplicate colors.** When two swatches share a color, the **first** one owns the pixels. A duplicate inside a range gets the red checker band and its pixels don't move. Use **Check Duplicate Colors** to list them, then give each one a slightly different color in Palette Ops.
- **Opacity and blend modes.** A pixel drawn at partial opacity, or changed by a blend mode, no longer matches a palette color exactly, so it doesn't cycle. Paint cycling areas at 100% opacity on Normal layers.
- **Editing the palette.** Ranges follow Palette Ops edits. Deleting a swatch shrinks the range around it, inserting a swatch inside a range grows it, and moving swatches drags the range along when its colors stay together.

> 🎨 **Try it: a waterfall in a minute**
>
> 1. Pick eight blues that run light → dark → light, and put them next to each other in the palette.
> 2. Paint a waterfall with vertical streaks that step through those blues from top to bottom.
> 3. Turn on Palette Ops, `Ctrl` +drag across the eight blues, and press `Shift` + `Tab`. The water falls.
> 4. `Ctrl` +Wheel over the range to set the speed. Hold `Shift` as well for fine control.
>
> Or open one of the finished scenes in `SAMPLES/COLOR CYCLING/`: waterfall, fire, tunnel, marquee, ocean-sunset, pinwheel, candles, hanukkah, halloween, skull, snake, christmas, new-year, valentine and st-patricks.

To share cycling art as a runnable QB64 program, a GrafX2 GIF, an animated GIF or a DeluxePaint LBM, see Chapter 10 — Color Cycling Formats.

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➡️ Next: Chapter 4 — Layer System Deep Dive

# Ch. 04 📚 Layer System Deep Dive

> **What you'll learn:** How to use DRAW's 64-layer system, opacity and visibility, the 19 blend modes, layer groups, multi-layer selection, alignment, and the unique **symbol layer** system for repeated artwork.

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## Layer Basics — Create, Delete & Reorder

> 🎯 **Goal:** Work with multiple layers.

DRAW supports up to **64 layers** per canvas. The Layer Panel on the right lists every layer top-to-bottom in render order — top of the list is on top of the canvas.

| Action | Shortcut |
|---|---|
| New Layer | `Ctrl+Shift+N` |
| Duplicate Layer | `Ctrl+Shift+D` |
| Delete Layer | `Ctrl+Shift+Delete` |
| Move Up / Down in stack | `Ctrl+PgUp` / `Ctrl+PgDn` |
| Move to top / bottom | `Ctrl+Shift+]` / `Ctrl+Shift+[` |
| Rename | Right-click → Rename |

You can also **drag a layer row** with the left mouse button to reorder. **Right-click** any layer for the full context menu (rename, duplicate, delete, group, blend mode, etc.).

> 🎨 **Try it — multi-layer sprite**
>
> 1. Layer 1: scene/background.
> 2. Layer 2: character outline.
> 3. Layer 3: character base colors (with Opacity Lock — see below).
> 4. Layer 4: highlights and effects.

# Opacity, Visibility & Opacity Lock

> 🎯 **Goal:** Control transparency and protect existing pixels.

Each layer row has three persistent controls:

- **Eye icon** — toggle visibility. **Alt+Click the eye** for **Solo mode**: hide every other layer. Drag horizontally across multiple eye icons to sweep the visibility state.
- **Opacity bar** — drag, or hover and use the mouse wheel, for 0–100%.
- **Lock icon** — **Opacity Lock**. When enabled, drawing tools can only modify pixels that are *already opaque*. Perfect for recoloring within a finalized silhouette without bleeding outside.

> 🎨 **Try it — opacity-locked recolor**
>
> 1. Draw a character on one layer (transparent background).
> 2. Toggle the lock icon on that layer.
> 3. Take a giant brush and slosh new color over the whole canvas. Notice it only paints inside the silhouette.

## Layer thumbnails

**Settings** → **Panels** → **Thumbnails** puts a small picture of each layer before its name: **Off**, **Tiny** (16 px), **Small** (32 px) or **Medium** (48 px). The rows grow to fit. In an animation a thumbnail shows the current frame. Two more settings sit under it:

- **Thumbnail Shows** — **Content** (the default) zooms in on what's on the layer: the box round its non-transparent pixels, so a small logo on a big canvas is still easy to see. An empty layer shows just the checkerboard. **Canvas** shows the whole canvas, the same framing for every layer.
- **Min Crop (px)** — for Content, the smallest area a thumbnail shows, in canvas pixels each way (default 16). It caps the zoom, so a single dot stays a dot instead of filling the thumbnail. Set it to 1 for no limit.

In `DRAW.cfg`: `LAYER_THUMBNAILS=OFF|TINY|SMALL|MEDIUM`, `LAYER_THUMBNAIL_FIT=CONTENT|CANVAS`, `LAYER_THUMBNAIL_MIN_CROP=16`.

# Blend Modes — All 19 Explained

> 🎯 **Goal:** Use blend modes creatively.

A layer's blend mode determines how its pixels combine with the composite below it. DRAW implements all the modes you would expect from a modern raster editor:

| Family | Modes |
|---|---|
| **Basic** | Normal · Multiply · Screen · Overlay |
| **Math** | Add (Linear Dodge) · Subtract · Difference |
| **Comparison** | Darken · Lighten |
| **Dodge / Burn** | Color Dodge · Color Burn |
| **Light** | Hard Light · Soft Light · Vivid Light · Linear Light · Pin Light |
| **Color / Difference variants** | Exclusion · Color · Luminosity |
| **Group only** | Pass Through (Chapter 4 / EP15) |

Common pairings:

- **Multiply** — for shadows and ambient occlusion. Multiplying any color by a darker color produces a richer darker color.
- **Screen / Add** — for highlights, glow, fire and lights. Screen is the gentler companion; Add can blow out very quickly.
- **Overlay** — for contrast and saturation boost passes.
- **Color** — apply hue and saturation while preserving the underlying luminosity. The classic photograph-tinting trick.
- **Luminosity** — the inverse: apply brightness while preserving the underlying color.

To cycle a layer's blend mode without opening the dropdown, `Shift` **+Right-click** on the layer row.

> 🎨 **Try it — three-layer lighting study**
>
> 1. Base sprite on layer 1, Normal mode.
> 2. Layer 2: Multiply mode, paint cool desaturated tones for shadows.
> 3. Layer 3: Screen (or Add) mode, paint warm highlights.

# Layer Groups, Symbols & Advanced Operations

> 🎯 **Goal:** Organize complex artwork.

## Groups

| Action | Shortcut |
|---|---|
| Create empty group | `Ctrl+G` |
| Group selected layers | `Ctrl+Shift+G` |
| Ungroup | `Ctrl+Shift+U` |

Groups can be nested arbitrarily and use a special **Pass Through** blend mode that lets each child blend naturally with what's below the group.

## Merge operations

- **Merge Down** — `Ctrl+Alt+E`
- **Merge Visible** — `Ctrl+Alt+Shift+E`
- **Merge Selected** — fold a multi-selection into one layer.
- **Merge Group to Single Layer** — flatten a group while preserving its visual result.

## Multi-layer selection

`Ctrl+Click` toggles individual selection. `Shift+Click` also works like `Ctrl` because Mac users need that. Once multiple layers are selected, every relevant operation — clear, fill, flip, scale, rotate — runs against the whole selection. There is also `Select All in Group`.

## Alignment & distribution

DRAW provides nine alignment commands (Left/Center/Right × Top/Middle/Bottom) and two distribution commands (horizontal, vertical). Combined with multi-layer selection, this is the fastest way to lay out drawn objects in pleasing ways.

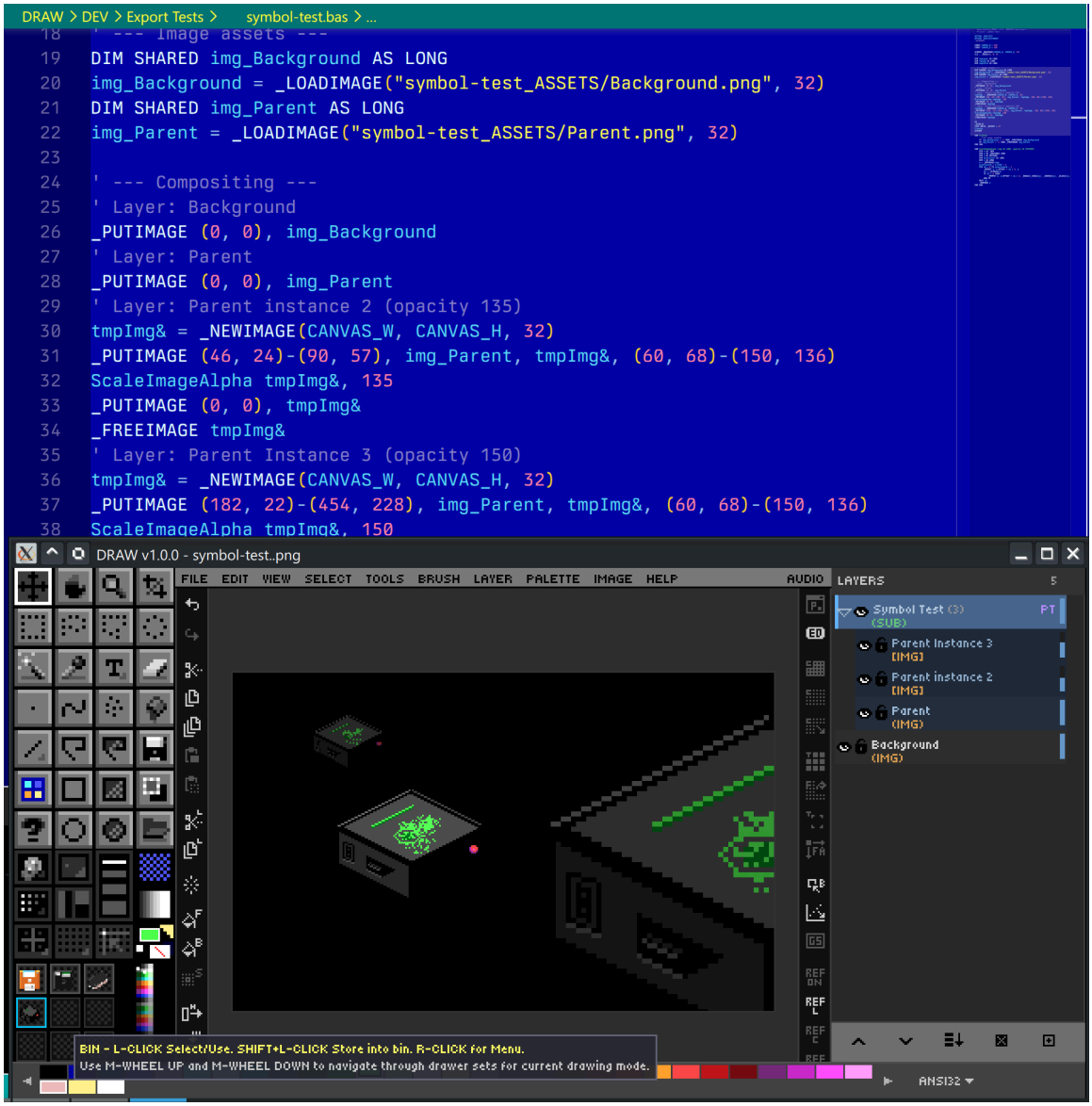
## Symbol Layers

A **Symbol Parent** layer can have one or more **Symbol Children**. Drawing on the parent automatically syncs to every child — but each child can be **independently scaled and repositioned**. Use this for repeated elements like coins, enemies, tiles, or recurring UI components.

When you finish a child or want to have a child no longer inherit changes from it's parent:

- **Rasterize** — bake the child into a normal pixel layer (severs the link permanently).
- **Detach** — break the sync but keep the child as an independent layer.

Create them via `Layer → New Symbol Layer` or `Layer → Convert to Symbol`.




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➡️ Next: Chapter 5 — Selection & Clipboard

# Ch. 05 ✂️ Selection & Clipboard

> **What you'll learn:** Every selection tool DRAW offers, how selections clip drawing tools, the full clipboard vocabulary, and the Stroke Selection effect.

---

## Selection Tools — Marquee, Freehand & Wand

> 🎯 **Goal:** Select regions for editing and manipulation.

A selection in DRAW does two things at once: it defines a region for clipboard operations, and it **clips every drawing tool** so paint only lands inside the marching-ants outline.

### Marquee variants

| Variant | Icon | What it does |
|---|---|---|
| Rectangle |  | Drag to draw a rectangular selection. |
| Ellipse |  | Oval / circular selection area. |
| Freehand |  | Draw the selection boundary by hand. |
| Polygon |  | Click each vertex; close with Enter. |
| Magic Wand |  | Click to select contiguous same-color pixels. |

The shared modifier vocabulary works across **every** selection tool:

- `Shift` **+drag/click** — *add* to the existing selection (union).
- `Alt` **+drag/click** — *subtract* from the existing selection.

After a selection exists you get **resize handles** on its edges; **drag inside the marquee** to translate the selection (with `Move` semantics); **arrow keys** nudge by 1 pixel (10 with `Shift`).

### Magic Wand specials

- `Shift+Click` — add to selection.
- `Alt+Click` — subtract.
- **Hold `F` and click** — instant flood fill with FG, no selection step. This works across ALL layers.
- **Hold `E` and click** — instant erase to transparent. This works across ALL layers.
- **Hold `W` and click** — sample from the merged composite of all visible layers, not just the active one.

### Selection commands

| Command | Shortcut |
|---|---|
| Select All | `Ctrl+A` |
| Deselect | `Ctrl+D` |
| Invert Selection | `Ctrl+Shift+I` |
| Expand / Contract | Image / Select menu |
| Selection from current layer | (non-transparent pixels) |
| Selection from selected layers | union mask |

# Copy, Cut, Paste & Clipboard Power

> 🎯 **Goal:** Master clipboard operations.

DRAW has more clipboard verbs than most editors because it cares about both layer-aware and pixel-aware semantics:

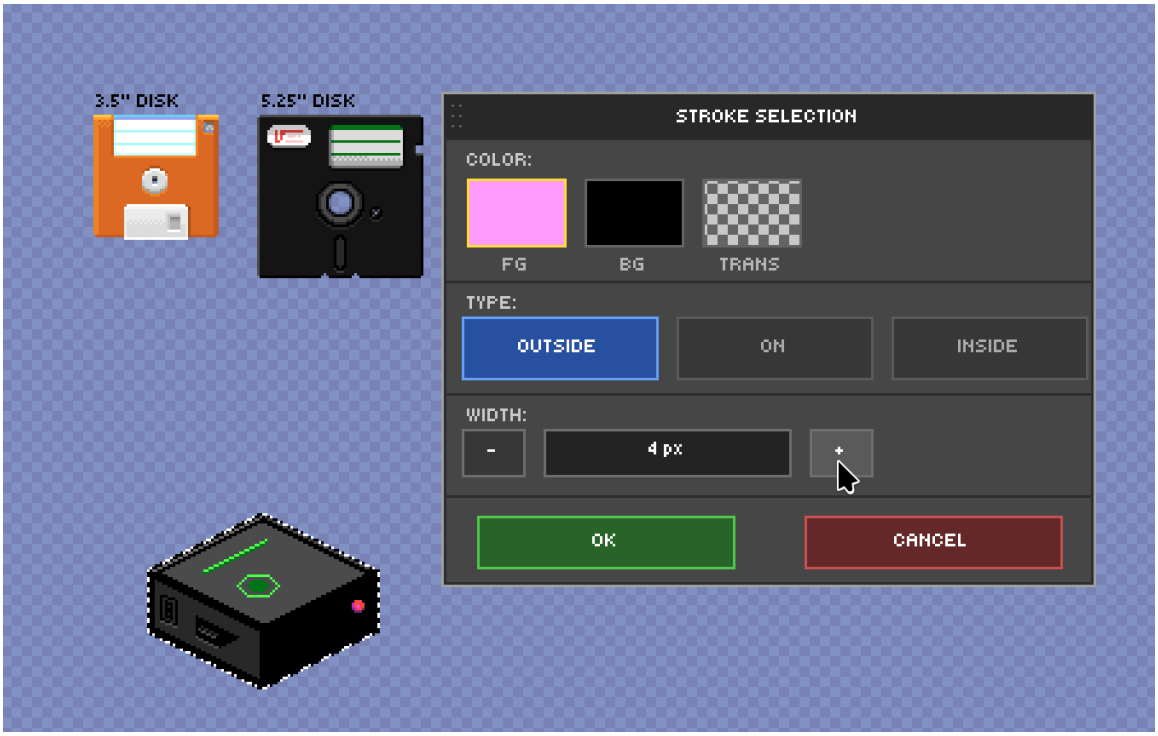
| Action | Shortcut | Notes |
|---|---|---|
| Copy | `Ctrl+C` | Selection → clipboard. |
| Cut | `Ctrl+X` | Copy + clear. |
| Paste | `Ctrl+V` | Floats the clipboard **centered on the canvas** and **auto-engages the Move tool** — drag to reposition (Alt+drag stamps copies), then deselect / switch tool / `Enter` to commit. |
| Paste in Place | `Ctrl+Alt+V` | Paste at the *exact* original position. |
| Copy Merged | `Ctrl+Shift+C` | Copies the visible composite, not just the active layer. |
| Copy to New Layer | `Ctrl+Alt+C` | One-step "duplicate selection onto a new layer above". |
| Cut to New Layer | `Ctrl+Alt+X` | Same, but removes from the source. |
| Paste from OS Clipboard | `Ctrl+Shift+V` | Brings in any image you copied from another application. |
| Clear / Erase | `Ctrl+E` or `Delete` | Clears the selection (or the whole layer if no selection — no prompt). |

### Stroke Selection

`Edit → Stroke Selection` adds a stroke around the selection boundary. You can configure:

- **Width** in pixels.
- **Position** — inside, outside, or centered on the boundary.
- **Color** — current FG, or a custom value.

This is the cleanest way to outline finished sprites in a uniform thickness.




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➡️ Next: Chapter 6 — Transforms & Image Adjustments

# Ch. 06 🔄 Transforms & Image Adjustments

**What you'll learn:** Quick flips, rotations and scales, the on-canvas Transform overlay (Scale / Rotate / Shear / Distort / Perspective), the Move tool with Smart Guides, and DRAW's image-adjustment dialogs (Brightness/Contrast, Hue/Sat, Levels, etc.).

---

## Quick Transforms — Flip, Rotate & Scale

🎯 **Goal:** Transform layers and selections quickly.

These are the no-dialog shortcuts you'll reach for hundreds of times a session.

| Action | Shortcut | Icon |
|---|---|---|
| Flip Horizontal | `H` |  |
| Flip Vertical | `Ctrl+Shift+H` |  |
| Rotate 90° CW | `>` |  |
| Rotate 90° CCW | `<` |  |
| Scale Up 50% | `Ctrl+Shift+=` |  |
| Scale Down 50% | `Ctrl+Shift+-` |  |

All of the above operate on the active selection if there is one, otherwise on the active layer (or the multi-selected layers). For canvas-level flips that move *every* layer at once, use the corresponding entries under the `Image` menu.

# Transform Overlay — Scale, Rotate, Shear, Distort, Perspective

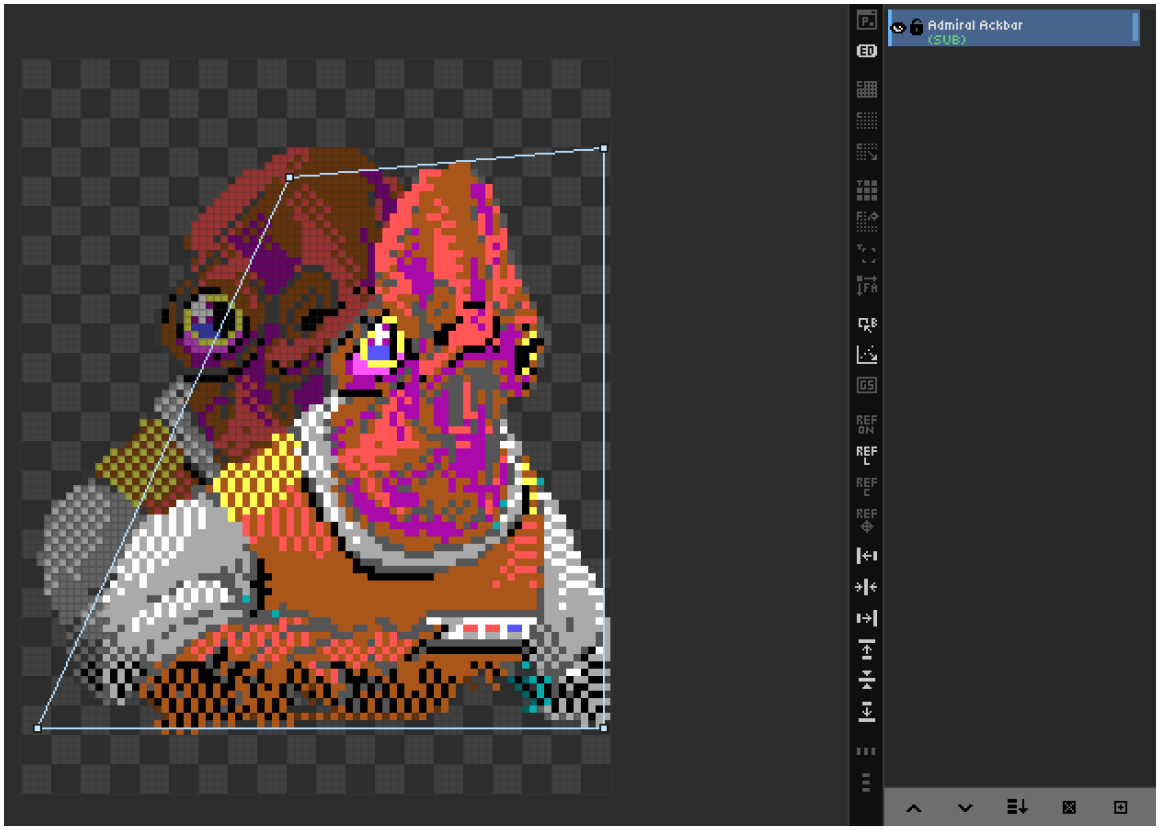
> 🎯 **Goal:** Use the interactive on-canvas transform tool.

Open it with `Edit → TRANSFORM…` (or via the Command Palette). Note that **Transform is not a toolbar tool** — it's a temporary overlay you commit or cancel.

The overlay supports five modes; switch between them with yhe menu bar items and use the corresponding hotkeys:

1. **Scale** — drag handles. `Shift` locks aspect ratio.
2. **Rotate** — drag *outside* the bounding box. `Shift` snaps to 15° increments.
3. **Shear** — drag the box edges to skew.
4. **Distort** — drag each corner independently.
5. **Perspective** — `Shift` enables a special wall / floor projection.

The overlay frame and handles are themeable in `THEME.CFG`. Press `Enter` to apply (full undo step is recorded), `Esc` to cancel without modifying anything.



# Move Tool & Smart Guides

> 🎯 **Goal:** Reposition layer content precisely.

The **Move** tool ( `V` ,  ) translates the active layer (or multi-selection). Modifiers:

- **Drag** — move.
- **Arrow keys** — nudge 1px (10px with `Shift` ).
- **Alt+Drag** — clone stamp; the original stays where it was and the dragged duplicate becomes a new layer.
- **Ctrl+Arrows** — move selection content 1px ( `Ctrl+Shift+Arrows` for a larger step; hold `Alt` to clone). *(Ctrl+Arrows resizes only while placing an imported/pasted image.)*
- **Shift+Click** — auto-select the topmost layer under the cursor.

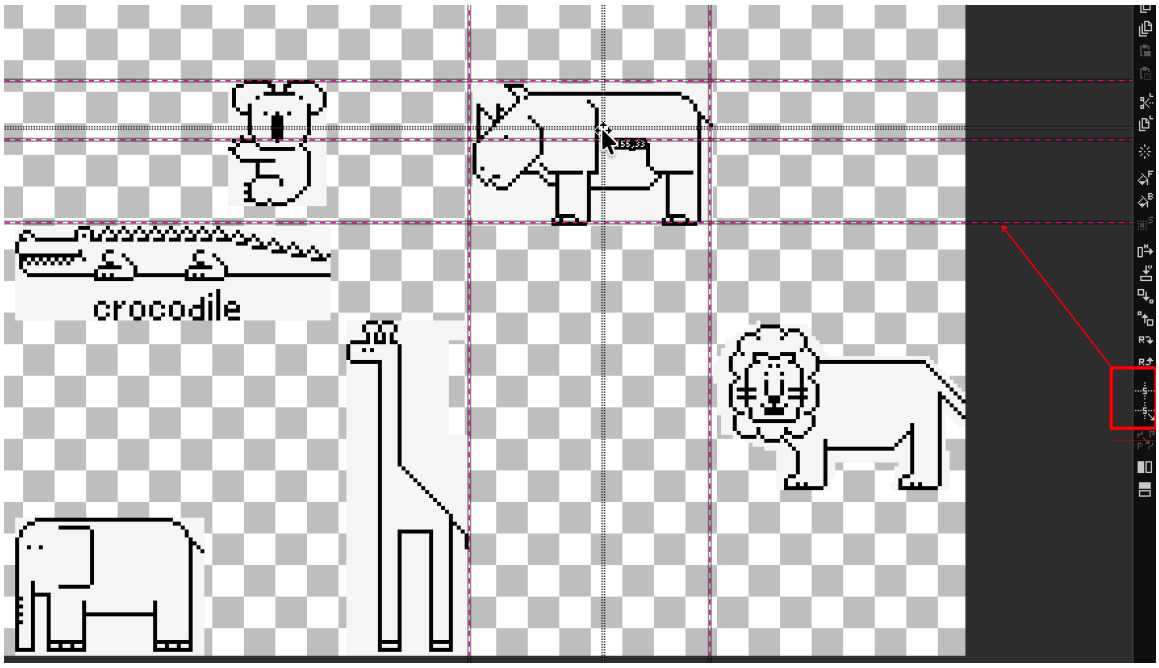
> Pro-tip: If you have an active selection, you can clone in Selection mode while holding `Ctrl+Alt`

## Smart Guides

When you move a layer, DRAW can render **Smart Guides** — temporary alignment lines that snap to canvas edges, canvas centers, and other layers' edges. This makes pixel-perfect alignment between sprites trivial.

| Toggle | Shortcut |
|---|---|
| Show Smart Guides | *Grid menu* |
| Snap to Smart Guides | *Grid menu* |

Smart-guide colors and opacity are themeable.



# Image Adjustments — Color Correction & Effects

> 🎯 **Goal:** Apply per-layer color adjustments.

DRAW has two flavors of adjustment: **dialog-based** (with live preview, scrollable parameters) and **one-shot** (instant, undoable).

### Dialog-based (live preview)

- **Brightness / Contrast** — Change the values and intensities of colors.
- **Hue / Saturation** — Rotate the colors on the color wheel and make them more or less colorful.
- **Levels** — Black point, White point, Gamma.
- **Color Balance** — Shadows / Midtones / Highlights.
- **Blur** — Gaussian, adjustable radius.
- **Sharpen** — adjustable intensity.

Hover any slider and use the **mouse wheel** for fine control.

### One-shot

- **Invert** (RGB negative)
- **Desaturate** (luminosity grayscale) (note you can use `Ctrl+Shift+Alt+G` to toggle Gray Scale Mode preview ON or OFF)
- **Posterize** (N levels, with optional dithering)
- **Pixelate** (block size)
- **Remove Background**

Every adjustment **preserves the alpha channel** and is undone in a single `Ctrl+Z`.

# The EFFECTS Menu — Photoshop / "Eye Candy" Style Effects

> 🎯 **Goal:** Stylize a layer with a large library of filters and generative effects.

Beyond the Image-menu adjustments, DRAW has a dedicated **EFFECTS** menu organized into **flyout submenu categories** (hover a category to open its list). Every effect opens a dialog with a **split ORIG / ADJ live-preview loupe**, **mouse-wheel sliders**, and an **OK / RESET / CANCEL** button row — **RESET** restores that effect's default parameters. Effects apply per-layer (multi-layer aware), preserve alpha where appropriate, and undo in a single `Ctrl+Z`.

- **ADJUST** → Gamma, Sepia, Threshold, Colorize, Gradient Map, Solarize, Duotone / Tritone
- **STYLIZE** → Glow, Film Grain, Vignette, Outline / Stroke, Edge Detect (Sobel), Grow / Shrink (dilate / erode), Chromatic Aberration, **Emboss** (with a **Light Angle**), Wind
- **LAYER FX** → **Drop Shadow** (Distance / **Angle** / Softness / Opacity), Perspective Shadow (Length / **Cast Angle**), **Bevel** (Height / Strength / **Light Angle** / Direction), Outer Glow, Inner Glow, Chrome / Metallic
- **DISTORT** → Wave / Ripple, Twirl, Pinch / Bulge (spherize), Kaleidoscope (auto mandala)

- **PIXELATE** → Crystallize, Stained Glass, Mosaic / Tessellate, Extrude, Pointillize
- **NOISE** → Add Noise, Median / Despeckle, Dust & Scratches
- **RENDER** → Clouds, Difference Clouds, Lens Flare, Terrain, Grid (with perspective), Sky (day / night / space)

### SHAPE → (Eye Candy "Shape", 20 effects)

Effects that work off the shape's alpha edge — backlights, glows, shadows, and elemental overlays: **Backlight, Corona, Cutout, Bevel, Chrome, Extrude, Outer Glow, Inner Glow, Drop Shadow, Perspective Shadow, Motion Trail, Rust, Snow, Fire, Smoke, Icicles, Drip, Glass, Electrify, Water Drops.**

### TEXTURE → (Eye Candy "Texture", 15 effects)

Procedural surfaces painted onto the shape, each with a **MIX** slider (0–100) to blend the texture with the original art instead of replacing it outright: **Wood, Marble, Brick Wall, Brushed Metal, Weave, Clouds, Swirl, Water Drops, Ripples, Animal Fur, Texture Noise, Stone Wall, Reptile Skin, Diamond Plate, Lightning.**

> 💡 Effects that draw *outside* the shape (Drop Shadow, Outer Glow, Fire, Smoke, Icicles, Drip, Backlight, Corona, Motion Trail, Snow flecks) work best on a **transparent layer** so there's room around the artwork — add one with `Ctrl+Shift+N`. Effects that use a colour (Backlight, Corona, Inner/Outer Glow, Gradient Map, Duotone, Drop/Perspective Shadow) read the current **FG / BG** colours.

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➡️ Next: Chapter 7 — Text System

# Ch. 07 📝 Text System

> **What you'll learn:** How DRAW's text tool works, the bundled and custom font support, rich per-character formatting, persistent text layers, and the unique **Character Mode** for ANSI-style art.

---

## Text Tool Basics — Fonts & Entry

> 🎯 **Goal:** Add text to your pixel art.

Press `T` to grab the Text tool, then click on the canvas to drop a text caret. Type to enter characters; `Enter` starts a new line; `Escape` commits the text and exits text-entry mode.

### Bundled fonts

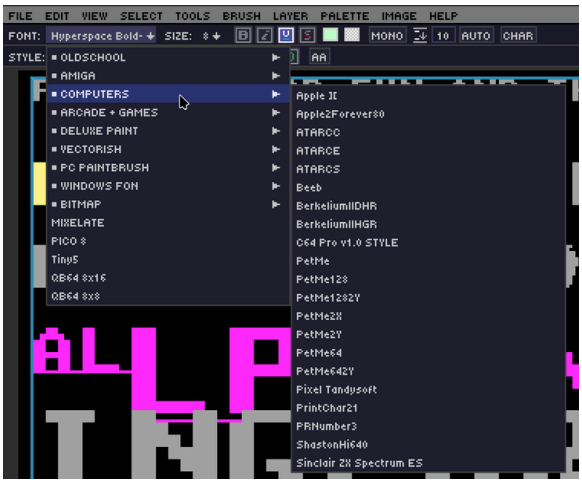
| Font | Shortcut | Description |
|---|---|---|
| VGA | `T` | Classic 8×8 monospace — the iconic DOS look. |
| Tiny5 | `Shift+T` | Minimal 5×5 — perfect for tight UI labels. |
| Custom TTF / OTF | `Ctrl+T`, or Left-Click in the font dropdown | Load any vector font from disk. |

### Bitmap fonts

DRAW ships with **620 color bitmap fonts** in the DPaint-style `.bmp` spritesheet format. Their pixels are preserved verbatim — no antialiasing, no recoloring — at a fixed native height per font. PSF, BDF, and Fontaption formats are also supported.

### The Text Property Bar

When the text tool is active, a property bar appears with the font dropdown, size, and the standard formatting toggles (bold, italic, underline, strikethrough). This bar is fixed in position under the menu bar and not movable or undockable.



# Rich Text — Per-Character Formatting

> 🎯 **Goal:** Apply advanced text formatting.

While editing a text layer, every character can carry its own properties. Selection works exactly like a desktop word processor:

| Action | Result |
|---|---|
| `Shift`+Arrow | Select characters one at a time. |
| Double-click | Select word. |
| Triple-click | Select line. |
| Quad-click | Select all. |
| `Ctrl+A` | Select all. |

Per-character properties:

- **Bold / Italic / Underline / Strikethrough**
- **Outline** — color + size.
- **Shadow** — color + X/Y offset (1–10px).
- **Per-character FG and BG colors**.
- **Letter spacing** and **line height**.

### Auto-wrap mode and style presets

Auto-wrap reflows the text inside the bounding box. Style presets let you save the entire formatting state of a selection as a named preset and apply it elsewhere with one click — Save / Load / Update / Delete buttons live in the property bar.

### Text-local undo

While in text-entry mode, **`Ctrl+Z` and `Ctrl+Y` operate on a separate 128-state text-local history**. This means typo experiments don't pollute your global history. Sound feedback on each keystroke is configurable in audio settings.

### Style Auto-Sync

As you move the text cursor, the formatting toolbar now updates live to reflect the style of the character at the cursor. This makes it effortless to inspect and modify mixed-style text — the toolbar always shows the current character's formatting.

# Text Layers & Character Mode

> 🎯 **Goal:** Master persistent text and ANSI-style art.

## Text layers

Once you commit a text-tool entry, it becomes a **text layer** — fully persisted in `.draw` files and re-editable forever. Click it again with the text tool to resume editing. From the Layer menu you can:

- Rasterize a single text layer (bake it into pixels).
- Rasterize all text layers (one-shot flatten of every text layer).
- Insert a new empty text layer ready for typing.

## Character Mode

Character Mode is for **ANSI / DOS / text-mode art**. Once enabled, the canvas is treated as a grid of cells with a virtual cursor:

> NOTE: DRAW is NOT a full fledged text mode art program, but this is a neat way to make GUI elements!

- Free grid navigation with arrow keys.
- `F1` – `F12` insert ANSI block characters (  ).
- "Font stickiness" — the chosen font follows the cursor as you navigate.
- The DOT and RECT drawing tools fill cells with the current glyph instead of pixels.
- `Alt+U` picks both the FG and BG colors from the character under the cursor.
- A character grid overlay snaps cursor positions cleanly.

### Character Map panel

Open with `Ctrl+M` for a 16×16 grid of all 256 glyphs in the active font.

- Click a glyph to insert it (or use it as a custom brush).
- Dock the panel left or right.
- Toggle Unicode / CP437 mapping from the **View menu** → **Charmap Unicode/CP437 Toggle** (or the command palette).

> 📸 **Screenshot needed — Character Map panel**
>
> - **Setup:** Switch to VGA font, enable Character Mode, open `Ctrl+M`.
> - **Action:** Hover one of the block-shading glyphs (e.g., `▒`).
> - **Capture:** Character Map panel docked right, glyph grid clearly visible.
> - **Save as:** `images/ch07-charmap.png`

> 🎨 **Try it — DOS-style title card**
>
> 1. Enable Character Mode.
> 2. Pick the VGA font.
> 3. Use `F1` – `F4` to lay block shading along the borders.
> 4. Switch FG/BG colors per cell with `Alt+U` to match an existing color set.

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➡️ Next: Chapter 8 — Grid, Symmetry & Drawing Aids

# Ch. 08 📐 Grid, Symmetry & Drawing Aids

**What you'll learn:** How to use DRAW's four grid geometries, three symmetry modes, angle-snap, the crosshair assistant, Pixel Perfect mode, and the pattern tile preview.

---

## Grid System — 4 Geometry Modes

🎯 **Goal:** Use grids for precise pixel art layout.

| Action | Key |
|---|---|
| Toggle grid visibility | `'` (apostrophe) |
| Toggle pixel grid (≥ 400% zoom) | `Shift+'` |
| Toggle snap-to-grid | `;` (semicolon) |
| Increase grid size (uniform W+H) | `.` |
| Decrease grid size (uniform W+H) | `,` |
| Toggle grid alignment (Corner ↔ Center) | `/` (slash) |
| Cycle geometry mode | `Ctrl+'` |
| Set grid size from brush / custom brush | `Ctrl+Shift+/` |

Grid sizes range from 2 to 256 pixels per cell.

### Independent axis sizing

Hold `G` and press an arrow key to resize width or height separately — useful for non-square grids (e.g., a 16×8 isometric tile cell):

| Key | Effect |
|---|---|
| `G + Right` | Increase grid **width** by 1px |
| `G + Left` | Decrease grid **width** by 1px |
| `G + Down` | Increase grid **height** by 1px |
| `G + Up` | Decrease grid **height** by 1px |

The `.` / `,` keys adjust width and height together; `G+Arrow` adjusts each axis independently.

### Geometry modes

1. **Square** (default) — standard tile grid.

2. **Diagonal** — 45° rotated diamonds.
3. **Isometric** — 2:1 pixel-art standard.
4. **Hexagonal** — flat-top hexagons.

### Alignment

Each geometry can be set to either **Corner alignment** (snap on intersections) or **Center alignment** (snap on cell centers). The choice depends on whether you want to draw *between* cells or *on* them.

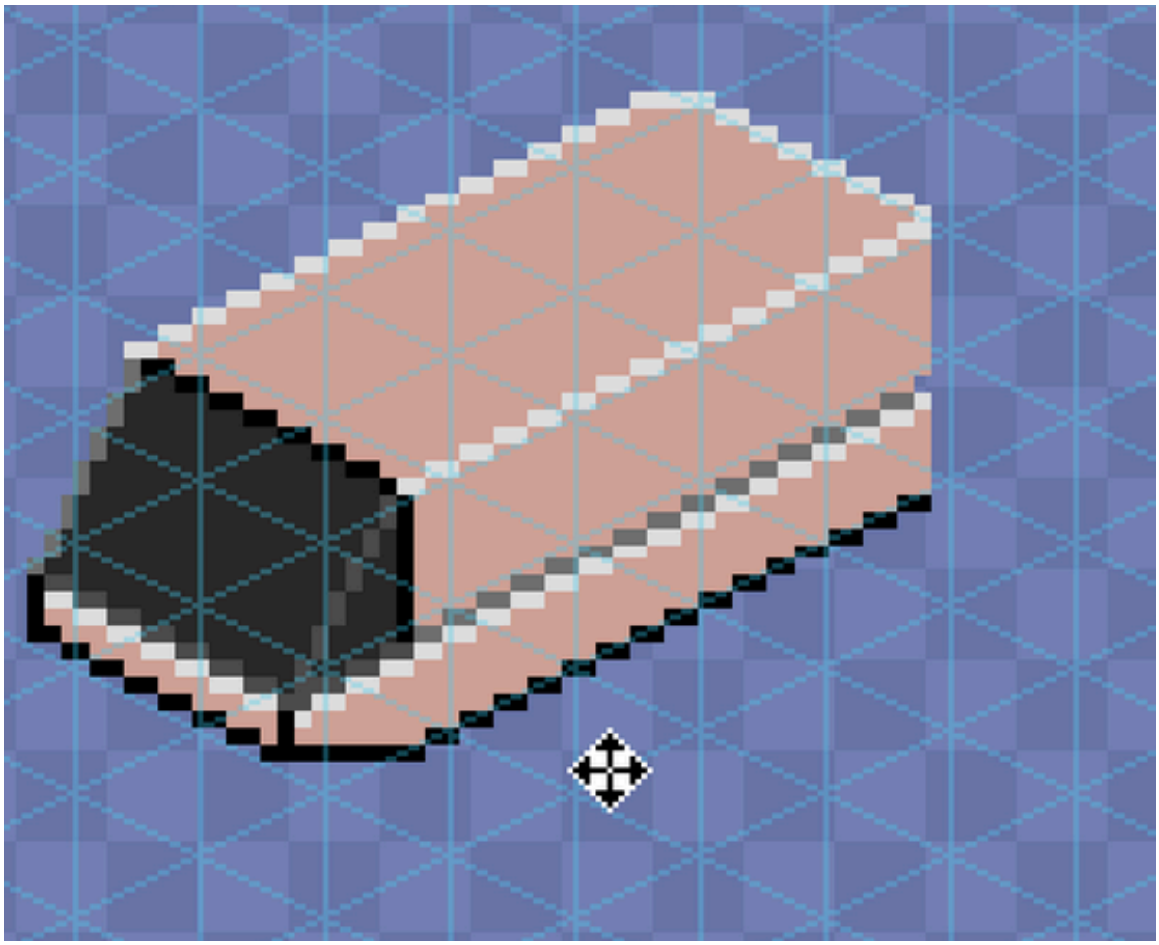
### Grid Cell Fill

When grid cell fill is enabled, the fill tool paints whole cells of the active geometry — squares for the square grid, diamonds and triangles for diagonal, and hexagons for hex.

`Ctrl+Shift` while drawing temporarily **bypasses snap**, so you can sketch freely without disabling the grid. All grid state is persisted in `.draw` files.

**🎨 Try it — isometric cube**

1. `Ctrl+'` to switch to Isometric mode.
2. `;` to enable snap.
3. Use the Polygon tool to outline the top, left, and right faces of a cube on three layers.
4. Apply Multiply/Screen blend modes for top vs. side shading.



# Symmetry Drawing — Mirror & Kaleidoscope

> 🎯 **Goal:** Create symmetrical art effortlessly.

Press `F7` to cycle symmetry modes:

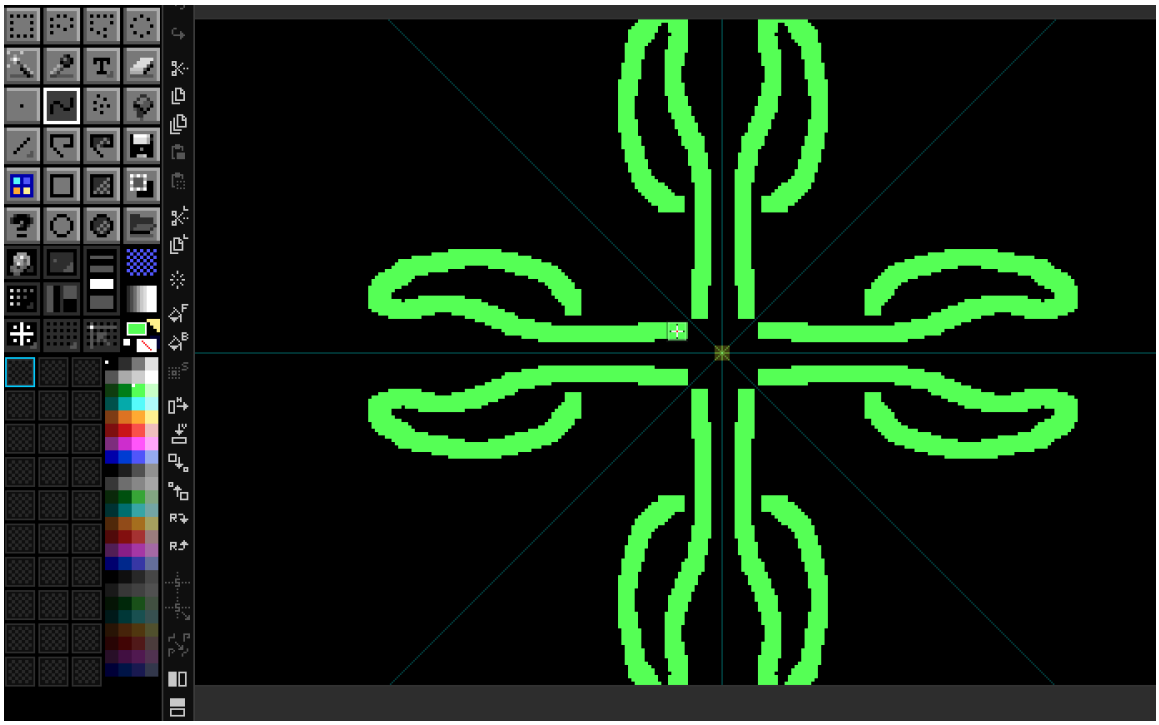
| Mode | Status bar | Copies | Shape |
|---|---|---|---|
| Off | `SYM:0` | 1 | — |
| Vertical | `SYM:1` (\|) | 2 | Bilateral |
| Cross | `SYM:2` (+) | 4 | Quad |
| Asterisk | `SYM:3` (*) | 8 | Kaleidoscope |

`Ctrl+Click` repositions the symmetry center anywhere on the canvas. DRAW renders subtle visual guides at the center.

Symmetry works with **every** drawing tool — Brush, Dot, Line, Rect, Ellipse, Polygon, Spray, Custom Brush. `F8` disables symmetry (or, in Fill Adjust contexts, opens the adjust overlay).

🎨 **Try it — 8-way mandala**

1. New 256×256 canvas.
2. `F7` × 3 until status bar reads `SYM:3`.
3. `Ctrl+Click` the canvas center.
4. Draw one curved stroke from center outward — eight reflections appear instantly.
5. Add layers and blend modes to taste.



# Drawing Aids — Angle Snap, Crosshair & Assists

> 🎯 **Goal:** Use helpers for clean, precise artwork.

## Angle Snap

Hold `Ctrl+Shift` while drawing a Line, Polygon, Brush stroke, or Dot connecting line. DRAW supports two angle-snap regimes:

- **Degree mode** — 15° / 30° / 45° / 90°.
- **Pixel-art mode** — integer-ratio angles like 1:1, 2:1, 1:2, 3:1, etc., which produce visually clean stairsteps without the half-pixel artefacts of 30° / 60° lines.

Choose the mode in `DRAW.cfg`.

## Crosshair Assistant

Hold `Shift` to display a full-screen crosshair through the cursor — invaluable for aligning distant elements. Color, opacity, width, and the optional outline stroke are all themeable.

## Pixel Perfect mode

`F6` toggles **Pixel Perfect** brush smoothing — DRAW retroactively removes L-shaped corners from your strokes for the cleanest possible 1-pixel outlines.

## Other view aids

- **Grayscale Preview** — `Ctrl+Alt+Shift+G`. View the composite as luminance only to check value structure.
- **Pattern Tile Mode** — *View → Pattern Tile Mode* (or the tile-mode button in the Advanced Bar). Renders the canvas as a 3×3 tiled preview so seamless textures show their seams immediately.
- **Canvas Border** — `#`. Toggles a thin border around the canvas (useful at high zoom over dark themes).

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➡️ Next: Chapter 9 — Custom Brushes & Drawer Panel

# Ch. 09 🪄 Custom Brushes & Drawer Panel

> **What you'll learn:** How to capture, transform, recolor and stamp custom brushes; the 30-slot Drawer panel for brush / pattern / gradient libraries; and DRAW's six dithering algorithms.

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## Custom Brushes — Capture, Transform & Paint

> 🎯 **Goal:** Create and use custom brushes.

A **custom brush** is any selection captured into the brush slot — DRAW remembers the pixels and treats them as the active stamp for every drawing tool that supports stamping.

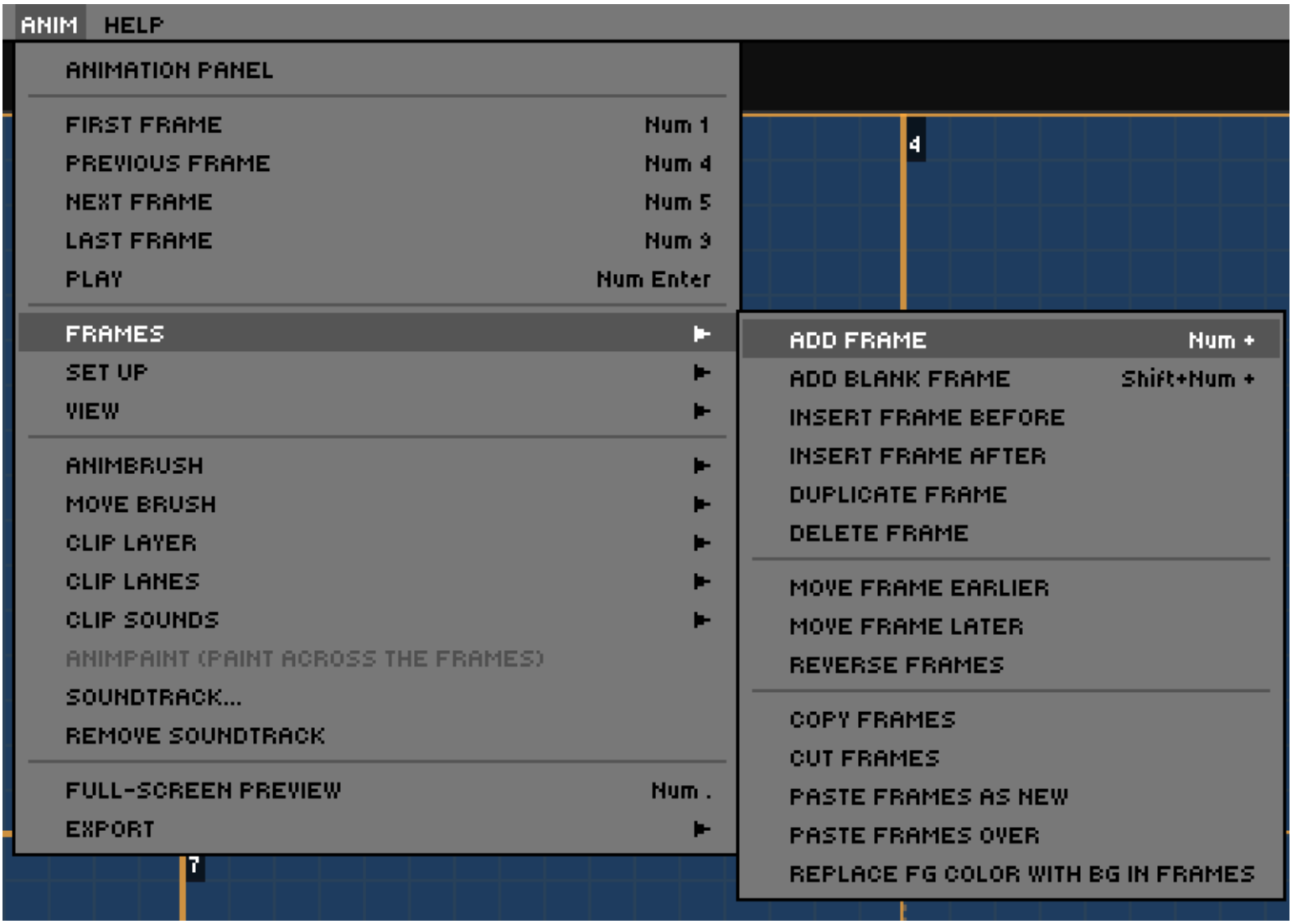
### Capturing

Make a selection, then capture it as a brush via the Brush menu, the Organizer button, or the Command Palette. There are two capture modes:

- **From Current Layer** — only non-transparent pixels.
- **From Selected Layers** — union mask across the multi-selection.

Non-rectangular shapes are preserved via the alpha channel.

### The ANIM menu




Everything for frames lives in the **ANIM** menu. The things you use all the time are on top - the Animation panel, First / Previous / Next / Last Frame, Play and Full-Screen Preview - and the rest is grouped in flyouts:

| Flyout | What's in it |
|---|---|
| **Frames** | Add, insert, duplicate, delete, move, reverse, copy / cut / paste frames |
| **Set Up** | Frame Grid, Detect Frames, Import PNG Sequence; reference / same-on-every-frame / linked layers; tags and frame notes |
| **View** | Lightbox, panes, X-sheet, Big Frame + X-Sheet, onion skin, Preview Shows Frame, clip strokes to the frame |
| **AnimBrush**, **Move Brush** | The two brushes below |
| **Clip Layer**, **Clip Lanes**, **Clip Sounds** | Clip layers, their keyframe lanes and their sounds (below) |
| **Export** | Animated GIF, APNG, PNG sequence, sprite sheet, BAS player, export scale |

AnimPaint and the soundtrack stay on the top level. Every item is also in the command palette ( `?` ), so you can type its name instead. The **Sprite**, **Animate** and **Scene** workspaces (Chapter 22) set DRAW up for each kind of animation work with one click on the workflow buttons at the top of the toolbox.

## AnimBrush - a brush with a picture per frame

On a sprite sheet (a document with frames), **ANIM → AnimBrush → Pick Up** makes a brush that has one picture - a *cel* - for each frame:

1. Draw a selection around the part you want inside one frame (a marquee, ellipse, lasso or magic wand; the shape is kept).
2. To take only some frames, Ctrl+click them in the lightbox strip first, or click a tag on the X-sheet (the brush is then named after the tag - you see the name in the status bar and on its bin). Otherwise the cels come from every frame.
3. Choose **Pick Up** (the current layer) or **Pick Up Merged** (every visible layer).

> 💡 **Quicker: hold `B` while you make the selection.** When you let go of the mouse the selection is picked up and you are on the Brush: an AnimBrush on a picture with frames, a plain brush (as `Ctrl+B`) on one without. A quick tap of `B` still just picks the Brush.

DRAW reads the same spot in every frame's cell - the gutter and margin between cells are skipped - so a walk cycle becomes a walking brush. The tool switches to the Brush, and the brush under the pointer is always the cel your next stamp will use; the status bar shows it too (`ANIMBRUSH 3/8 FOLLOWS FRAME`: cel 3 of 8).

**Step modes** (ANIM → AnimBrush):

- **Cel follows the frame** (the default) - on frame 5 the brush uses frame 5's cel. Stamp the same spot on each frame and the flame burns there. The frame you picked the brush up from gets cel 1; `Alt+\` makes the current frame cel 1 instead.
- **Next cel each click** - every press moves on one cel, so a row of torches on one frame don't flicker in step. On a picture without frames the brush always works this way.
- **Every frame at once** - one press stamps the brush on every frame, each with its own cel, cut at the edge of each frame's cell. Ctrl+Z takes the whole thing away in one step.

**Order** runs the cels Forward, Reverse, Ping-Pong (1 2 3 4 3 2 ...) or Once (holds the last cel). `Alt+]` and `Alt+[` step to the next / previous cel. `Ctrl+B` drops the AnimBrush; `Ctrl+B` over a selection still makes an ordinary one-picture brush. Switching to Pattern or Gradient mode puts the AnimBrush away with its cels, and Color mode brings it back.

Everything in **Transform controls** and the recolor / outline modes below works on an AnimBrush too: Rotate 90° (`>` / `<`), Scale 2x (`Ctrl+Home` / `Ctrl+End`) and Outline (`Shift+O`) change **every** cel, not just the one you see; Flip, Scale and Recolor are applied to whichever cel a stamp uses. With the **Eraser** an AnimBrush erases each frame's own cel shape (with *Every frame at once*, the whole animation in one press). In the lightbox, a stamp in another frame's pane lands in that frame and uses that frame's cel.

## Move Brush - a brush that travels across the frames

**ANIM → Move Brush** stamps the brush once in each frame along a path, so a ball rolls or a bird flies through the animation without placing every frame by hand. It works with any custom brush; with an AnimBrush the bird flaps as it flies.

1. Pick up a brush (`Ctrl+B` over a selection, or an AnimBrush).
2. Choose **ANIM → Move Brush → Start**. The status bar shows `MOVE BRUSH`.
3. Drag the path in a frame - on the sheet, or in a lightbox pane. A see-through ghost of every stamp shows along it, with a dot for each frame. Drag again to try another path.
4. Press `Enter` to stamp it (`Esc` cancels). The first frame gets the brush at the start of the path, the last frame at its end, the frames between along the way - each in its own cell. `Ctrl+Z` takes the whole move away in one step.

Which frames: the ones selected in the strip (Ctrl+click two or more), else the frame you drag in through the last frame. In the same menu:

- **Ease** - *None* spaces the stamps evenly; *In* starts slow and speeds up; *Out* starts fast and slows down; *In-Out* is slow at both ends.
- **Next Cel Every Frame / 2 / 3 / 4** - with an AnimBrush, how often it moves on to its next cel (a slow walk: every 2 frames).

**Turn, size, fade and tint over the move.** While Move Brush is on, the Animation panel shows the move's settings in place of its frame rows (it opens for the move if it was shut). Every number is a box you can drag, wheel, click to type, or middle-click for its default:

- **MOVE X / Y / Z** - X and Y are where a straight path ends. Type them instead of dragging: with no path yet the move starts in the middle of the frame. **Z** moves the brush away from you (or toward you, with a minus) in perspective: it shrinks as it heads for the middle of the frame.
- **TURN X / Y / Z** - degrees over the move. Z spins the brush flat (360 = once round); X and Y tilt it in 3D, like a coin flipping.
- **SIZE**, **OPACITY**, **TINT** - each has a start and an end value: grow from 20 % to 100 %, fade in from 0 to 100, or turn toward the FG colour.
- The **ease** applies to all of them. **PLAY** shows the ghost one step at a time; **RESET** puts Z, turn, size, opacity and tint back.
- **PATH: DRAWN** makes the stamps follow the shape you drag instead of a straight line.

So a coin spinning in place is: pick up the coin, Start, type **TURN Y 360**, Enter.

## AnimPaint - painting across the frames

**ANIM → AnimPaint** turns the brush into a recorder: press in a frame, drag, let go, and each frame from there on gets the brush where the pointer was at that frame's moment - at the animation's FPS, each frame lasting its hold. Draw a slow wiggle and a firefly wanders; draw quickly and it darts. The stroke lands when you let go (Ctrl+Z takes it back in one step) and AnimPaint stays on for the next stroke; Esc stops it. The size, turn, opacity and tint settings above apply here too.

## Clip layers - an AnimBrush that keeps playing

A stamp is pixels: once it lands, changing your mind about timing means erasing and stamping again. A **clip layer** keeps the AnimBrush itself, so a torch flickers, a coin spins or a character walks on the spot for as long as the animation runs, and you can change when it starts, how fast it plays or where it sits at any time.

1. Pick up an AnimBrush.
2. Choose **ANIM → Clip Layer → Place AnimBrush as Clip**, then click in a frame. A new layer appears above the current one, named after the AnimBrush with `[C]` in front. Cel 1 is in the frame you clicked, where you clicked, and the clip plays to the last frame. `Esc` or a right-click cancels before you click.
3. The Animation panel opens on the clip's settings (they replace the frame rows while a clip layer is the current layer; the `|< < PLAY > >|` row stays so you can scrub):
   - **LOOP / PING / ONCE / HOLD / REV** - play the cels round and round, back and forth, once (then nothing), once and stay on the last cel, or backwards.
   - **IN / OUT** - the first and last frame it shows in. `END` means to the last frame, and stays that way when you add frames.
   - **CEL / EVERY** - the cel it starts on, and a new cel every N frames (2 = on twos).
   - **X / Y** - where the cel sits in each frame's cell.
   - **FLIP X / FLIP Y** mirror every cel. **BAKE** - see below.

   Each number is a box: drag it, wheel over it, click to type, or middle-click for its default. The same settings are in **ANIM → Clip Layer**, and `Image → Flip Horizontal / Vertical` flip the clip too.

**Moving it:** the Move tool ( `V` ) drags the clip in a lightbox pane and every frame follows; the arrow keys nudge it 1 px. **Frames:** adding, inserting, deleting or reordering frames leaves the clip layer alone and moves its IN / OUT with the frames around it.

**What you can't do to it:** a clip layer's pixels are rebuilt from the clip, so painting on it, Cut, Paste, Clear, Scale, Rotate, Transform and Effects are refused with a message in the status bar, and Merge Down onto it is refused. **Bake Clip to Pixels** (ANIM → Clip Layer, the panel's BAKE button, or the layer's right-click menu) turns it into a normal layer with the pixels it shows; `Ctrl+Z` brings the clip back. Merging a clip layer down into the layer below also keeps just its pixels.

Every change is one undo step (a drag on a number box too). The clip is saved in the `.draw` file with a copy of its cels, so it doesn't need the Drawer slot it came from; every export - GIF, APNG, PNG sequence, sprite sheet, BAS - shows it frame by frame. A DRAW too old to know clip layers opens the file with the clip as a normal layer of pixels.

### More ways to make a clip

- **Make Clip from Selected Frames** (ANIM → Clip Layer): draw a selection around something that already moves across your frames - select the frames in the lightbox strip first, or leave them all - and it becomes a clip. The pixels are cut from the layer and the clip shows them in the same frames at the same place, so nothing looks different, but now you can move it, slow it down or loop it.
- **Drop an animated GIF or APNG** onto the canvas while the document has frames: it becomes a clip starting in the frame under the pointer. (A `.draw` file still opens as a document, and a PNG sprite sheet still becomes a brush.)
- **Update Clip from Bin / from AnimBrush** swaps a clip's cels for the Drawer bin it came from (after you changed that bin) or for the AnimBrush you are holding. Every clip using the same cels changes with it.

### The clip on the X-sheet

On the X-sheet a clip layer's row is a bar from its IN to its last frame, with the cel each frame shows. Drag either end to change IN or OUT, or drag the middle to slide the whole clip along the frames. Each drag is one undo step.

### Sharing cels between clips - clip sources

The cels of a clip are its **source**. Duplicating a clip layer gives you a second clip on the same source, and the Animation panel's **SOURCE** row (under the clip settings) works on it:

- `<` / `>` - switch the clip to the previous / next source in the document.
- **RENAME** - name the source (every clip on it shows the new name).
- **LAYERS** - select every layer using this source.

**ANIM → Clip Layer → Clip Sources...** lists every source, its cels and the layers using it. **Remove Unused Clip Sources** clears the ones no layer uses (they come back if you undo your way to a layer that used them).

### Editing a clip's cels

**ANIM → Clip Layer → Edit Clip Cels** puts the clip's cels on a new layer (named after them, e.g. **Cels of torch**): cel 1 on frame 1, cel 2 on frame 2 and so on, where the clip sits. The clips using those cels are hidden while you work. Paint on that layer with any tool, then choose **Apply Cel Edits**: every clip on that source plays the new cels, and the layer goes away. One `Ctrl+Z` undoes the whole Apply. **Discard Cel Edits** just removes the layer. The clip needs as many frames as it has cels, and its cels must fit in a frame.

### Sounds on a clip

A clip can carry its own sounds, and they travel with it wherever it plays:

- **Sound on This Cel...** - plays whenever the cel showing in the current frame comes up (a footstep on the walk cycle's contact cel).
- **Sound When the Clip Starts...** - once, on its IN frame.
- **Sound Each Time the Clip Starts Over...** - on IN, and again each time its cels loop.
- **Remove the Clip's Sounds** - all of them.

The same three are the **SOUND** row's `CEL` `START` `LOOP` buttons in the Animation panel; the row under it shows one sound at a time - `<` / `>` step through them, `X` removes that one. Every clip on the same source plays the sounds. A hidden clip layer is silent. On the X-sheet, a clip with sounds has a **SOUND** row under it that marks where each one plays (read-only - edit the sounds in the panel). The sounds play with the animation and go into the BAS export's audio. **Bake Clip to Pixels** asks whether to keep them: **Yes** ( `Enter` ) turns them into ordinary soundtrack clips, **No** ( `Esc` ) drops them. The `.draw` file keeps the sounds' file paths, the same way the soundtrack does.

#### Moving a clip - keyframe lanes

A clip can move, grow, turn, fade or change colour over its frames. Each of these is a **lane**: **X**, **Y**, **SCALE X**, **SCALE Y**, **ROTATE**, **OPACITY**, **TINT** (toward the clip's tint colour), **PIVOT X** / **PIVOT Y** (where it scales and turns about), and for its sounds **VOLUME** and **PAN**. A lane holds **keys** - a frame and a value - and moves smoothly from one key to the next.

In the Animation panel's clip rows:

- **LANE** - `<` / `>` pick the lane; the box shows its value on the current frame. Type, drag or wheel the box and the value is **keyed on this frame** (a new key, or the one already there). A `*` after the lane's name means this frame has a key.
- **KEYS** - how many keys the lane has, **KEY** (key the value here), **DEL** (remove the key here) and the **ease** of the key at or before this frame: LINEAR, IN, OUT, IN-OUT (the Move Brush's eases) or HOLD (stays put until the next key).
- **MOD** - a wobble added on top of the lane: OFF, SINE, TRI, SQUARE, SAW (repeating) or NOISE (a seeded jitter - the same every time it plays). Its boxes are AMOUNT (in the lane's unit), PERIOD (frames) and PHASE (frames; the SEED for noise). A firefly bobbing: Y, SINE, amount 3, period 12. A flame flickering: OPACITY, NOISE, amount 20, period 2.

Keys count from the clip's IN, so a clip slid along the X-sheet keeps its moves. Before the first key the lane holds the first key's value, after the last the last one's. Positions are whole pixels, so a slow move steps one pixel at a time.

Other ways to key:

- The **Move tool** on a clip that has X or Y keys keys the current frame instead of moving the whole clip (drag or arrow keys). Without X / Y keys it moves the clip as before.
- The **X-sheet** shows a **LANES** row under a clip with keys: a diamond on each frame with a key, filled where the lane picked in the panel has one. Drag a diamond along the row to move that frame's keys to another frame.
- **ANIM → Move Brush → Apply as Clip** makes the Move Brush's move a clip layer instead of stamping it: the brush (an AnimBrush's cels, or a plain brush) plays over the move's frames, with the move in its lanes - a straight path as two keys with the move's ease, a drawn path as a key per frame, plus its SIZE, OPACITY, TINT and TURN Z. Z and the 3D turns have no lane and are left out (Apply still stamps them).
- **ANIM → Clip Lanes**: **Clear This Lane**, **Clear All Lanes**, **Copy Lanes** / **Paste Lanes** (onto another clip layer), **Tint Toward the Foreground Colour** (the colour the TINT lane goes toward).

VOLUME and PAN set each of the clip's sounds to the lane's value on the frame it starts (a sound keeps its level while it plays). Every lane is one `Ctrl+Z` per change, is saved in the `.draw` file, and is in the pixels that onion skin, playback, Bake and every export use.

### The AnimBrush pack and example scenes

DRAW ships with `animbrush-pack`, a Drawer brush set of 15 AnimBrushes: torch, campfire, smoke, star, firefly, flag, bat, ghost, owl, waterfall, splash, ripple, grass, butterfly and bird. Pick it from the Drawer's brush set list; hovering a bin plays it, and the number in its corner is its cel count.

To build a scene from it:

1. **File → New Animation...** - choose a frame count every loop divides into (24 fits loops of 2, 3, 4, 6, 8 and 12 cels, so nothing jumps when the scene wraps).
2. Draw the background in frame 1 and turn on **ANIM → Set Up → Layer Same on Every Frame**.
3. `Num 0` (the lightbox), then **ANIM → AnimBrush → Every Frame at Once**.
4. For each thing that animates in place: a new layer ( `Ctrl+Shift+N` ), click its bin, press `Alt+]` a few times so it is out of step with its neighbours, and click where it goes.
5. For each thing that travels: **Move Brush**, drag its path, `Enter`.

Three finished scenes built this way are in `SAMPLES/ANIMBRUSH/scenes/` : a castle at night, a campfire in the woods and a waterfall valley, every element on its own layer. Each brush is also in `SAMPLES/ANIMBRUSH/brushes/` as a strip with frame data, for **AnimBrush → Load...**.

## Animated scenes with sound

`SAMPLES/ANIMSCENES/` has nine longer scenes (48–64 frames, 12 fps), each with its **own AnimBrush pack** and a **soundtrack** (a music track and a sound-effects track, timed to the frames). They show what the Move Brush's numbers do across many frames - position on straight and drawn paths, SIZE, MOVE Z (perspective), TURN X / Y / Z, OPACITY and TINT:

| Scene | What moves |
|---|---|
| `space` | parallax star layers, a ship curving down to the planet (Z + turn + tint), a cut to the surface through black, a lander touching down in dust |
| `alien-attack` | an invaders-style loop: a marching formation, a tracking tank, shots, flipping bombs, a mothership blown apart (size + spin + tint + fade) |
| `brick-breaker` | a simulated rally: the ball on a drawn path, bricks bursting, a capsule flipping into the paddle |
| `bakery` | a sunlit table: steam puffs that grow and fade, dust motes on drawn loops, a sunbeam that brightens and warms |
| `beach` | flapping umbrellas, rolling surf, a spinning beach ball, a gull sailing toward you (size) and one flying away (Z), a sunbather turning red (tint) |
| `dungeon` | flickering torches and skull candles, spiders on silk, a ghost fading in and out, a bat flying at you |
| `fireworks` | rockets, bursts that grow, fade and cool from white (size + opacity + tint), a tilted ring (turn X), a pinwheel, a finale |
| `platformer` | a robot running and jumping on drawn arcs, coins flipping (turn Y), the fall into the pit (tumble + Z + red + fade), GAME OVER |
| `stormy-night` | the extra pack: rain and lightning, a TV, a radio, a computer, a snow globe, a dog, a cat and two monsters |

Open `NAME.draw` , `Num 0` for the lightbox and `Num Enter` to play it with its sound. The pack is `NAME-pack.dset` (Drawer → Load Set); `NAME-moves.txt` lists every Move Brush the scene used with its numbers, so you can rebuild any element yourself: pick the bin, **Move Brush → Start**, drag the path, type the same values and `Enter` .

> 💡 For composing a scene like these, **ANIM → View → Big Frame + X-Sheet** shows the lightbox as one big pane (the current frame) under the X-sheet: you see the whole picture large while every layer's frames and the sound stay in view. Choose it again (or `Num *`, which steps 1 / 3 / 5 / 7 panes) to go back.

### Transform controls

| Action | Key |
|---|---|
| Flip Horizontal | `Home` |
| Flip Vertical | `End` |
| Scale Up | `PgUp` |
| Scale Down | `PgDn` |
| Reset Scale | `/` |

### Modes that work *with* a custom brush

| Tool | Effect |
|---|---|
| Brush / Spray | Stamp the brush at every step. |
| Line / Rect / Ellipse / Polygon | Stamp along the geometry. |
| Fill | **Tiled fill** of the brush as a seamless pattern. |
| Eraser | Stamp transparency in the brush shape. |

### Recolor and outline modes

- **Recolor** ( `F9` ) — paints the brush in the current FG color, preserving its alpha.
- **Outline** ( `Shift+O` ) — adds a BG-colored outline around the brush silhouette.

### Export

`F12` exports the current brush as a standalone PNG (alpha preserved) — perfect for sharing brushes between projects.

> 🎨 **Try it — leaf scatter**
>
> 1. Draw a small 8×8 leaf on a transparent layer.
> 2. Marquee around it; capture as custom brush.
> 3. Switch to Spray, increase brush size, set FG to a green family.
> 4. Spray a tree's foliage onto a new layer in seconds.

# Drawer Panel — 96 Reusable Slots

> 🎯 **Goal:** Organize and reuse brushes and patterns.

The **Drawer panel** is a docked grid of 96 slots per mode that hold reusable brushes, patterns, or gradients. Each slot is one click away from being the active stamp.




## Drawer modes

| Mode | Shortcut | Use for |
|---|---|---|
| Brush | `F1` | Stamping and painting. |
| Gradient | `F2` | Color transitions for fills. |
| Pattern | `F3` | Seamless tiled fills. |

## Storing and managing slots

| Action | Result |
|---|---|
| `Shift`+Left-click an empty slot | Store the current brush there. |
| Right-click any slot | Context menu — load, save, clear, replace, export, etc. |
| Drag-and-drop slots | Reorder. |
| Drop image files onto the panel | Batch import as slot brushes. |

**Load an AnimBrush from a file. ANIM → AnimBrush → Load...** makes an AnimBrush from an animated GIF or APNG (its frames, as a player shows them), a PNG or `.draw` with frames (its cells), or a plain picture that is a row or column of squares (a strip). The cels are cropped to the area anything is drawn in, keeping each cel's place. Drop such a file on a

Drawer bin and it lands there as an AnimBrush bin (the bin under the pointer if it is empty, else the first free one); a picture without frames dropped on a bin is an ordinary brush, as before.

**AnimBrush bins.** `Shift`+click a bin while you hold an AnimBrush and the bin keeps every cel: it shows cel 1 with a small film-strip badge holding the cel count, and while you hover it the bin plays (at the animation's FPS) - so does the Preview's Bin Quick Look. Click the bin to get the AnimBrush back, cels and all. Dragging the bin moves its cels with it; storing a plain brush over it makes it one picture again. AnimBrush bins are saved in `.dset` sets and in `.draw` files; an older DRAW opens those with the bin's first cel only.

### `.dset` import / export

The drawer state can be saved as a `.dset` file and reloaded later — a clean way to share brush packs and pattern libraries between projects or with other artists. There's a sample set in `DEV/brush-set.dset`.

DRAW 2.8 has **96 slots** per mode (earlier versions had 30). Sets and `.draw` files keep working both ways: an older DRAW opens a new file with its first 30 slots, and a new DRAW opens an old file with the slots after 30 empty.

### Mini palette

The Drawer can show a **mini palette** beside its bins for quick FG / BG selection (left / right click). It is **off by default**: turn it on in **Settings** → **Panels** → **Drawer** → **Mini Palette**. With it off, the bins take its space (more columns and rows). For a full swatch panel, see **View** → **Swatches**.

### Narrow and wide toolboxes

The drawer fits the toolbox column's width (drag the toolbox's inner edge, or set its columns in a workspace):

- **Wider than the default 4 columns:** up to 6 bin columns, so fewer rows.
- **Without the mini palette** (the default): as many bin columns as fit the width, up to 6.
- **Narrow:** fewer bin columns. When even one bin can't sit beside the mini palette (if it's on), the palette moves under the bins.
- **Not every row fits:** `Shift` **+wheel** over the bins scrolls them. The plain wheel still switches `.dset` sets.
- **The bins fill the width:** they grow to span the drawer, so no strip is left at the right.
- **See a bin big:** hover it and the Preview panel shows it enlarged (Bin Quick Look, see Chapter 11).
- **Zoom:** `Ctrl` **+wheel** over the bins zooms them: wheel up gives fewer, bigger bins per row (down to 1), wheel down gives more, smaller ones (up to 12). DRAW remembers the zoom ( `DRAWER_BIN_COLS` in DRAW.cfg). To go back to the automatic size, right-click a bin and choose **Bin Size: Auto**.

### 1-bit patterns

Patterns can be stored with **opaque backgrounds** (1-bit black/white) so that flooding with a 1-bit pattern under a colored FG produces classic Aldus PageMaker / DPaint dither patterns.

### Paint modes

The drawer ties into three paint modes that affect Fill and Brush behaviour:

- **Normal** — solid current FG color.
- **Pattern** — tile the active drawer slot.
- **Gradient** — interpolate between palette stops via the active gradient slot.

### Dithering algorithms

When painting gradients (or applying Posterize) you can pick from:

- **Ordered** — Bayer 2×2, 4×4, 8×8.
- **Floyd-Steinberg** — error diffusion.
- **Atkinson** — classic Mac dither.
- **Stucki** — heavier diffusion.
- **Blue Noise** — perceptually uniform.

➡️ Next: Chapter 10 — File I/O & Export

# Ch. 10 💾 File I/O & Export

**What you'll learn:** Every way to get artwork in and out of DRAW — the native `.draw` format, nine raster export formats, the famous QB64 source-code export, color-cycling formats (QB64 program, GrafX2 GIF, animated GIF, DeluxePaint LBM), and the Extract Images sprite-decomposition tools.

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## Opening & Saving — Formats Explained

🎯 **Goal:** Understand all file format options.

### Open / Import

| Format | Shortcut / menu |
|---|---|
| Image (PNG / BMP / JPG / GIF) | `Ctrl+O` (a sprite sheet PNG with its `.json` frame data beside it - Aseprite, TexturePacker, DRAW - opens with its frames; otherwise ANIM → Set Up → Detect Frames slices it at its gaps) |
| Project ( `.draw` ) | `Alt+O` |
| Aseprite ( `.ase` / `.aseprite` ) | File → Open Aseprite (two or more frames open as animation frames on a sheet, with their durations, tags and linked cels) |
| Photoshop ( `.psd` ) | File → Open Photoshop |
| DeluxePaint ( `.lbm` / `.iff` , ILBM or PBM) with cycle ranges | `Ctrl+O` , or File → Color Cycling → Open DeluxePaint / GrafX2 File |
| GrafX2 GIF with cycle ranges | `Ctrl+O` (the ranges come along automatically) |
| Import Image (oversized, interactive placement) | File → Import Image |
| Drag-and-drop file onto window | Windows |
| Command-line argument | `DRAW path/to/file.draw` |

### Save options

| Command | Shortcut | Notes |
|---|---|---|
| Save | `Ctrl+S` | Silent resave to current path. See *Saving an image you opened* below. |
| Save As... | `Ctrl+Shift+S` | Prompt for new name and format. |

| | | |
|---|---|---|
| Flatten & Save | `Ctrl+Alt+Shift+S` | Write the merged image back to the file you opened (or last saved), in the same format — no layers kept, no `.draw`. Asks for a name only for an untitled document, a `.draw` project, or a format DRAW cannot write (PSD, Aseprite, LBM, ANSI). |
| Export Selection | `Ctrl+Alt+S` | Save just the selected region. |
| Export Layer as PNG | Layer menu | One-click per-layer PNG. |
| Export Brush as PNG | `F12` | Current custom brush. |
| Export Again | `Ctrl+Shift+E` | Repeat the last File → Export As with the same format and file name — no dialog. Opening or creating another document forgets it. |

### Saving an image you opened

Whatever way you open a file — File → Open, Recent Files, the command line (`DRAW art.png`), drag and drop or the image browser — `Ctrl+S` saves back to it:

| Opened | `Ctrl+S` writes |
|---|---|
| `art.draw` | `art.draw` |
| `art.png` | `art.png`, with your layers hidden inside (it is still a normal PNG everywhere else) |
| `art.gif` / `.bmp` / `.jpg` / `.tga` / `.qoi` / `.ico` / `.hdr` | **Save .draw Beside Images** on (default): `art.draw` next to it, `art.gif` untouched.<br>Off: `art.gif` itself, flattened, in its own format (an animated GIF opened as frames is written back as an animated GIF). |
| `art.psd` / `.aseprite` / `.lbm` / `.ans` | **on:** `art.draw` next to it. **Off:** asks where to save (DRAW cannot write these formats). |

The setting is in **Settings** → **General** → **Saving**, or `SAVE_DRAW_SIDECAR=TRUE/FALSE` in `DRAW.cfg` (`--option SAVE_DRAW_SIDECAR=FALSE` for one run).

`Ctrl+Alt+Shift+S` (**File** → **Flatten & Save**) always writes the flattened image back to the image file, whatever the setting — use it when you want a plain `art.png` with no layers inside.

The **Recent Files** list holds up to 10 entries with quick-jump shortcuts `Alt+1` … `Alt+0`.

`File → New from Template…` reads from `ASSETS/TEMPLATES/` and gives you a starter canvas with appropriate dimensions, palette, and grid for common targets (NES sprite, Game Boy tile, Lospec sheet, etc.).

`ASSETS/TEMPLATES/ANIMATION/` holds animation starters - 16×16, 32×32, 64×64 and 128×128 with 8 blank frames, 320×200 with 4 - that open with their frames already set up, so the ANIM menu, the lightbox (`Num 0`) and the frame keys work straight away. On any other picture, `ANIM → Frames → Add Frame` (`Num +`) makes frames 1 and 2; until then the frame commands are greyed out.

# The `.draw` Format — PNG with Superpowers

> 🎯 **Goal:** Understand the native DRAW project format.

A `.draw` file is a **valid PNG file** with a custom `drAw` chunk embedded inside. Every image viewer on the planet can render the flattened preview. Only DRAW can read the rich data inside the chunk and reconstruct the editable project.

What `.draw` preserves:

- All layers, blend modes, opacities, names, groups, symbol parents/children.
- Active palette state and the auto-generated `[DOCUMENT]` palette.
- Tool states, grid settings, snap, crosshair config.
- Reference image configuration and overlay opacity.
- Text layer data (re-editable!).
- Extract Images settings.
- Character Mode state.

The format is versioned for forward compatibility — newer DRAW builds still read older `.draw` files, and they extend the chunk safely when new features arrive.

# Export As — 9 Image Formats

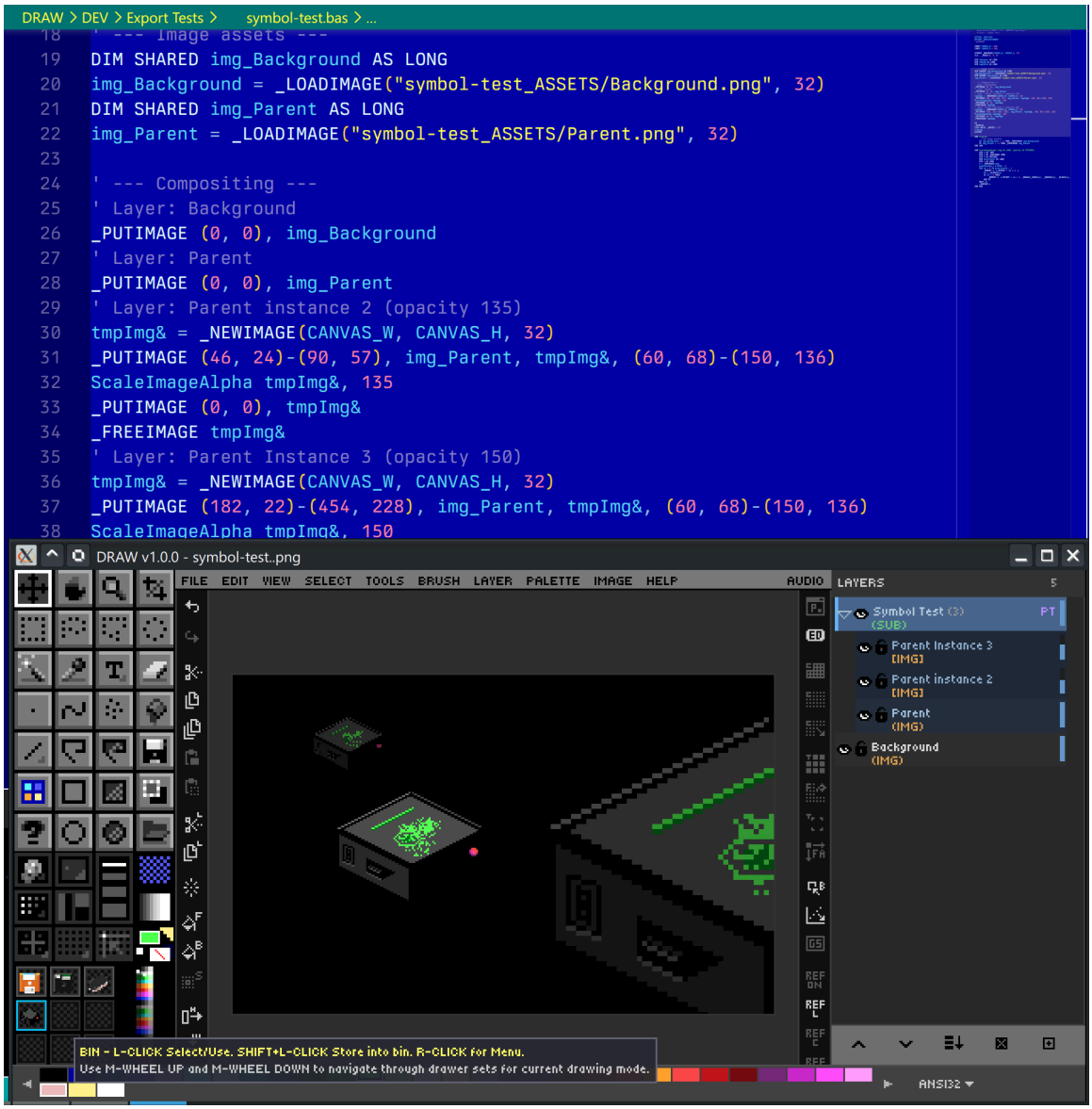
> 🎯 **Goal:** Export artwork in various formats.

| Format | Use case |
|---|---|
| **PNG (native `.draw`)** | Embeds the full project in a PNG; viewable anywhere, re-editable in DRAW. |
| **PNG (plain)** | Standard PNG, no metadata. |
| **GIF** | Static, palette-aware. |
| **JPEG** | Lossy compression for previews. |
| **TGA** | Truevision; popular in game dev. |
| **BMP** | Windows Bitmap — for retro tooling. |
| **HDR** | High Dynamic Range. |
| **ICO** | Windows Icon (multi-resolution). |
| **QOI** | Modern lossless Quite OK Image format. |

## QB64 Source Code Export

The signature feature: export your artwork as a self-contained `.bas` program. The file declares a screen, sets pixels in order, and runs as a standalone QB64-PE compile. Open it, hit compile, and your sprite paints itself onto the screen. This is wonderful for:

- BASIC tutorials.
- Embedding small graphics in QB64-PE programs without shipping image files.
- Showing how a piece of pixel art was constructed pixel by pixel.



# Color Cycling Formats — Share the Motion

> 🎯 **Goal:** Get color-cycling art out of DRAW in a form other people and programs can play, and bring cycling art in from DeluxePaint and GrafX2.

Cycle ranges (Chapter 3) are saved in every `.draw` file. To take them anywhere else, use **File → Color Cycling**:

| Command | Writes | Plays in |
|---|---|---|
| **Export Cycling QB64 Program (.bas)...** | One self-contained `.bas` file | QB64-PE: `qb64pe -x art.bas` |
| **Export GIF + Cycle Ranges (GrafX2)...** | A still `.gif` with GrafX2 `CRNG` ranges | GrafX2, and DRAW (ranges load back); other viewers show the still image |
| **Export Animated GIF of the Cycle...** | An animated `.gif` of one full loop | Any browser, chat app or image viewer |
| **Export DeluxePaint ILBM (.lbm)...** | IFF ILBM (planar) with `CRNG` chunks | DeluxePaint (Amiga), GrafX2, PyDPainter, Pro Motion, and DRAW |
| **Export DeluxePaint PBM (.lbm, PC)...** | IFF PBM (chunky), as saved by DeluxePaint II Enhanced on the PC | DeluxePaint for DOS, GrafX2, and DRAW |
| **Open DeluxePaint / GrafX2 File...** | — | Opens `.lbm` / `.iff` / `.gif` with the file's palette and ranges |

## The cycling QB64 program

The exported `.bas` needs no image files. A single `DATA` block, deflated and base64-encoded, holds the palette, the ranges, your art as a 32-bit image, and an 8-bit overlay containing only the cycling pixels. The program draws the art, then rotates the overlay's palette with `_PALETTECOLOR`. Each range runs on its own clock, in the same speed units and directions DRAW uses. It opens at the largest whole-number zoom that fits your screen. **Space** pauses, `+` / `-` change the speed, **R** restarts and **Esc** quits. The comments at the top list every range and explain the technique, so it also works as a lesson in palette animation.

## Formats and what they keep

- **Speeds** use DeluxePaint's `CRNG` unit (16384 = 60 steps per second), so ranges move from DRAW to `.lbm` and GIF and back without drifting.
- **Ping-pong and paused ranges** are DRAW extras that `CRNG` can't express. DRAW writes ping-pong as forward and a paused range as inactive, then adds a small private block ( `DRAWCYCL` in GIFs, `DRCY` in LBMs) that other programs skip and DRAW reads back exactly.
- **Transparency** becomes the GIF transparent color, or the LBM transparent color when the palette has a free slot.
- **Off-palette pixels** are mapped to the nearest palette color that is *not* in a cycle range, so they never start cycling by accident. The log reports how many there were.
- **Importing** a DeluxePaint or GrafX2 file creates a new document with the file's exact palette and pixel indices. A swatch must own its pixels, so when the file repeats a color that matters, DRAW nudges the later copy by one or two levels. Inactive ranges and ranges with speed 0 import as paused. Old 4-bit Amiga palettes are rescaled. Extra-Halfbrite images become 64 colors. HAM images aren't supported.

- **Saving** an opened `.lbm` or `.gif` with `Ctrl+S` writes a companion `.draw` next to it. The original file is never overwritten (see *Saving an image you opened* for the setting that changes this for `.gif`).

### The animated GIF

Frame 1 is the whole picture. Each later frame holds only the box around the cycling pixels plus that step's colors, so the files stay small. The loop is the exact least common multiple of your ranges' periods. Speeds whose periods divide evenly loop in a few seconds; very mismatched speeds are cut at 60 seconds or 1000 frames, with a note in the log. Times are rounded to GIF's 1/50-second steps.

### Batch conversion from the command line

Every format above is also available without opening a window:

```
DRAW art.png --palette art.gpl --cycle 16-31:8 --cycle 40-47:12:ping --export art.draw
DRAW art.draw --export art.bas           # .bas  .gif  .lbm (ILBM)  .bbm or .pbm.lbm (PBM)
DRAW art.draw --export-anim art-anim.gif
DRAW old.lbm  --export old.draw          # DeluxePaint → DRAW, ranges included
```

`--cycle LO-HI[:STEPS_PER_SECOND[:fwd|rev|ping]]` adds a range and can be repeated. DRAW exits after the work is done, with status 0 on success. `DEV/tools/make-cycle-examples.sh` builds the scenes in `SAMPLES/COLOR CYCLING/` this way.

A sheet with frames exports its animation the same way. `--export` and `--export-anim` can be repeated, so one run writes every format:

```
DRAW walk.draw --export-anim walk.gif --export-anim walk.apng --export-anim walk.json
DRAW walk.draw --export-anim walk.png          # PNG sequence walk_0001.png ... (one per tick at the FPS)
DRAW walk.draw --export-anim walk.bas          # QB64 player, the frames in DATA
DRAW walk.draw --export-anim walk.sheet.bas    # the player + walk.sheet.png it cuts the frames from
DRAW walk.draw --export walk.png --option ANIM_EXPORT_SCALE=2 --export-anim walk.gif
DRAW walk.draw --tag jump --export-anim jump.gif   # just the frames tagged "jump"
DRAW --sequence frames/run_0001.png --export-anim run.gif   # a numbered PNG sequence as frames
```

`.gif` and `.bas` mean the frames only with `--export-anim` (with `--export` they are the color cycling formats); `.apng`, `.json` and `.sheet.bas` work with either flag, and `--export walk.png` writes the picture flattened. Each export prints one line - `wrote PATH (summary)` or `FAILED PATH: why` - and the exit status is 1 when any of them failed. `--tag NAME` limits the animated exports to that tag's frames (the sprite sheet still has every frame and lists the tags); a name the document doesn't have fails the run before anything is written. `--sequence` opens the file named with all its numbered siblings (`run_0001.png`, `run_0002.png` ... or `run1.png` ... `run10.png`) as frames, in number order; a file the same as the one before it becomes a held frame (ANIM → Set Up → Import PNG Sequence... does the same from the menu). A document with a soundtrack (ANIM → Soundtrack...) exports its BAS players with the sound copied beside them as `walk-sound.ogg` (named after the `.bas`), and the program plays it in step with the frames; keep the two together. The batch runs in DRAW's own window for a moment, so on a server run it under Xvfb.

# ANSI Art — Import & Export

> 🎯 **Goal:** Bring classic ANSI/BBS art into DRAW as pixels, and turn your pixel art into shareable ANSI.

DRAW reads and writes the ANSI-art family of formats, so you can drop lo-fi mockups into DRAW, refine them as pixels, and hand them back out as `.ans`.

## Importing ANSI (File → Import ANSI...)

Supported inputs:

| Kind | Extensions | Notes |
|---|---|---|
| **Text ANSI** | `.ans` `.asc` `.nfo` `.diz` | SGR escape sequences — 16-color, **iCE**, **256-color** ( `38;5;n` ), and **24-bit truecolor** ( `38;2;r;g;b` ). UTF-8 block/box art is auto-detected and mapped to CP437. |
| **BIN** | `.bin` | Raw character+attribute pairs; width from SAUCE ( `BinaryText` ). |
| **XBIN** | `.xb` `.xbin` | Full XBIN: custom 16-color palette, embedded font, and RLE compression. |
| **TundraDraw** | `.tnd` | 24-bit truecolor binary stream. |

The **Import ANSI** dialog previews the art and its detected format, and lets you choose how it rasterises onto the canvas:

- **WIDTH** — `8 PX` (square) or `9 PX` (authentic VGA 9-dot; box-drawing characters join across cells).
- **ASPECT** — `SQUARE` or `DOS 1.2` (the classic non-square CRT stretch).
- **SCALE** — `1×` – `8×` nearest-neighbor.
- **Cell height** follows the SAUCE font: `IBM VGA` → 8×16, `VGA50` / `EGA43` → 8×8.

If an **XBIN** carries an embedded **font** or a custom **palette**, the dialog offers to install them into your libraries:

- **Import font** writes the font to `FONTS/ANSI Imports/` so it appears as a bitmap font for the Text tool.
- **Import palette** writes a `.GPL` into your user palettes and refreshes the picker.

You can also open ANSI art straight from the command line: `./DRAW.run artwork.xb`.

## Exporting ANSI (File → Export ANSI...)

Export converts the canvas to **half-block** ANSI (each character cell shows two stacked pixels via `▀`). The dialog gives a live source-vs-render preview plus:

| Control | Choices |
|---|---|
| **FROM** | `DOCUMENT` (flattened) · `LAYER` (current) · `SELECTION` (marquee region) — auto-selects Selection when one is active |
| **COLOR** | `16` · `iCE` · `256` · `RGB` (24-bit) — auto-detected from the art |
| **CELLS** | `PER-PIXEL` (one canvas pixel per cell — maximum detail) · `1:1 8PX` (one 8-px cell — round-trips terminal art at native size) |

| | |
|---|---|
| **FONT** | `VGA 8x16` · `VGA50 8x8` (written to the SAUCE record) |
| **SIZE** | Column count (auto, or stepped) |

A byte-exact **SAUCE** record (dimensions + font) is appended. Use **Selection** as the source to mock a scene in pixels and export just one section to ANSI.

> 💡 **Round-trip tip:** `PER-PIXEL` keeps every pixel but renders ~8× larger in a terminal/viewer; `1:1 8PX` maps 8×16-pixel blocks to one cell, so exported terminal art re-imports at its original size.

# Extract Images — Sprite Sheet Decomposition

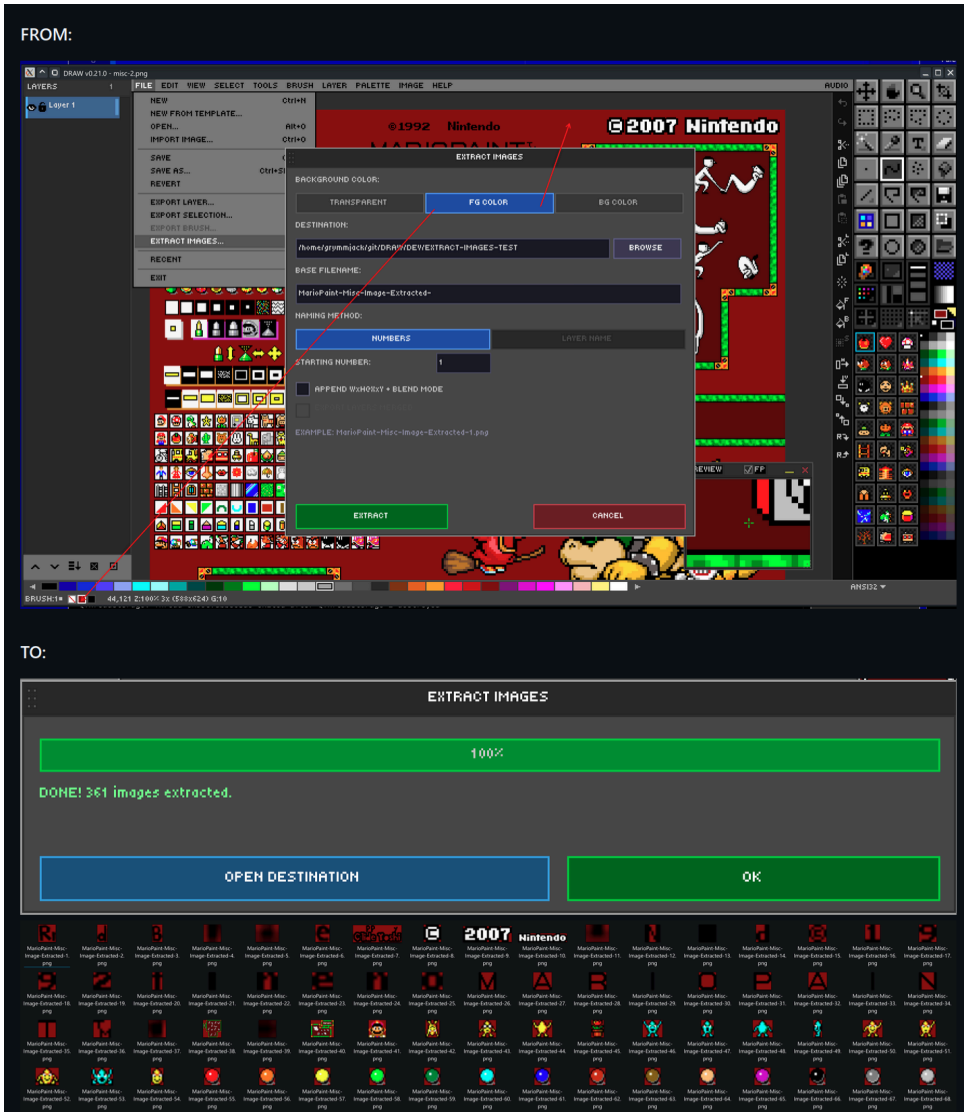
> 🎯 **Goal:** Extract sprites from sheets or compositions.

DRAW can decompose a finished image into individual sprite PNGs. Three extraction methods are available:

- **Flood-fill connected regions** — auto-detect distinct sprites by transparent gutters. Best for unsorted sprite sheets.
- **Per-layer extraction** — each layer becomes one PNG. Best when your sheet was built layer-by-layer.
- **Merged extraction** — flatten visible layers and extract regions from the result.

Background options for each output sprite: **Transparent**, **FG color**, or **BG color**. Output is a folder of separate PNG files; the settings you pick are persisted in the `.draw` file so you can re-export later with one click.

There is a related command, **Extract From Grid...**, which slices a regular grid (e.g., 16×16 cells) into named PNGs — perfect for tilesets — and **Extract To Layers From Grid...**, which slices a grid into named *layers* in a new project.




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➡️ Next: Chapter 11 — Canvas & View Controls

# Ch. 11 🖥️ Canvas & View Controls

> **What you'll learn:** How to navigate any canvas — zoom, pan, the floating Preview Window, pattern-tile preview, reference images, fill / clear shortcuts, and canvas resize / crop.

---

## Zoom, Pan & Navigation

> 🎯 **Goal:** Navigate canvases of any size efficiently.

### Zoom

| Action | Shortcut |
|---|---|
| Zoom in/out continuously | `Ctrl` +Mouse wheel |
| Zoom in / out 1 step | `Ctrl+=` / `Ctrl+-` |
| Reset to 100% | `Ctrl+0` |
| Zoom to canonical level | Hold `Z` and press `1-8` for 100%-800%, `9` for 1600%, and `0` for 3200% |
| Zoom tool | `Z` (  ) |
| Zoom to dragged region | `Z` then drag |
| Zoom out (Zoom tool) | `Alt` +Click |

Zoom snaps to canonical levels: 25%, 50%, 75%, 100%, 200%, 300%, 400%, 600%, 800%, 1600%, 3200%

### Pan

You can pan **with any tool active**, no need to switch:

- Middle-mouse-drag.
- `Spacebar` +drag.
- **Double-middle-click** — instantly resets pan and zoom to fit the canvas.

Scrollbars appear automatically for canvases larger than the viewport. The canvas border ( `#` ) outlines the edge of the canvas at any zoom.

# Preview Window, Tile Mode & View Options

> 🎯 **Goal:** Use preview and tiling for workflow efficiency.

## Preview Window

`F4` toggles the **Preview Window**, a floating, resizable, repositionable secondary view. It runs in one of three modes:

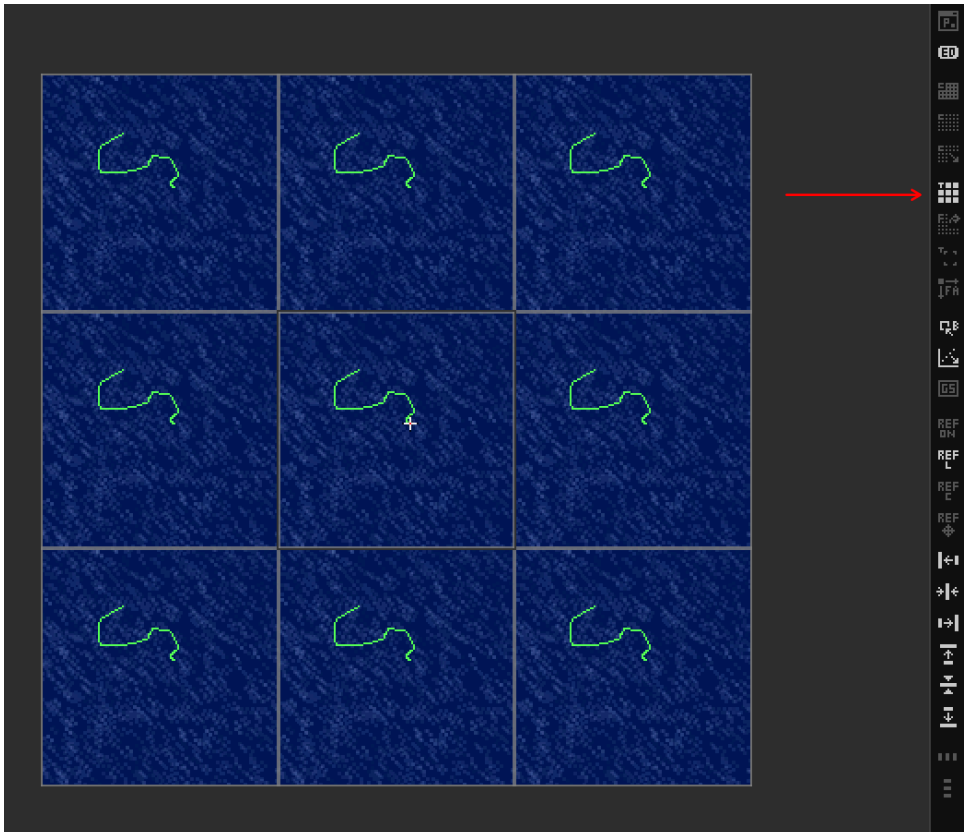
1. **Follow** — a magnifier that tracks the cursor.
2. **Floating Image** — a static reference display (e.g., the source you're tracing).
3. **Bin Quick Look** — hover any drawer slot to see its content big in the preview. A small brush grows by whole steps (a 7×5 brush shows at 7×5 × 4, say), so pixel art stays crisp; a big one shrinks to fit. Patterns tile the pane. Works on bins you scrolled to as well. Turn it off in **Settings → Panels → Preview → Bin Quick Look**.

Independent zoom and pan inside the preview, plus `Alt` **+Click to color-pick** without switching tools. The preview also remembers up to 10 recent images for quick swaps.



## Pattern Tile Mode

**View** → **Pattern Tile Mode** (or the tile-mode button in the Advanced Bar) toggles a 3×3 tiled preview (up to 512×512 source). Seams in seamless textures jump out instantly because they appear nine times.



## Other view options

- **Grayscale Preview** — `Ctrl+Alt+Shift+G`. Inspect value structure without color distraction.
- **Reference Image** — `Ctrl+R`. See Chapter 15.

## Canvas fill & clear

- `Delete` — clear (Edit → Clear). Removes the active selection's pixels (or the whole layer if no selection). No prompt.
- `Backspace` — fill with FG color. Instant, no prompt.
- `Shift+Backspace` — fill with BG color. Instant.

## Canvas crop & resize

- **Crop tool** ( ) — interactive with handles; commit with Enter.
  - Drag handles **inward** to trim the canvas to the selected region.
  - Drag handles **outward past a canvas edge** to **grow** the canvas in that direction — new pixels are transparent on all layers. This is a quick alternative to the canvas resize dialog when you just need to add space on one side.
  - Arrow keys nudge the crop box 1px; `Shift+Arrow` nudges 10px.
- **Image** → **Canvas Resize…** — dialog with anchor selection and pixel/percent toggle.

➡️ Next: Chapter 12 — UI Customization & Settings

# Ch. 12 ⚙️ UI Customization & Settings

> **What you'll learn:** How to configure DRAW to your taste — the eight-tab settings dialog, theming, panel docking, and the various ways `DRAW.cfg` works under the hood.

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## Settings Dialog — All 8 Tabs Explained

> 🎯 **Goal:** Configure DRAW to your preferences.

Open the settings dialog with `Ctrl+,` (comma) or `Edit → Settings`. The dialog has eight tabs covering every persistent option.

| Tab | What it controls |
|---|---|
| **General** | Display scale, fullscreen toggle, FPS limit, UI scaling, tooltips and the **Help Card** (on/off, which edge of the canvas area it docks to, hover delay). **Tooltip Delay** and **Help Card Delay** work the same way: 0–5000 ms (0 = instant). The **−**/**+** buttons and the mouse wheel step by 50 ms, and you can click the number to type an exact value, **Enable AI Features**. |
| **Grid** | Default grid size, geometry, alignment, snap state, crosshair appearance. |
| **Palette** | Default palette, recent-palettes list size, Lospec UI visibility. |
| **Panels** | Default visibility for Toolbox, Layers, Edit Bar, Advanced Bar, Preview, Character Map, Drawer, Color Mixer; Layers panel options, including **Thumbnails** (Off / Tiny / Small / Medium), **Thumbnail Shows** (Content / Canvas) and **Min Crop (px)**; the animation **Lightbox Strip**: **Strip Layout** (Tray / Row), **Min Cell (px)** and **Cell Size (px)**, and **X-Sheet Rows** (Text / Tiny / Small / Medium: the layer rows and the SOUND rows, whose waveform grows with them). |
| **Audio** | SFX and music enable/disable, master volume, mute. |
| **Fonts** | Default font and size, paths to scan for TTF/OTF. |
| **Appearance** | Various color scheme configurations. |
| **Directories** | Where DRAW looks for templates, palettes, music. |

### Enable AI Features

AI image generation is **off by default**, and while it is off DRAW surfaces nothing about it — no AI menu, no commands, no layer markers, no status text. Some people want nothing to do with generative tooling, and a disabled-but- visible feature is still in the way.

Tick **Enable AI Features** in the General tab to turn it on. That checkbox is the only place AI is mentioned while disabled. See Ch. 21 — AI Image Generation.

### User fonts

Drop your own fonts into DRAW's **user data directory** and they show up in the font pickers — no reinstall, no config editing. The base folder is `<data>/FONTS/` (Linux `~/.local/share/DRAW/FONTS/`, with the platform data-dir equivalent on macOS/Windows). Where each type goes:

| Font type | Extensions | Where to put it |
|---|---|---|
| **TTF / OTF** | `.ttf` `.otf` | `FONTS/` root **or** any subfolder |
| **Bitmap** | `.fxx` /F16/F14/F08, `.psf`, `.bdf`, `.pcf`, `.fon` | a **subfolder** of `FONTS/` |
| **Color bitmap (CBF)** | DPaint-style `.bmp` spritesheets | a **subfolder** of `FONTS/` |
| **TheDraw** | `.TDF` | `FONTS/THEDRAW/` — plain `.TDF` files, no `.TDX` index needed |

A subfolder's name becomes its group label in the font dropdown. Each source has a checkbox in the **Fonts** settings tab — **User TTF/OTF Fonts**, **User Bitmap Fonts**, **User TDF Fonts** — mirrored by `FONTS_INCLUDE_USER`, `FONTS_INCLUDE_USER_BITMAP`, and `FONTS_INCLUDE_USER_TDF` in `DRAW.cfg`. Toggles take effect on the next launch.

## `DRAW.cfg`

`DRAW.cfg` is a plain-text key/value file that lives next to the executable. You can hand-edit it any time. There are also **OS-specific** variants — `DRAW.linux.cfg`, `DRAW.macOS.cfg`, `DRAW.windows.cfg` — that override the base file when present.

CLI flags:

- `--config /path/to/your.cfg` — use a non-default config file.
- `--config-upgrade` — reconcile your existing config with any new defaults introduced by an upgrade. Recommended after each release.
- `--option KEY=VALUE` — override any config key from the command line (repeatable; beats the `.cfg`). E.g. `--option FONTS_INCLUDE_USER_TDF=TRUE`.
- `--options-list` — print every config key with its default and description, then exit.
- `--cfg` — print the path of the config file DRAW is actually using (after `--config`, OS-specific files and `--portable` are taken into account), then exit. Handy as `vim $(DRAW --cfg)`.
- `--dirs` — list every DRAW folder: config, data, cache, crash logs, templates, brush/pattern/gradient sets, palettes, fonts, and the current theme's images, sounds, music, cursors and fonts.
- `--dir-NAME` — print one of those folders as a bare path, e.g. `cd $(DRAW --dir-theme-sounds)` or `ls $(DRAW --dir-crash-logs)`. Names: `cfg data cache crash-logs templates brushes patterns gradients palettes fonts theme theme-images theme-sounds theme-music theme-cursors theme-fonts`.

`DRAW --help` is colored in a terminal (plain when piped, with `--no-color`, or when `NO_COLOR` is set). If `ASSETS/DRAW.ans` exists, the ANSI-art logo in it is printed above the help.

# Theming — Icons, Colors & Sounds

> 🎯 **Goal:** Customize DRAW's look and feel.

A **theme** is a folder under `ASSETS/THEMES/` that contains:

- A `THEME.CFG` file with all UI colors.
- A directory of icon PNGs (replaceable per theme — see `ASSETS/THEMES/DEFAULT/IMAGES/`).
- A `SOUNDS/` folder of WAV/OGG files (per-theme SFX).
- A `MUSIC/` folder of tracker tunes that play on startup or via the Audio menu.
- A `FONTS/` folder of bitmap and vector fonts.
- A `splash.png` for the launch animation.

Anything you can theme — UI palette, transform-overlay frame, smart-guide colors, layer panel highlights — lives in `THEME.CFG`. **No recompile is required**; DRAW reloads themes at runtime.

## Display & Toolbar scale

- **Display Scale** — 1× through 8×. Suits HiDPI monitors.
- **Toolbar Scale** — 1× through 4×. Independent of display scale, so you can have small toolbar icons on a HiDPI display.
- **Panel Scale** — the Animation, Color Mixer, Advanced Color Picker, 3D Color Space, Pen Pressure and Swatches panels each have their own size: **Auto** (the default) follows the UI Scale, or fix one at 1× through 8×. Right-click the panel's title bar → **Scale**, or Settings → Panels → Panel Scale.

# Panel Layout & Docking

> 🎯 **Goal:** Arrange panels for your workflow.

## Dockable panels

| Panel | Toggle |
|---|---|
| Toolbox | `Tab` |
| Layer Panel | `Ctrl+L` |
| Edit Bar | `F5` |
| Advanced Bar | `Shift+F5` |
| Character Map | `Ctrl+M` |
| Preview Window | `F4` |
| Drawer | (Organizer / View menu) |
| Color Mixer | View → Color Mixer |

Each can be docked **left or right** by `Ctrl+Shift` +clicking on the panel itself.

Beyond left and right, panels can be **stacked** in one column, **tabbed**, **collapsed** to their title, **floated** in their own window and docked back. Drag a panel by its handle (its title strip, the layers header, or the thin grip that appears on a panel's top edge), and right-click a handle for Move to Left/Right Edge, Collapse, Float, Reset Arrangement and — on the Animation, Color Mixer, color picker, 3D Color Space, Pen Pressure and Swatches panels — **Scale** (Auto or 1×–4×). The floating windows (Preview, Color Mixer, the color pickers, Pen Pressure, Swatches, Browser) dock by dragging their title bar onto a dock target. Docked, the Animation, Color Mixer, color picker, 3D Color Space, Pen Pressure and Swatches panels fill their slot: their rows and sliders stretch to the column's width, and in a column with room to spare their big area — the color field, the wheel and its shade rows, the 3D view, the response graph — grows taller. Floating, they go back to their own size. See **Chapter 22 — Moving, stacking, tabbing and floating panels**.

## UI master toggles

- `F11` — toggle **all** UI (canvas-only mode).
- `Ctrl+F11` — keep only the menu bar.
- `F10` — toggle the status bar.

DRAW also supports **auto-hide** while drawing: panels fade out so they don't obscure your work, then return when the cursor leaves the canvas.

## Workspaces

A **workspace** is a layout preset that DRAW overlays on your own setup. It can set which panels show, the toolbox, the menus, the bars and extra keys. Leaving it puts everything back, and nothing it changes is saved to `DRAW.cfg`. Switch with **View** → **Workspace**, `Ctrl+Shift+W`, or the `[WS: name]` status badge. Inside a workspace, `F11` shows what it hid. See **Chapter 22 — Workspaces & Screenshot Annotation**.

### Cursor system

The cursor system uses your OS-native cursor for UI hovers and a custom-painted cursor for tool-specific feedback (crosshair on dot, brush footprint on brush, etc.). This is automatic and themeable.

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➡️ Next: Chapter 13 — Audio: Music & Sound Effects

# Ch. 13 🔊 Audio: Music & Sound Effects

> **What you'll learn:** DRAW's per-theme sound effect bank and tracker-music player, plus how to swap, mute, and customize the audio experience — including MIDI playback and SF2 SoundFont configuration.

---

## Audio System — SFX, Music & Customization

> 🎯 **Goal:** Configure the creative audio experience.

### Sound effects (21 categorized slots)

DRAW emits short SFX for distinct UI actions — menus, tools, selection, fill, clipboard ops, layer ops, text entry, sliders, drag-and-drop, and more. The sounds live in the active theme's `SOUNDS/` folder as WAV or OGG files. Replace any file with one of the same name to retheme that action without touching code.

You can:

- Globally enable / disable SFX.
- Adjust master SFX volume.
- Mute (independent of disabling).

### Background music

DRAW plays **tracker** modules and **MIDI** files natively:

| Format | Description |
|---|---|
| `.mod` | Amiga ProTracker / FastTracker. |
| `.xm` | FastTracker II Extended Module. |
| `.it` | Impulse Tracker. |
| `.s3m` | Scream Tracker 3. |
| `.rad` | Reality Adlib Tracker. |
| `.mid` / `.midi` | Standard MIDI Format (SMF). |
| `.rmi` | RIFF MIDI (Windows MIDI container). |

The **Audio** menu (rightmost in the menu bar) hosts:

- **Auto-shuffle** — when the current track ends, DRAW picks a random next track from the active music folder.
- **Next track** — `}`
- **Previous track** — `{`
- **Random track** — picks at any time. Use the **Random Track** submenu to restrict by format: `.MOD`, `.IT`, `.XM`, `.RAD`, `.MID`, or `.RMI`.
- **Volume up / down** — ±10% increments, independent of SFX volume.
- **NOW PLAYING** — read-only display of the current track name and format (e.g. `NOW: jester - stardust [MOD]`). Name is truncated to 32 characters.
- **Explore Music Folder** — opens the music directory in your OS file manager.

All audio settings persist in `DRAW.cfg`.

---

## MIDI Playback & SoundFont Configuration

> 🎯 **Goal:** Get the best possible MIDI sound quality.

DRAW uses QB64-PE's built-in **OPL3 FM emulator** for MIDI/RMI playback out of the box — no extra files required. For richer, more realistic MIDI audio you can supply your own **SoundFont2** (`.sf2`) file.

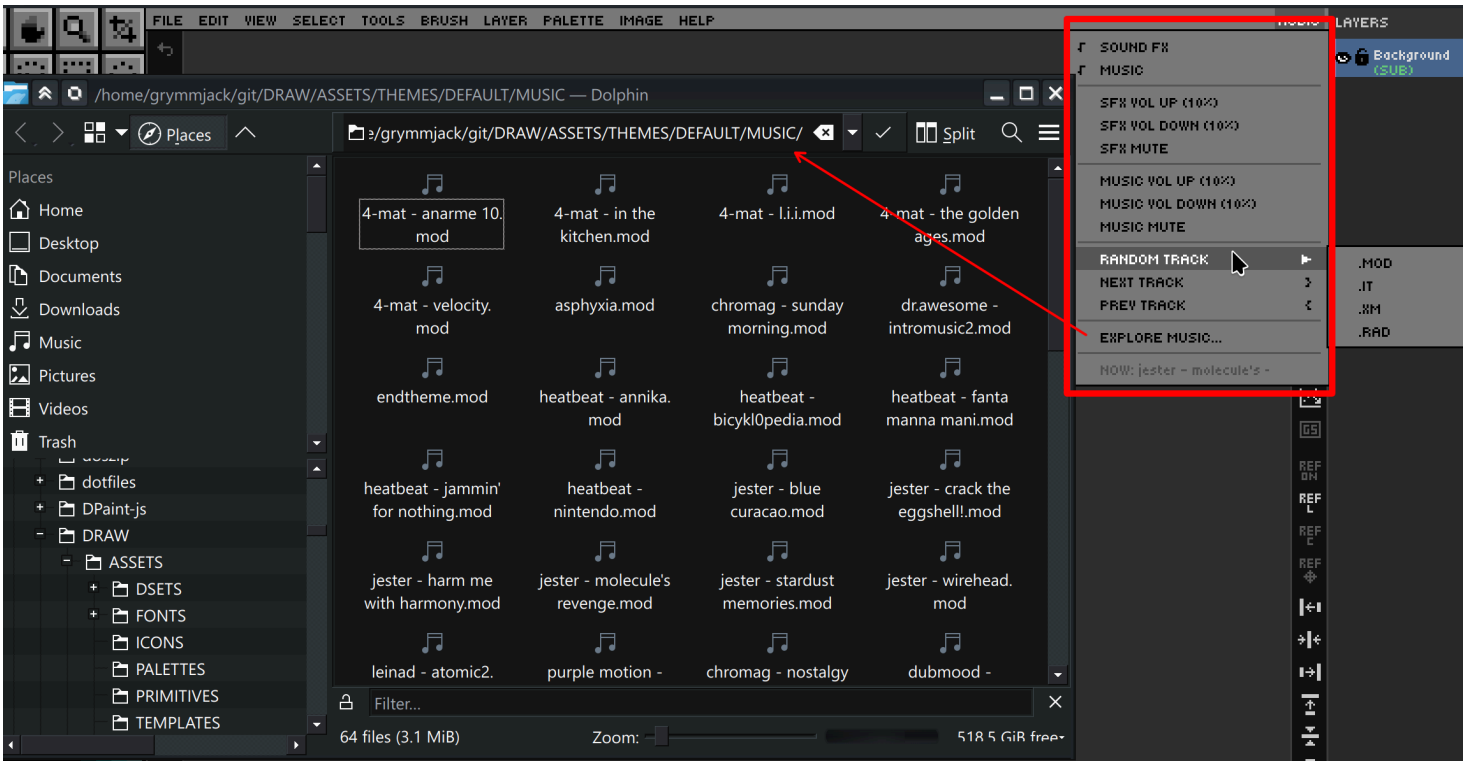
### Configuring an SF2 SoundFont

1. Open **Settings** (Edit → Settings or `Ctrl+,`).
2. Navigate to the **Audio** tab.
3. Scroll to the **MIDI Soundfont** section.
4. Click the `...` browse button to locate a `.sf2` file on your system.
5. Click **Apply** or **OK** — the font is loaded before every MIDI/RMI track.
6. To revert to built-in OPL3 FM synthesis, click the **Clear** button next to the path field.

> **Tip:** Free general-MIDI soundfonts such as *GeneralUser GS* or *SGM-V2.01* work well and are widely available online.

### How MIDI playback works

- `_MIDISOUNDBANK` is called with the SF2 path immediately before `_SNDOPEN` for each MIDI/RMI track.
- If the SF2 path is empty or the file no longer exists, DRAW automatically falls back to the built-in OPL3 emulator.
- The `MIDI_SF2_FILE` key in `DRAW.cfg` stores the path so it persists across sessions.
- Drop `.mid`, `.midi`, or `.rmi` files into `ASSETS/THEMES/DEFAULT/MUSIC/` and they will be included in random-track auto-shuffle alongside tracker modules.




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➡️ Next: Chapter 14 — Pixel Art Analyzer

# Ch. 14 🔍 Pixel Art Analyzer

> **What you'll learn:** How DRAW's built-in Pixel Art Analyzer detects the five most common pixel-art mistakes and how to fix each one.

---

## Pixel Art Analyzer — Find & Fix Common Issues

> 🎯 **Goal:** Improve pixel art quality with automated analysis.

The **Pixel Art Analyzer** scans the active layer (or selection) for five well-known pixel-art problems and visualizes each one in an overlay so you can fix them efficiently.

### Detection categories

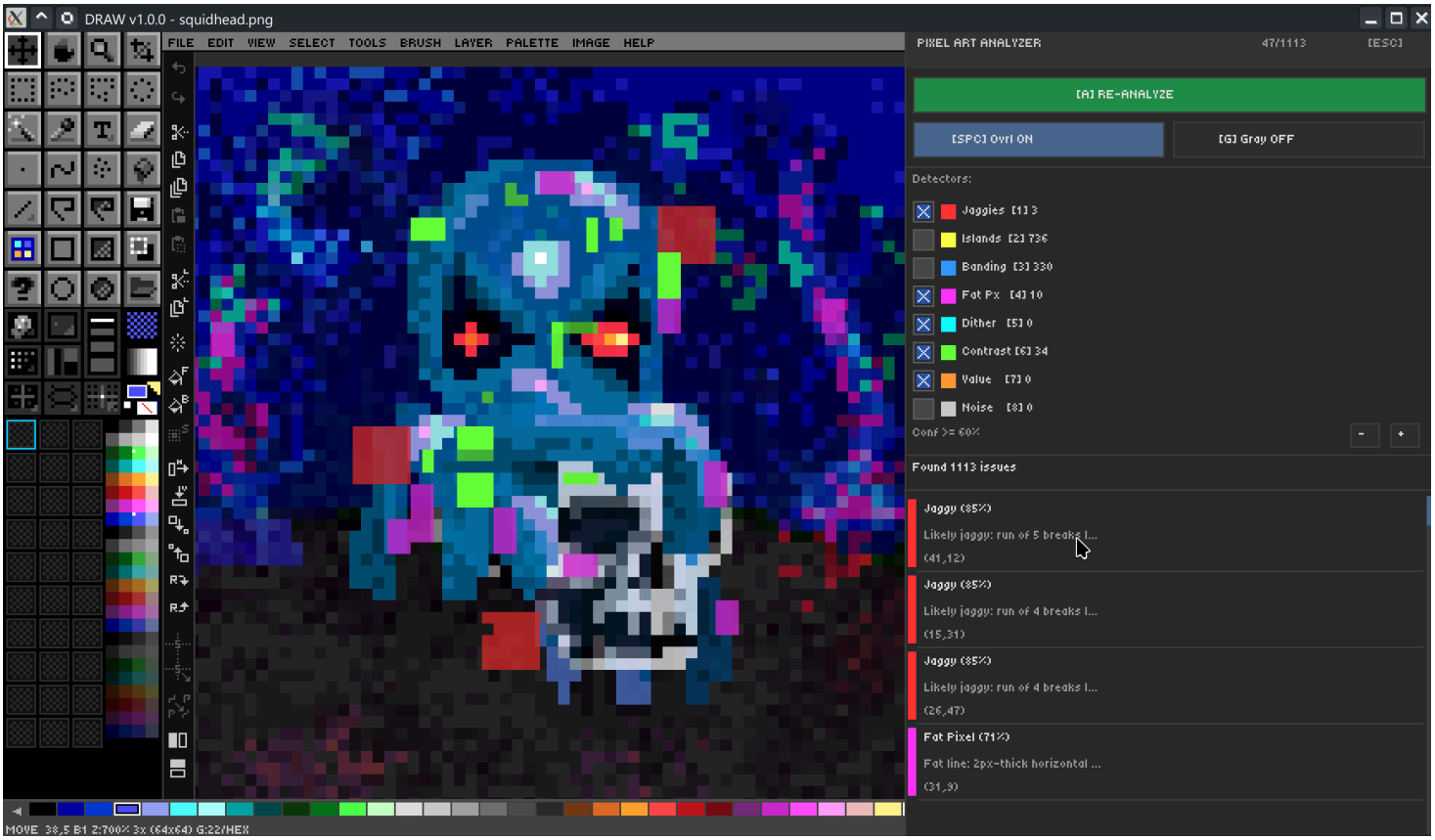
| Category | What it finds | Why it's a problem |
|---|---|---|
| **Orphan pixels** | Single isolated pixels surrounded by transparency or different-colored pixels. | Visual noise and unintentional speckling. |
| **Jagged lines** | Stairstepped lines with uneven step lengths. | Reads as rough or amateurish; clean stairsteps should be regular. |
| **Banding** | Repeating horizontal/vertical color stripes that should be a smooth gradient. | Looks like a quantization artefact, not intentional shading. |
| **Pillow shading** | Concentric layered shading that follows the silhouette outline. | Produces a "puffed up" 3D effect that ignores light direction. |
| **Doubles** | Two adjacent identical-color pixels along an outline that should be 1px. | Throws off the line weight and stair-step rhythm. |

### How it works

The analyzer uses a precompute engine for fast incremental analysis. After the first scan it caches the per-pixel classifications so re-analysis after a fix is near-instant. Results are presented in an interactive dialog with totals per category and a clickable list that pans/zooms to each occurrence.

### 🎨 Try it — before/after cleanup

1. Open a sprite that you know is rough.
2. Run `Image → Pixel Art Analyzer`.
3. Step through orphan pixels first; remove or merge each one.
4. Re-analyze. Move to jagged lines next, then doubles, banding, and pillow shading last.
5. Save before-and-after copies for your portfolio.




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➡️ Next: Chapter 15 — Reference Image & Import

# Ch. 15 🖼️ Reference Image & Import

> **What you'll learn:** How to load a reference image for tracing or study, and how to import oversized images with interactive placement, rotation, and flipping.

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## Reference Image — Tracing Made Easy

> 🎯 **Goal:** Use reference images for tracing and study.

Press `Ctrl+R` to toggle a reference image. The reference renders **behind every layer** with adjustable opacity, so it never interferes with the artwork you're producing.

| Action | Shortcut |
|---|---|
| Toggle reference image | `Ctrl+R` |
| Adjust reference opacity | `Ctrl+Shift` +Wheel (5–100%) |
| Reposition / zoom / nudge | Reference toolbar buttons |

Reference image state is persisted in the `.draw` file along with its opacity, position, and zoom — so a project's tracing setup is never lost.

> 🎨 **Try it — pixel art from a photo**
>
> 1. Load a photo as a reference image.
> 2. Set opacity to 50%.
> 3. Pick a limited palette (e.g., Endesga 32).
> 4. Trace key shapes on a layer above with the line and polygon tools.
> 5. Fill regions with flood fill.
> 6. Hide the reference ( `Ctrl+R` ) to evaluate the standalone result.

# Image Import — Oversized Placement & Transform

> 🎯 **Goal:** Import and position external images.

`File → Import Image` is different from `File → Open Image`. Open *replaces* your canvas; Import places the incoming image as a **floating overlay** that you position interactively before it commits as a layer.

While in import mode:

- Pan and zoom *inside* the imported image to crop tightly.
- Rotate 90° CW / CCW with the import toolbar buttons.
- Flip horizontal / vertical.
- Drag the destination box's resize handles to scale the placement.
- Live preview reflects every transform.

This is the cleanest way to bring an external sprite into a project at the right size and orientation, without permanent destructive resampling.

### Aseprite & Photoshop

DRAW reads `.ase` / `.aseprite` and `.psd` files directly via `File → Open Aseprite` and `File → Open Photoshop`. Layers are preserved (where they map cleanly to DRAW's blend mode set) and re-exposed as native DRAW layers.

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➡️ Next: Chapter 16 — Keyboard Shortcuts & Command Palette

# Ch. 16 ⌨️ Keyboard Shortcuts & Command Palette

> **What you'll learn:** How to find any command instantly via the Command Palette, and the most important shortcuts grouped by activity.

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## Command Palette — 200+ Commands at Your Fingertips

> 🎯 **Goal:** Access any command instantly.

Press `?` to open the **Command Palette**. Start typing — DRAW does fuzzy-matching across command names so you can type `flip` to find both *Flip Horizontal* and *Flip Canvas Vertical*. Each match shows its hotkey on the right; arrow keys to highlight, Enter to run.

Use `Help → Cheat Sheet` for **Quick Reference Mode**, which lists every command in the application without filtering — handy for learning what's available.

The Palette is grouped into categories that match the menus:

- Tools
- File
- Edit
- View
- Color
- Brush
- Layer
- Canvas
- Assist
- Grid
- Symmetry
- Select
- Help
- Image
- Audio

# Keyboard Shortcuts Cheat Sheet

> 🎯 **Goal:** Memorize the essential shortcuts.

The full, authoritative list is in `SHORTCUTS.md` at the repository root — generated directly from the input registry, so it never drifts from the app, and it covers **both keyboard and mouse**. Below are the essentials — what you'll use thousands of times a session.

## Tool selection (single key)

| Key | Tool |
|---|---|
| `B` | Brush |
| `D` | Dot |
| `L` | Line |
| `R` | Rectangle ( `Shift+R` filled) |
| `C` | Ellipse ( `Shift+C` filled) |
| `P` | Polygon ( `Shift+P` filled) |
| `F` | Fill |
| `K` | Spray |
| `I` | Picker |
| `E` | Eraser |
| `M` | Marquee |
| `W` | Wand |
| `V` | Move |
| `Z` | Zoom |
| `T` | Text |
| `Q` | Bezier |
| (Toolbar) | Smart Shapes (Polygon, Pie/Donut, Rounded Rect, Tab, Pill, Pac-Man, 3D Dice, Bevel Rect, Arrow, 3D Text) |

## Essential combos

| Combo | Action |
|---|---|
| `Ctrl+S` / `Ctrl+O` / `Ctrl+N` | Save / Open / New |
| `Ctrl+Z` / `Ctrl+Y` | Undo / Redo |
| `Ctrl+C` / `Ctrl+X` / `Ctrl+V` | Copy / Cut / Paste |
| `Ctrl+A` / `Ctrl+D` / `Ctrl+E` | Select All / Deselect / Clear |
| `Ctrl+L` | Toggle Layer Panel |
| `[` / `]` | Brush Size − / + |
| `\` | Brush Shape (circle ↔ square) |
| `X` | Swap FG / BG |
| `0` – `9` | Paint Opacity 100% / 10–90% |
| `'` `;` `.` `,` | Grid: toggle / snap / size+ / size− |
| `/` | Toggle grid alignment (Corner ↔ Center) |
| `G+Arrow` | Adjust grid width/height independently |
| `}` / `{` | Music: next / previous track |
| `F4` – `F9`, `F11` | Panel and mode toggles |
| `Tab` | Toggle Toolbar |
| `?` | Command Palette |

## Smart Shapes & 3D Tool Modifiers

| Key | Function |
|---|---|
| `Up/Down/Left/Right` | Adjust shape parameters (segments, bevel, mouth, Z depth, etc.) |
| `Mouse Wheel` | Adjust primary parameter |
| `Shift` | Constrain aspect ratio |
| `Left-click drag` | Wireframe mode (FG color) |
| `Right-click drag` | Solid mode (BG color fill, FG color wireframe) |
| `4` / `6` / `8` / `0` / `1` / `2` | Switch dice type (D4/D6/D8/D10/D12/D20) |
| `W` / `A` / `S` / `D` / `Q` / `E` | Orbit light |
| `=` / `-` | Change light elevation |
| `L` + `0..9` | Light intensity presets |

## Bezier Tool

| Key / Action | Function |
|---|---|
| `Q` | Activate Bezier tool |
| Click | Drop corner anchor |
| Click+drag | Drop smooth anchor, shape handle |
| `H` | Toggle handle visualisation |
| `Backspace` | Remove last anchor |
| `Enter` | Commit curve |
| `Escape` | Cancel |

### Line Tool — End Caps

| Key | Function |
|---|---|
| `s` | Cycle start cap (while dragging) |
| `e` | Cycle end cap (while dragging) |

## Hold-Key Combos (Non-Modifier)

The following combos require holding a non-modifier key (not Ctrl/Alt/Shift) and clicking, pressing, or dragging another input. These are distinct from standard modifier shortcuts and are easy to miss!

| Hold Key | + Action | Effect |
|---|---|---|
| `F` | Left Click | Global Fill — flood-fills all visible non-locked layers with FG color (contiguous) |
| `F` | Right Click | Global Fill — flood-fills all visible non-locked layers with BG color (contiguous) |
| `Shift+F` | Left Click | Replace-all Global Fill — replaces all matching pixels across all visible layers with FG color |
| `Shift+F` | Right Click | Replace-all Global Fill — replaces all matching pixels across all visible layers with BG color |
| `E` | Click | Flood Erase — erases contiguous same-color pixels to transparent (no selection step) |
| `W` | Click | Magic Wand — select from merged canvas (all visible layers) |
| `G` | Arrow Key | Resize grid width/height independently |
| `L` | Digit (0-9) | Set Smart Shape 3D light intensity preset |
| `R` | Click | Picker — sample from reference image instead of visible layers |

> See also: Tool-specific sections for additional drag-time and context-sensitive combos (e.g., Smart Shapes, Line End Caps, etc.).

For everything else, use the Command Palette or read the chapter where the feature is introduced.

# Customize Controls — Rebind Anything (New in 2.1.0)

> 🎯 **Goal:** Make DRAW's controls match your muscle memory.

Every shortcut in DRAW is now **rebindable** — keyboard *and* mouse. Open **Edit** → **Customize Controls** (or search "Customize Controls" in the Command Palette).

**Rebinding a control**

1. Type in the **FIND** box to filter the list by command name.
2. Click **SET...** on a row to open the capture dialog.
3. Press the new key, click a **mouse button**, spin the **wheel**, or **tilt** the wheel left/right — whatever you press is captured. Toggle the Ctrl/Shift/Alt checkboxes to require modifiers.
4. If the input is already bound elsewhere, DRAW **warns and steals** it (the old binding is unbound). Click **OK** to apply — changes take effect immediately.

Rows you've changed from the factory default show a marker and a "= changed from default" legend. **RESET ALL** restores every control to the shipped defaults instantly.

**Mouse controls you can bind**

| Control | Default | Notes |
|---|---|---|
| Vertical wheel | Zoom in/out (cursor-centered) | Rebindable; can be set to canvas pan or brush size |
| Wheel **tilt** (left/right) | Brush size − / + | Tilt right = bigger; rebind to **Pan Left/Right** for horizontal scroll |
| `Ctrl` + wheel | Brush size | |
| Thumb buttons (Back / Forward) | Undo / Redo | |
| Right-click (Zoom tool) | Zoom **out** | Left-click zooms in |
| Pan drag button | Middle / Space+drag | Rebindable |

There are also dedicated **Pan Up / Down / Left / Right** actions you can bind to the wheel or tilt for scroll-to-pan.

**Presets & export**

- **Load preset** applies a full keymap in one click: *Aseprite, Photoshop, GIMP, Krita, Illustrator, Procreate, MS Paint, Pro Motion, DeluxePaint*.
- **Export as...** writes your live keymap (including overrides) to **Markdown, HTML, or PDF**.
- Bindings persist across sessions in `DRAW.bindings`; the CLI flag `--load-preset <name>` applies one at launch.

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➡️ Next: Chapter 17 — Undo, Redo & History

# Ch. 17 ↩️ Undo, Redo & History

> **What you'll learn:** How DRAW's unified history system works, why text-local undo is separate, and why you can experiment fearlessly.

---

## History System — Fearless Experimentation

> 🎯 **Goal:** Understand the unified undo/redo system.

DRAW has **one timeline** for everything you do — drawing, transforms, layer ops, palette ops, even the import overlay. Press `Ctrl+Z` to step backward and `Ctrl+Y` to step forward.

### How history is recorded

History records **on mouse release**, automatically. There is no manual save-state command, and there is a guard against double-recording on the same frame. Each entry carries a human-readable label — *"Brush stroke on Layer 3"*, *"Move 12px on selected layers"*, *"Apply Transform"*, for a future history panel and comments in QB64PE Project Export.

### Multi-layer awareness

When an operation affects multiple layers (a multi-layer select-and-flip, a Smart Erase across visible layers, a group merge, etc.), DRAW records a single composite history step. Smart Erase additionally tracks per-layer erasure so undo / redo behaves intuitively per layer.

### Text-local undo

The text tool has its own 128-state local history while you are editing a text layer. `Ctrl+Z` and `Ctrl+Y` operate on that local history during text entry — typo experiments don't pollute the global timeline. As soon as you commit ( `Esc` ), the entire edit becomes a single global history entry.

### Try things

The single biggest productivity unlock in DRAW is: **try things**. Symmetry, palette ops, blend modes, custom-brush fills, and transform overlays are all undoable. There is no "save first" anxiety — `Ctrl+Z` always works.

> 💡 **Tip:** When you make a destructive-looking change (palette ops, image adjustments), don't try to remember the parameters. Just `Ctrl+Z` and `Ctrl+Y` to flip-flop and visually compare.

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➡️ Next: Chapter 18 — Real-World Pixel Art Workflows

# Ch. 18 🎓 Real-World Pixel Art Workflows

**What you'll learn:** Step-by-step recipes for the most common pixel-art tasks: a 16×16 sprite, a seamless tile, isometric building, mandala, ANSI art, sprite-sheet assembly, and converting a photo into pixel art.

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## Workflow — Game Sprite (16×16 Character)

🎯 **Goal:** Create a complete game character sprite.

1. `Ctrl+N` — new canvas, 16×16, grid = 1, snap on.
2. **Choose a limited palette** — NES or PICO-8 are great starting points.
3. **Layer 1: Silhouette** — draw the outline in a single dark color.
4. **Layer 2: Base colors** — turn on Opacity Lock and fill regions.
5. **Layer 3: Shading** — set blend mode to Multiply, paint cool desaturated tones in shadow areas.
6. **Layer 4: Highlights** — set blend mode to Screen, dab warm light tones on edges facing your imaginary light source.
7. **Run the Pixel Art Analyzer** — fix orphans, doubles, jaggies, banding.
8. **Export** — File → Export As → PNG (plain) for engines, or save as `.draw` to keep the layers editable.

# Workflow — Seamless Tile Texture

> 🎯 **Goal:** Create a tileable background pattern.

1. `Ctrl+N` — new canvas, 32×32 or 64×64.
2. **View** → **Pattern Tile Mode** — enable Pattern Tile Mode. The 3×3 preview will reveal seams instantly.
3. Draw your base pattern.
4. Watch the seams in the preview. Fix them at the canvas edges by painting through the wrap.
5. Capture the result as a custom brush.
6. On a fresh test canvas, flood-fill with the brush to verify seamlessness.
7. Export as a plain PNG tile.

# Workflow — Isometric Pixel Art

> 🎯 **Goal:** Create isometric buildings and objects.

1. `Ctrl+'` — set grid geometry to Isometric.
2. `;` — enable snap.
3. The grid guides ensure perfect 2:1 ratio. (Also try Edit → Angle Snap: Pixel Art)
4. Use the Polygon tool for angled surfaces (top, left, right faces of a cube).
5. Put each face on its own layer.
6. Use Multiply / Screen blend modes for light/shadow on each face.
7. Group the layers per object so you can move it as a unit.

# Workflow — Symmetrical Mandala / Pattern

> 🎯 **Goal:** Create complex symmetrical artwork.

1. `F7` **× 3** — enable 8-way (asterisk) symmetry.
2. `Ctrl+Click` the canvas center.
3. Draw one slice — eight reflections appear instantly.
4. Layer subsequent strokes on different layers with different blend modes.
5. Vary opacity for depth.
6. Use custom-brush stamps (small flowers, dots, geometric shapes) for ornamentation.

# Workflow — ANSI / Text Art with Character Mode

> 🎯 **Goal:** Create text-mode art using block characters.

1. Press `T`, switch to the VGA font.
2. Enable Character Mode.
3. Open the Character Map (`Ctrl+M`).
4. Use `F1` – `F12` for the ANSI block-shading characters ( ▒ ▓ ▌ ▀ ▄ ▐ ).
5. Navigate with the virtual cursor.
6. `Alt+U` picks colors from any existing character cell.
7. The DOT and RECT tools fill cells with the active glyph instead of pixels.
8. Toggle CP437 mode (View menu → Charmap Unicode/CP437 Toggle) for the classic DOS feel.

# Workflow — Sprite Sheet Assembly & Export

> 🎯 **Goal:** Assemble sprites into a sheet and re-extract.

1. Create individual sprites on separate layers.
2. Use grid + snap for consistent spacing.
3. Use **layer groups** per animation frame.
4. Use **Align & Distribute** for perfect layout.
5. Export the full sheet as PNG.
6. Use **Extract Images** to decompose the sheet into individual PNGs (Chapter 10).
7. Save the project as `.draw` to preserve layers for future edits.

# Workflow — Photo to Pixel Art

> 🎯 **Goal:** Convert a photograph into pixel art.

1. `Ctrl+R` — load the photo as a reference image.
2. Choose a limited palette (Endesga 32 is a great default).
3. Set reference opacity to ~60% with `Ctrl+Shift` +Wheel.
4. Trace key shapes on a new layer with line and polygon tools.
5. Fill regions with flood fill.
6. Add detail with the brush and dot tools.
7. Apply Image Adjustments (Levels, Hue/Sat) for color tweaking.
8. Run **Posterize** if you want pixel-art-friendly value steps.
9. Hide the reference and evaluate the standalone result.

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➡️ Next: Chapter 19 — Tips, Tricks & Advanced Techniques

# Ch. 19 💡 Tips, Tricks & Advanced Techniques

**What you'll learn:** Ten time-saving tips that experienced DRAW users rely on every day, plus advanced layer techniques for non-destructive workflows.

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## 10 Time-Saving Tips You Need to Know

🎯 **Goal:** Boost productivity with pro tips.

1. `Alt` **+Click is a temporary picker — *with any tool that can drop pixels*.** No need to switch to the Picker tool for a quick eyedrop.
2. **Middle-click drags pan — *with any tool*.** Stop reaching for the Pan tool just to nudge the view.
3. **Hold `E` for a temporary eraser.** Release `E` and you're back to the previous tool. Lifesaving for cleanup mid-stroke.
4. `Shift` **+Right-click draws a connecting line** from the previous click to the current cursor — works in Brush, Dot, Line.
5. `?` **(Command Palette) is faster than menu diving.** If it takes more than two clicks to find a command, you're doing it wrong.
6. **Opacity Lock + a giant brush = instant recolor.** No selection needed. Just lock the layer and slosh.
7. **Custom brush + tiled fill = instant pattern fills.** Capture a small motif as a brush, then flood fill an area.
8. `Ctrl+Shift+C` **is Copy Merged.** Flatten the visible composite to clipboard without merging your actual layers.
9. **Double-middle-click = instant pan/zoom reset.** Get back to a fitting view in one gesture.
10. `.draw` **is a PNG you can open in any viewer.** Show off your art on Discord without exporting first — your project file *is* the preview.
11. **Move shapes while dragging:** Hold SPACE while dragging any geometric or Smart Shape tool (Line, Rectangle, Ellipse, Polygon, Smart Shapes) to temporarily reposition the in-progress shape with the mouse, instead of resizing it. Release SPACE to resume resizing from the new anchor. Grid snap is honored. (Photoshop-style move-during-drag.)

# Advanced Layer Techniques

> 🎯 **Goal:** Push the layer system to its limits.

### Non-destructive color adjustment workflow

Build a stack with the artwork at the bottom, a Multiply layer for shadows above it, a Screen layer for highlights, and a Color-mode layer at the top for the overall colorway. Tweak the upper layers without ever touching the base. When the look is right, optionally merge down — but consider keeping the layers for easy revisions.

### Group opacity for complex transparency

A group's opacity slider acts on the entire group as if it were a single rasterized layer. This is how you fade an entire compound element (e.g., a complete UI panel made of many layers) without losing the per-element relationships.

### Pass Through mode

A group set to **Pass Through** lets each child blend with what's *below the group*, not just within. This is essential for blend-heavy layers (Multiply / Screen / Add) inside a group, where Normal-mode group compositing would otherwise isolate them.

### `Alt`+Drag with the Move tool — clone stamping

Holding `Alt` while dragging with the Move tool clones the dragged content into a new layer at the destination. Your original stays put.

### Solo mode

`Alt`+Click any layer's eye icon to **solo** it — every other layer is hidden. Click again to restore. Indispensable for diagnosing a stack you've lost track of. Also handy is using the Move tool, and `Shift+Click` to select the layer under the pointer.

### Multi-layer selection for bulk operations

Select multiple layers (`Ctrl`+Click / `Shift`+Click) and run any of: clear, fill, flip, rotate, scale, transform overlay, or alignment. The history records a single composite step.

### Layer alignment for sprite-sheet layout

Align Left / Center / Right and Distribute Horizontally are the fastest way to organize a row of sprites — far faster than nudging by hand even with snap.

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➡️ Next: Chapter 20 — Appendix: Quick Reference

# Ch. 20 📋 Appendix: Quick Reference

> **What you'll learn:** Where the canonical cheat sheet lives, plus a one-page summary of the most-used keyboard shortcuts, the 56 bundled palettes, and all 19 blend modes.

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## Canonical reference

The complete, always-up-to-date list of keyboard **and mouse** shortcuts lives in `SHORTCUTS.md` at the repository root — generated directly from the input registry. The summary below is curated for quick visual scanning — when in doubt, the canonical file wins. All of these are rebindable via **Edit** → **Customize Controls** (see Chapter 16).

## Keyboard shortcut summary

### Tool keys

| Key | Tool | Key | Tool |
|---|---|---|---|
| `B` | Brush | `M` | Marquee |
| `D` | Dot | `W` | Magic Wand |
| `L` | Line | `V` | Move |
| `R` | Rect ( `Shift+R` filled) | `Z` | Zoom |
| `C` | Ellipse ( `Shift+C` filled) | `T` | Text |
| `P` | Polygon ( `Shift+P` filled) | `I` | Picker |
| `F` | Fill | `E` | Eraser |
| `K` | Spray | | |

### Modifier vocabulary

| Modifier | Default meaning |
|---|---|
| `Shift` | Constrain / add to selection / draw from center / fast-scroll / Smart Erase. |
| `Ctrl` | Perfect aspect (square / circle) / command keystrokes. |
| `Alt` | Subtract from selection / temporary picker / clone (with Move). |
| `Ctrl+Shift` | Angle snap / bypass grid snap. |

### Panel toggles

| Key | Panel |
|---|---|
| `Tab` | Toolbar |
| `F4` | Preview Window |
| `F5` | Edit Bar |
| `Shift+F5` | Advanced Bar |
| `Ctrl+L` | Layer Panel |
| `Ctrl+M` | Character Map |
| `F10` | Status Bar |
| `F11` | All UI |
| `Ctrl+F11` | Menu Bar only |

### Canvas / view

| Action | Shortcut |
|---|---|
| Zoom in / out | `Ctrl+=` / `Ctrl+-`, Mouse Wheel (Zoom tool: right-click zooms out) |
| Brush size − / + | `[` / `]`, `Ctrl`+Wheel, wheel tilt left/right |
| Reset zoom | `Ctrl+0` |
| Reset pan + zoom | Double-middle-click |
| Pan | Middle-drag, `Space`+drag (or bind the wheel/tilt to Pan in Customize Controls) |
| Pattern tile | View menu / Advanced Bar |
| Grayscale preview | `Ctrl+Alt+Shift+G` |
| Reference image | `Ctrl+R` |
| Canvas border | `#` |

### Color

| Action | Shortcut |
|---|---|
| Swap FG / BG | `X` |
| Reset FG / BG | *Palette menu → Default Colors* |
| BG = transparent | `Shift+Delete` |
| Paint opacity | `0` – `9` |
| Temp picker FG / BG | `Alt`+L-click / R-click |

## Brush

| Action | Shortcut |
|---|---|
| Size − / + | `[` / `]` |
| Shape | `\` |
| Preview footprint | hold `` ` `` |
| Pixel Perfect | `F6` |
| Recolor mode | `F9` |
| Outline mode | `Shift+O` |

## Selection / clipboard

| Action | Shortcut |
|---|---|
| Select All / Deselect | `Ctrl+A` / `Ctrl+D` |
| Invert | `Ctrl+Shift+I` |
| Copy / Cut / Paste | `Ctrl+C` / `X` / `V` |
| Paste in Place | `Ctrl+Alt+V` |
| Copy Merged | `Ctrl+Shift+C` |
| Copy / Cut to New Layer | `Ctrl+Alt+C` / `X` |

## Layers

| Action | Shortcut |
|---|---|
| New / Duplicate / Delete | `Ctrl+Shift+N` / `D` / `Delete` |
| Group / Ungroup | `Ctrl+G` / `Ctrl+Shift+U` |
| Group selected | `Ctrl+Shift+G` |
| Move up / down | `Ctrl+PgUp` / `Ctrl+PgDn` |
| Move to top / bottom | `Ctrl+Shift+]` / `Ctrl+Shift+[` |
| Merge Down / Visible | `Ctrl+Alt+E` / `Ctrl+Alt+Shift+E` |

## History / Edit

| Action | Shortcut |
|---|---|
| Undo / Redo | `Ctrl+Z` / `Ctrl+Y` |
| Clear (selection or layer) | `Delete` or `Ctrl+E` |
| Fill FG / BG | `Backspace` / `Shift+Backspace` |

### Grid / Symmetry

| Action | Shortcut |
|---|---|
| Toggle grid | `'` |
| Pixel grid | `Shift+'` |
| Snap to grid | `;` |
| Grid size − / + | `,` / `.` |
| Cycle geometry | `Ctrl+'` |
| Cycle symmetry | `F7` |

### Swatches panel (View → Swatches; keys with the pointer over the panel)

| Action | Shortcut |
|---|---|
| Swatch → FG / BG | L-click / R-click |
| Select a range / toggle one / box select | `Shift`+click / `Ctrl`+click / drag from an empty place |
| Select all / none | `Ctrl+A` / `Ctrl+D` or `Esc` |
| Move / copy the selection | Drag a selected swatch / `Ctrl`+drag |
| Edit the color | Double-click a swatch |
| Add the FG / delete / lock | `Ins` / `Del` / `L` |
| Find the FG's swatch | `F` |
| Undo / redo swatch edits | `Ctrl+Z` / `Ctrl+Y` |
| The edit menu | ☰ or `Shift`+R-click |

See Ch. 3 — Color & Palette.

### Animation brushes

| Action | Shortcut |
|---|---|
| AnimBrush: pick up / load | *ANIM → AnimBrush → Pick Up / Load...* |
| Previous / next cel | `Alt+[` / `Alt+]` |
| Current frame gets cel 1 | `Alt+\` |
| Drop the AnimBrush | `Ctrl+B` |
| Store in a Drawer bin (every cel) | `Shift`+click the bin |
| Move Brush: start / apply / cancel | *ANIM → Move Brush → Start*, drag the path, `Enter` / `Esc` |

See Ch. 9 — Custom Brushes & Drawer Panel.

### AI (only when AI features are enabled)

| Action | Shortcut |
|---|---|
| Cancel AI generation / batch | `Ctrl+Alt+K` |

See Ch. 21 — AI Image Generation.

### Workspaces & capture

| Action | Shortcut |
|---|---|
| Workspace switcher | `Ctrl+Shift+W` |
| Show what a workspace hid / hide it again | `F11` (inside a workspace) |
| Move a panel (new column / stack / tab / float) | Drag its handle (title strip, layers header, top-edge grip) |
| Panel menu: Move to Left/Right, Collapse, Float / Dock Back, Scale (floating-panel family), Reset Arrangement | Right-click a handle |
| Collapse / expand a slot | Click its chevron, or double-click its title; click a collapsed title to open it |
| Dock a floating window | Drag its title bar onto a dock target |
| Capture Screen | `Ctrl+Shift+P` (or `DRAW --capture`) |
| Annotate: rect / filled / ellipse / line / arrow / text | `R` / `Shift+R` / `O` / `L` / `A` / `T` |
| Annotate: highlighter / redact / numbered callout / restart numbers | `H` / `X` / `N` / `Shift+N` |
| Annotate: copy picture / quick save / done | `Ctrl+C` / `Ctrl+S` / `Ctrl+Enter` |

The Annotate keys work only in the Annotate workspace. See Ch. 22 — Workspaces & Screenshot Annotation.

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## All 56 bundled palettes

Located in `ASSETS/PALETTES/`. The full set includes:

- **Console classics** — NES, PICO-8, Commodore 64, Game Boy (DMG and Pocket), MSX, Amiga, Atari.
- **Arcade / DOS** — CGA, EGA, VGA.
- **Modern curated** — Endesga 32 / 64, DawnBringer 16 / 32, AAP-64, Sweetie 16, Resurrect 64.
- **Lospec greatest hits** — and 40+ more available via DRAW's Lospec integration.

## All 19 blend modes

| Family | Modes |
|---|---|
| Basic | Normal · Multiply · Screen · Overlay |
| Math | Add (Linear Dodge) · Subtract · Difference |
| Comparison | Darken · Lighten |

| | |
|---|---|
| Dodge / Burn | Color Dodge · Color Burn |
| Light | Hard Light · Soft Light · Vivid Light · Linear Light · Pin Light |
| Color | Exclusion · Color · Luminosity |
| Group only | Pass Through |

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# Chapter 21 — 🤖 AI Image Generation

> 🎯 **Goal:** Generate pixel art from a text prompt using an external generator of your choosing, and work with the results as ordinary layers.

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## Off by default

AI features are **disabled until you ask for them**, and while disabled DRAW shows nothing about them at all — no AI menu, no commands, no layer markers, no status text. Enable them in **Settings** → **General** → **Enable AI Features**.

Turning the feature off again hides it completely, including on projects that already contain AI layers: those layers keep working and simply render as ordinary layers.

## DRAW does not talk to a model

DRAW runs an **external command-line generator that you configure**. It builds a command, runs it, and imports the PNG that comes back. You choose the engine, the models, and whether anything leaves your machine.

The bundled example is pixelmon, a local ComfyUI + SDXL pipeline. A test engine ( `DEV/ai-echo.sh` ) is also included so you can exercise the whole pipeline in about a second.

Generators live in `DRAW.ai.cfg` in your config directory. **AI** → **Tools...** edits them; the first time you use an AI command with nothing configured, DRAW offers to write a starter file.

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## Generating

**AI** → **New AI Layer...** opens the generate dialog.

| Field | Notes |
|---|---|
| **Tool** | Which configured generator to run |
| **Style** | Optional style guide — see below |
| **Preset** | A saved prompt; picking one loads it into the prompt box |
| **Prompt** | Multi-line, word-wrapped, with selection and clipboard |
| **Size** | `[IMAGE SIZE]` fills in the canvas size; `[IMAGE xN]` appears only when your display is scaled |
| **Position** | 3×3 preset grid, or type X/Y directly |
| **Seed** | Same seed reproduces a result; `[RANDOM]` re-rolls |
| **Count** | Above 1 generates a batch — see below |
| **SURPRISE ME** | Random style, random saved prompt, fresh seed |

Generation runs **in the background**. DRAW stays fully usable, and the status bar shows an animated progress line with the elapsed time. Press `Ctrl+Alt+K` to abort — this terminates the generator process, it does not merely stop watching it.

### Working with a selection

With a selection active, the dialog starts with the selection's **size** and its **top-left corner**. The result is scaled to fit inside the selection with its aspect ratio preserved, so nothing is skewed: the limiting dimension fills the selection exactly and the other is centred.

Enlarging is restricted to **whole-number factors**, because scaling pixel art by a fractional amount resamples it into mush. A small result inside a large selection therefore sits centred at 1×, 2× or 3× rather than being smeared to fill.

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## Batch generation

Set **Count** above 1 (up to 20) to generate several variations at once. They land in a **layer group**, and every item shares identical settings — same tool, style, prompt, size and position — differing **only by seed**. A batch is a set of variations on one idea, not a set of unrelated images.

Seeds run from the seed shown in the dialog upward, so re-running a batch with the same starting seed reproduces the same set.

Items are generated one at a time. The status bar shows progress as `[2/10]`, and `Ctrl+Alt+K` stops the remainder — anything already generated is kept.

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## AI layers

A generated layer is marked `[AI]` in the layer panel. Hovering it shows the full prompt, with the engine, model, style and seed beneath.

Right-click an AI layer for:

- **Edit Prompt...** — change the stored prompt without regenerating
- **Regenerate** — re-run it, pre-filled with the prompt and seed it was made with

The prompt, style, tool, seed and size are **saved in the `.draw` file**, so a layer can be regenerated — or reproduced exactly — after reloading a project.

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## Styles and prompts

**AI → Style Editor...** manages style guides. A style can:

- wrap your prompt with a **prefix** and **suffix**
- add its own **CLI arguments** (e.g. pixelmon's `--style ega`)
- be **scoped to one tool**, since a style name means nothing to a different generator
- **lock a seed**, so its look is reproducible

> ⚠️ Command-line flags belong in a style's **Args** field, not its prefix or suffix. Prefix and suffix are *prompt text* — a flag placed there is passed inside the quoted prompt and the generator never sees it as a flag. DRAW migrates obviously-misplaced flags automatically and notes it in the log.

**AI → Prompt Editor...** manages saved prompt presets — plain text starting points you can pick from the generate dialog.

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## Argument macros

A tool's arguments are a **template**. DRAW expands `{macros}` before running it, so one configuration adapts to whatever you are working on.

Click **[?]** in the tool or style editor for a live reference listing every macro alongside **its current value**, so you can see exactly what your tool will receive.

| Group | Macros |
|---|---|
| Run | `{prompt}` `{style}` `{seed}` `{outdir}` `{outname}` |
| Document | `{if}` `{id}` `{iw}` `{ih}` |
| Selection | `{sw}` `{sh}` |
| Layer | `{numlayers}` `{lidx}` `{lname}` `{lw}` `{lh}` `{lx}` `{ly}` |
| Brush / grid | `{bw}` `{bh}` `{gw}` `{gh}` |
| Colour | `{pal}` `{numpal}` `{fg}` `{bg}` `{pmode}` |
| Context | `{tool}` `{px}` `{py}` `{font}` `{fsize}` `{fontfile}` |
| Steering | `{limg}` `{dimg}` `{bimg}` |

Three rules apply to every macro:

1. **Values are whitespace-trimmed.**
2. **An UPPERCASE name yields an UPPERCASE value** — `{PAL}` gives `ANSI32`, `{pal}` gives it as-is. `{pal:lower}` and `{pal:slug}` are also available.
3. **`sc*` variants give the screen-scaled value** — `{sciw}` is the image width multiplied by the display scale. Every dimension and position macro has one.

DRAW supplies `{outdir}` as a fresh directory per run and imports the newest PNG it finds there, so tools that name their own output files work unchanged.

### Steering images

`{limg}`, `{dimg}` and `{bimg}` export the **current layer**, the **flattened document** and the **custom brush** as PNGs, for generators that accept a reference image. `{dimg}` is the artwork only — never DRAW's interface.

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## The log

Every run is recorded in `ai.log` in your config directory: the full command line, the exit code, the tool's own output, and the imported file or the reason it failed. When a generation misbehaves, that file will tell you why. A failure dialog offers to open it for you.

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## Exercises

1. Enable AI features and let DRAW write the starter config. Generate once with the bundled `echo (test)` engine to confirm the pipeline end to end.
2. Make a selection, then generate into it. Note that the dialog pre-fills the selection's size and corner.
3. Create a style with a suffix of `pixel art, bold outlines` and generate the same prompt with and without it.
4. Run a batch of 4 and compare the variations — same prompt, four seeds.
5. Open the **[?]** macro reference and find three macros whose current values change when you switch tool or palette.

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➡️ Next: Chapter 22 — Workspaces & Screenshot Annotation

# Chapter 22 — 🧰 Workspaces & Screenshot Annotation

> 🎯 **Goal:** Make DRAW fit the job in front of you. A workspace switches the whole UI to a preset (fewer panels, a smaller toolbox, other keys), and **Capture Screen** turns DRAW into a screenshot annotator.

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## Workspaces

A **workspace** is a named layout preset. It decides:

- which panels show and where they dock;
- which toolbox buttons exist, in what order, and how many per row;
- which menus and menu items appear;
- which edit bar, advanced bar and organizer buttons show;
- single-key shortcuts that only work while it is active;
- a few starting options.

A workspace is an **overlay** on your own setup. Switching to one notes how everything is now, and leaving it puts every panel, dock and option back exactly as it was. Nothing a workspace changes is written to `DRAW.cfg`. DRAW only remembers *which* workspace you were in, so it reopens there.

### Built-in workspaces

| Workspace | What it is |
|---|---|
| **Default** | The stock panel layout. No overlay. Always there; can't be deleted. Moving or resizing panels in it starts a workspace of your own (see below). |
| **Simple** | Everyday drawing: a 2-column toolbox of 10 tools, File / Edit / View / Help menus, no advanced bar or power panels. |
| **Animate** | Frame-by-frame animation: the lightbox (strip, transport bar, the frames before and after side by side) with onion skin and the frame grid, and the Animation panel; the character map, 3D color space and browser are put away. On a picture without frames, `Num +` makes frames and the lightbox opens on them. |
| **Sprite** | One sprite or a short loop: the Preview (showing the current frame), the Animation panel and the Drawer; the frame grid on, no lightbox or onion skin; the Brush; no EFFECTS / AUDIO menus. |
| **Scene** | A scene of clips and layers: Big Frame + X-sheet (the lightbox as one large pane, the X-sheet under it), the Animation panel, the Drawer and the layers; onion skin off; the Move tool; every menu. |
| **Annotate** | Screenshot markup: a 1-column annotation toolbox docked left, a trimmed edit bar, the palette strip, single-key annotation tools, Rect in red, zoomed to fit. Capture Screen lands here. |

### Switching

| How | |
|---|---|

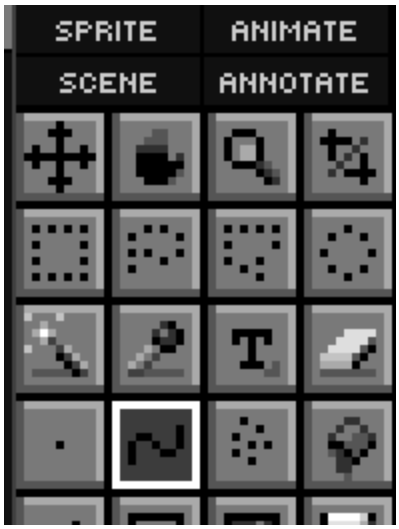
| | |
|---|---|
| **View → Workspace ▸** | Every workspace, with a check on the active one, plus Switcher, Configure, Open Folder and Reload |
| `Ctrl+Shift+W` | The workspace switcher: the command palette filtered to `Workspace:`. Type a few letters (`ann`) and press `Enter`. |
| **Command palette** (`Ctrl+P` / `?`) | Type `Workspace:` |
| `[WS: name]` **badge** | At the right end of the status bar; click it to open the switcher |
| `DRAW --workspace annotate shot.png` | Start in a workspace **for this run only** (not remembered) |
| `DRAW --workspaces` | List the workspaces (the remembered one is starred) and exit |
| **Workflow buttons** | The labelled buttons at the top of the toolbox (below) |

**There is always a way back.** Whatever a workspace hides:

- `Ctrl+Shift+W` and the command palette still work.
- `F11` shows everything the workspace hid, and a second `F11` hides it again. Outside a workspace, `F11` toggles all UI as before.
- Hidden commands keep their keys. Hiding means "not shown", not "disabled".

Things you toggle while in a workspace (opening the Color Mixer, say) last for that session. To make them part of the workspace, save the layout (below).

## Workflow buttons




The strip at the top of the toolbox has one button per workspace - by default **SPRITE**, **ANIMATE**, **SCENE** and **ANNOTATE**. Click one to switch to that workspace; it lights while you are in it. Click the lit one again to go back to Default.

Right-click a button to change the strip:

- pick the workspace it opens (every workspace is listed, yours too);
- **Move Left / Move Right**;
- **Add a Button** (after this one; it takes the first workspace not already on the strip);
- **Remove This Button**;
- **Hide Workflow Buttons**. **View → Workflow Buttons** brings the strip back.

The command palette has **Workflow Button 1** to **8** (type `workflow button 2`). The strip is the same in every workspace: it is saved in `DRAW.cfg` as `WORKFLOW_BUTTONS=sprite,animate,scene,annotate` (and `WORKFLOW_BAR=1` or `0`). A button's label is the workspace's `SHORT=` name, or its name when it has none.

## Arranging and saving a layout

Panels dock on the **left and right** edges of the window. The menu bar always stays on top, and the status bar and color strip always stay at the bottom.

**Drag a docked panel's inner edge** (the edge facing the canvas) to resize it. The pointer turns into a ↔ arrow over the edge. The panel snaps as you drag:

| Panel | Dragging changes |
|---|---|
| **Toolbox** | Button columns (1 to 8) |
| **Edit bar**, **advanced bar** | Icon columns (1 to 4) |
| **Layers** | Width in pixels (60 to 400) |

When you let go, the new size is saved:

- **In a workspace:** into that workspace. A built-in gets your own copy, which **RESET** in the configurator removes.
- **In Default:** DRAW switches to the **Current** workspace and keeps it there (see The Current workspace). Default itself keeps the stock layout.

Everything in the toolbox column follows its width:

- **Toolbox buttons:** more columns are added when the buttons don't fit the window height. A 1-column toolbox of 28 tools in a short window becomes 2 columns.
- **Organizer:** its four widget stacks fit into 1 to 4 columns.
- **Drawer:** as many bin columns as fit. In a narrow toolbox, the mini palette moves under the bins, and `Shift` **+wheel** over the bins scrolls them.
- **Empty space:** when a workspace hides the organizer and drawer, the column under the buttons is drawn as panel, not as a black gap.

## Moving, stacking, tabbing and floating panels

Every panel can be arranged by hand, the way Photoshop, GIMP or Krita allow: the toolbox, organizer, drawer, layers, edit bar, advanced bar and character map, and the Preview, Color Mixer, Advanced Color Picker, 3D Color Space, Pen Pressure and Browser windows.

**The handle** is what you grab:

- a panel with a title strip (a tab, a collapsed slot, a docked window): the strip;
- the layers panel: its header;
- a floating edit bar or advanced bar: the slim grip strip on top (they are too narrow for a title);
- the toolbox, organizer, drawer and bars, which have no title: hover the panel's **top edge** and a thin grip appears.

| Do this | And |
|---|---|
| **Drag a handle** | A blue preview shows where the panel will land, with a label: *new column*, *stack here*, *add as a tab* or *float here* |

| | |
|---|---|
| ...drop on a **screen edge**, or between two columns | It gets a column of its own there |
| ...drop on the **top or bottom third** of a panel | It stacks above or below that panel in the same column |
| ...drop on the **middle** of a panel | It becomes a **tab** next to it |
| ...drop anywhere else | It **floats** in its own window |
| **Drag the line between two stacked panels** | Moves height between them (the pointer turns into ↕) |
| **Click a tab** | Shows that panel |
| **Drag a tab strip past its tabs** | Moves the whole group (every tab) |
| **Click the chevron**, or **double-click a title** | Collapses the slot to its title strip; its height goes to the others |
| **Click a collapsed title** | Opens it again (click one of its tabs to open on that tab) |
| **Double-click a floating panel's title** | Docks it back where it was |
| **Drag a floating panel's right or bottom edge, or its corner** | Resizes it; a bar adds icon columns as it gets wider |
| **Right-click a handle** | Move to Left Edge / Move to Right Edge, Collapse to Title / Expand, Float / Dock Back, Reset Arrangement, and Scale (Auto or 1×–4×) on the Animation, Color Mixer, color picker, 3D Color Space, Pen Pressure and Swatches panels |

**The Browser can also dock along the top or bottom** of the canvas area (the one exception to left/right docking): drag its title bar to the top or bottom of the canvas and a full-width band appears. Drag the band's inner edge to make it taller or shorter; the canvas re-centres in the room left. Its right-click menu has *Dock Along the Top* / *Dock Along the Bottom*.

The floating windows (Preview, Color Mixer, Advanced Color Picker, 3D Color Space, Pen Pressure, Browser) keep their own title bar. **Drag the title bar onto a dock target** to dock one; docked, its title bar is the handle, and dropping it anywhere off the docks floats it again at its old size.

**Overlapping floating windows** stack like windows on your desktop: click any floating panel or window and it comes to the front, drawn whole over the others, and only the one in front takes clicks where they overlap. **View → Snap Panel Edges** (on by default) makes a dragged floating panel snap to the edges of the others and keeps them from overlapping; hold it still against a snap for about 1.5 seconds and it lets go, so you can drop it on top of another one. Turn it off to move floating panels freely. A floating toolbox or drawer stops growing once its buttons or bins all show, so no empty space is left beside or under them.

**Lock Layout** (View → Lock Layout, `Ctrl+Shift+L`) freezes the whole arrangement, so a shaky pen can't drag, dock or resize a panel by accident. Clicks, tabs and buttons still work.

**Small windows:** when a column is too short, its lowest panels collapse to their titles; when the window is too narrow, the outermost columns hide (the toolbox's column last) so the canvas keeps some room. Nothing is changed in your arrangement; widen the window and they come back.

**Reset Arrangement** (right-click any handle) returns to the classic layout, where each panel's side follows View → Layout and `Ctrl+Shift+Click`.

Arrangements save like widths: into the active workspace (a built-in gets your copy). **Default always keeps the stock layout**: the first time you move, dock, float or resize a panel there, DRAW keeps the layout in the **Current** workspace and switches to it - nothing to name (see The Current workspace). Switching tabs or collapsing a slot in Default is not saved. Set `STOCK_DEFAULT_LAYOUT=0` in `DRAW.cfg` to have Default keep your changes (`DOCK_*` keys) instead. In a workspace file arrangements are the `[DOCK]` and `[FLOAT]` sections (see Workspace files).

## The Current workspace

**Current** is always listed right under **Default** in **View → Workspace ▸** and the switcher. It is where changes made in Default go: move, dock, float or resize a panel, change a column count or customize the toolbars while in Default, and DRAW switches to Current with the change in it. Default itself never changes, so you can always go back to the stock layout and return to Current later. Once you are in Current, every change is saved into it straight away (`current.workspace` in your workspaces folder).

**On exit**, if Current is the workspace you are in and it changed during the session, DRAW offers to keep it under a name:

- **Workspace:** type a name to save it as a workspace of its own. Leave it blank (or press `Esc`) to skip - Current is kept either way.
- **Make this my default workspace** (on): DRAW starts in the saved workspace next time. Off: it starts in Current, as you left it.
- **Don't show me this prompt again:** turns the question off (`WORKSPACE_SAVE_PROMPT=0` in `DRAW.cfg`).

You can switch to any workspace and come back to yours later. "Current" and "Default" are reserved names.

**Save what's on screen** from **View → Workspace ▸**:

| Item | Does |
|---|---|
| **Save Current Layout As...** | Asks for a name and saves a new workspace exactly as you see it: which panels show, dock sides, columns and widths. When you're in a workspace, its toolbox buttons, menus and keys come along. DRAW switches to it. |
| **Update Workspace From Layout** | Saves the current layout into the active workspace (a user copy for a built-in) |

The configurator's **Toolbox** and **Bars** tabs also have a **Customize Toolbars...** button that opens the window on the selected workspace.

Use the configurator only for what you can't arrange by hand: which toolbox buttons and menus exist, the keys, and the starting options.

## The configurator

**View → Workspace → Configure Workspaces...** (also in the command palette).

- **Left:** every workspace. The active one is checked; `*` marks your own files.
- **Tabs:**

| Tab | Edits |
|---|---|

| | |
|---|---|
| **Panels** | Keep / Show / Hide for each of the 16 panels and bars, Keep / Left / Right dock edges, and the layers width (0 = keep yours). *Keep* means "leave it however you have it". |
| **Toolbox** | Columns (0 = keep yours, from `TOOLBOX_COLUMNS`), the button order (typed), and a checkbox per toolbox button. Customize Toolbars is the easier way. |
| **Menus** | A checkbox per menu on the bar, and hidden menu items (action ids or labels) |
| **Bars** | Columns and button checkboxes for the edit bar and advanced bar, and organizer widget checkboxes |
| **Keys** | The workspace's single-key overlay ( `key = tool` ), plus an add row |
| **Options** | Name, description, `[START]` tool / color / zoom, and `DRAW.cfg` overrides for this workspace |
| **Dock** | The `[DOCK]` / `[FLOAT]` arrangement line by line; **Use current** takes it from the screen, **Default** removes it |

- **Buttons:**

| Button | Does |
|---|---|
| **NEW** | Start a blank workspace |
| **DUPLICATE** | Copy the selected one |
| **RENAME** | Change its display name (the file keeps its id) |
| **DELETE** | Remove one of yours |
| **RESET** | Drop your copy of a built-in |
| **SAVE** | Write the file |
| **USE** | Save and switch to it |
| **CLOSE** | Leave; unsaved edits are discarded |

Editing a built-in and saving makes **your own copy** in your workspaces folder. It wins over the built-in until you RESET it. Default is not editable, since it means "no overlay": DUPLICATE it to start one of your own.

## Customizing toolbars

**View → Customize Toolbars...** (also in the command palette, or right-click the toolbox / edit bar / advanced bar handle → **Customize Buttons...**) opens a floating window for changing what the toolbox and the two bars show, in any order:

| Part | What it does |
|---|---|
| **Tabs** | **Toolbox**, **Edit Bar**, **Advanced Bar**: the panel you are editing |
| **Available** (left) | Every tool and command not on that panel yet: the toolbox tools, then the commands by category, then menu commands. Type in **Search...** to filter it. |
| **On the ...** (right) | The panel's buttons, in order. Click one to select it. |
| → | Add the selected Available item after the selected button (or at the end) |

| | |
|---|---|
| ← | Remove the selected button |
| **↑ / ↓** | Move the selected button |
| **+ DIV** | Insert a divider after the selection: the next button starts a new row |
| **+ GAP** | Insert an empty cell |
| **RESET** | Put the panel's default buttons back |
| **COLUMNS – / +** | The panel's columns (toolbox 1-8, bars 1-4) |
| **DONE** | Close the window (changes are already saved) |

You can also double-click an Available item to add it, press **Delete** to remove the selected button, and drag rows between the two lists: drag within the right list to reorder, from Available into the right list to add, or from the right list onto Available to remove.

**Anything can go anywhere.** A tool on a bar works like its toolbox button. A command in the toolbox runs when clicked. A command without an icon shows up to three letters of its name in a small outlined tile (**NLY** for New Layer), and its tooltip gives the full name and shortcut. A theme can give those commands real icons: put `new-layer.png` (16×16) in the theme's `IMAGES/COMMANDS/` folder, and optionally an 11×11 version in `IMAGES/COMMANDS/TOOLBOX/` for toolbox cells.

**Edit the real panels while the window is open.** As long as Customize Toolbars is showing, the toolbox and bars don't run anything when clicked:

- **Click** a button to select it. The window switches to that panel and the button is outlined on the panel and highlighted in the list.
- **Drag** a button to move it. A white line shows where it will land, on the same panel or on another one, so you can move it from the toolbox to the edit bar.
- **Drop it off every panel**, for example on the canvas, to remove it.
- **Drag an Available item** onto a panel to add it right there.

Close the window and the buttons work normally again.

**Changes are saved into a workspace**, never into `DRAW.cfg`:

- In a workspace, every change is saved into it at once. A built-in workspace gets your own copy. The window shows **SAVES TO:** and the workspace name.
- In Default, the first change goes into the **Current** workspace (Default plus your layout and the change), which DRAW switches to - nothing to name.

Switching back to Default puts the stock toolbox and bars back.

## Workspace files

Workspaces are plain `.workspace` text files, INI style:

- **Built-ins:** `ASSETS/WORKSPACES/`.
- **Yours:** the folder `DRAW --dir-workspaces` prints, also reachable from View → Workspace → Open Workspaces Folder. A file of yours with a built-in's name replaces it.
- After editing by hand, use **Reload Workspaces**.

```
[WORKSPACE]
NAME=Annotate
DESCRIPTION=Mark up screenshots
SHORT=ANNOTATE           ; the workflow button's label (optional)
BASED_ON=default         ; inherit another workspace, override only what you list

[CHROME]                 ; SHOW | HIDE | KEEP
MENUBAR=SHOW
TOOLBOX=SHOW
TOOLBOX_DOCK=LEFT        ; *_DOCK = LEFT | RIGHT
LAYER_PANEL=HIDE
PALETTE_STRIP=SHOW       ; independent of STATUS_BAR
LAYER_PANEL_WIDTH=180    ; px, 60-400

[TOOLBOX]
COLUMNS=2                ; 1-8; more are added when the buttons don't fit the height
BUTTONS=move,marquee,rect,|,ellipse,line,text,picker,crop,undo,cmd:701   ; - = empty cell, | = new row,
                         ; bar names and cmd:<action id> put commands in the toolbox
ICON.crop=my-crop.png    ; theme TOOLBOX folder, a path, or your workspaces folder

[MENUS]
ROOTS=FILE,EDIT,VIEW,IMAGE,HELP
HIDE_ITEMS=209,NEW WINDOW ; action ids or labels

[EDIT_BAR]
ORDER=undo,redo,|,brush,eraser,cmd:701   ; the buttons in this order: bar names, toolbox
                         ; names (tools), cmd:<action id> (any command), | divider, - gap
COLUMNS=2                ; icon columns, 1-4

[ADVANCED_BAR]
BUTTONS=preview,grayscale ; the older form: only these, in the default order
COLUMNS=1

[ORGANIZER]
WIDGETS=color-mixer,brush-size,grid

[KEYS]                   ; only while this workspace is active
r=rect
R=rect-filled            ; upper case = Shift
ctrl+s=quick-save
ctrl+enter=done

[OPTIONS]                ; any DRAW.cfg key, this workspace only
ANGLE_SNAP_DEGREES=45

[START]                  ; applied each time you switch to it
TOOL=rect
FG=#FF2040
ZOOM=FIT                 ; or a percent, 100 = 1:1

[DOCK]                   ; the arrangement, one line per column, outermost first
LEFT.1=AUTO; toolbox | organizer | drawer
RIGHT.1=WIDTH:180; layers@2 | *preview+colormixer | !browser
                         ; AUTO or WIDTH:px; | stacks, + tabs, * active tab,
                         ; @n height share, ! collapsed to its title
```

```
[FLOAT]                 ; floating panels: x,y,w,h in window pixels
pen=640,120,220,300
```

You rarely type `[DOCK]` yourself: arrange the panels by hand and use **Update Workspace From Layout**. A workspace without `[DOCK]` uses the classic layout, with each side from the `*_DOCK` keys.

**Names you can use**

| Section | Names |
|---|---|
| `[CHROME]` panels | `MENUBAR` `TOOLBOX` `LAYER_PANEL` `EDIT_BAR` `ADVANCED_BAR` `STATUS_BAR` `PALETTE_STRIP` `PREVIEW` `COLOR_MIXER` `CHARMAP` `COLOR_SPACE_3D` `ADV_COLOR_PICKER` `PEN_PANEL` `BROWSER` `ORGANIZER` `DRAWER` |
| `[CHROME]` docks | `TOOLBOX_DOCK` `LAYER_PANEL_DOCK` `EDIT_BAR_DOCK` `ADVANCED_BAR_DOCK` `CHARMAP_DOCK` |
| `[DOCK]` / `[FLOAT]` panels | `toolbox organizer drawer layers editbar advbar charmap preview colormixer advcolorpicker colorspace3d pen browser` |
| Toolbox buttons | `move hand zoom crop marquee marquee-free marquee-poly marquee-ellipse wand picker text eraser dot brush spray fill line polygon polygon-filled save bezier rect rect-filled export-sel smart-shapes ellipse ellipse-filled open` |
| Menu roots | `FILE EDIT VIEW SELECT TOOLS BRUSH LAYER PALETTE IMAGE EFFECTS AI HELP AUDIO` |
| Edit bar | `undo redo cut copy copy-merged paste paste-in-place cut-to-layer copy-to-layer clear fill-fg fill-bg stroke flip-h flip-v scale-down scale-up rotate-cw rotate-ccw smart-guides smart-guides-snap edge-mode flip-canvas-h flip-canvas-v` |
| Advanced bar | `preview edit-bar char-map char-grid char-grid-snap tile-mode grid-cell-fill grid-brush-size fill-adjust brush-edges angle-snap grayscale ref-on ref-load ref-clear ref-adjust align-left align-center align-right align-top align-middle align-bottom distribute-h distribute-v` |
| Organizer | `color-mixer canvas-ops brush-size pattern-mode palette-ops transform-ops gradient-mode symmetry grid grid-snap color-mode` |
| `[KEYS]` / `[START]` `TOOL` | Any toolbox name above, plus `arrow highlight redact callout callout-reset quick-save done copy-image capture`, or a numeric action id |

`[KEYS]` takes one key (`a`, `R` = Shift+R, `1`), `ctrl+key`, `alt+key`, `enter`, `space` or `tab`. Workspace keys win over DRAW's own while the workspace is active. They never fire while you type text, in a dialog or menu, or in the middle of a stroke. Menus and the command palette show the workspace's keys, so **Ellipse Tool** reads `O` in Annotate.

---

## Screenshot capture

**File → Capture Screen...** (`Ctrl+Shift+P`, also in the command palette), or `DRAW --capture` from a desktop shortcut.

1. DRAW hides its window, waits a moment, grabs the whole screen and comes back.

2. The **region picker** shows the frozen screen:

| Do | To |
|---|---|
| Drag | Draw a box |
| Drag inside the box | Move it |
| Drag a handle | Resize it |
| `Arrows` | Nudge 1 px ( `Shift` = 10) |
| `Enter` / double-click | Use the box (no box = whole screen) |
| `Space` | Whole screen |
| `Esc` / right-click | Cancel |

   A loupe and the pixel coordinates follow the pointer, and the box shows its size.

3. The region becomes a **new document** in the **Annotate** workspace. A region over 4096 px asks whether to scale it down or crop it.

The document you were working on is **not** lost. If it has a name or unsaved changes, it is set aside in DRAW's cache (no "discard changes?" question), and **Done** ( `Ctrl+Enter` ) brings it back.

### Annotate tools (keys work in the Annotate workspace)

| Key | Tool |
|---|---|
| `R` / `Shift+R` | Rectangle / filled rectangle |
| `O` | Ellipse |
| `L` | Line |
| `A` | **Arrow**: a line with an arrowhead that stays on for every line until you pick another tool. `S` / `E` while dragging change the caps. |
| `T` | Text |
| `H` | **Highlighter**: a yellow brush on a "Highlight" layer at 45% opacity (created the first time), so it shows on light and dark art alike |
| `X` | **Redact**: drag a box; it is pixelated as soon as you let go. One undo step per box. |
| `N` / `Shift+N` | **Numbered callout**: each click places a filled circle with the next number, in the FG color. `Shift+N` restarts at 1. |
| `C` | Crop (to the selection, or interactive) |
| `V` | Move |
| `Ctrl+C` | Copy the selection, or the whole picture, to the clipboard |
| `Ctrl+S` | **Quick save** a PNG to the screenshots folder, no dialog. The first save is named `DRAW-YYYYMMDD-HHMMSS.png` ; later saves overwrite it. |
| `Ctrl+Enter` | **Done**: quick save + copy to the clipboard + go back to the document and workspace you had before the capture. When DRAW was started by `DRAW --capture` , Done **closes DRAW** instead (Shottr-style) |

These tools are also under **Tools → Annotate** and in the command palette ( `Annotate:` ), so they work in any workspace.

### Capture settings (Settings → General → Screen Capture)

| Setting | Key | Default |
|---|---|---|
| Capture With | `CAPTURE_BACKEND` | `AUTO` : Windows built-in; macOS `screencapture` ; KDE `spectacle` ; GNOME `gnome-screenshot` ; sway/Hyprland `grim` ; X11 `maim` / `scrot` / `import` |
| Capture Command | `CAPTURE_COMMAND` | For *Custom Command*; `{file}` is the PNG DRAW reads back, e.g. `flameshot full -p {file}` |
| Capture Delay | `CAPTURE_DELAY` | 250 ms after hiding DRAW |
| Hide DRAW While Capturing | `CAPTURE_HIDE_DRAW` | on |
| Capture Workspace | `CAPTURE_WORKSPACE` | `annotate` (empty = stay) |
| — | `CAPTURE_SAVE_DIR` | Quick-save folder (empty = `Pictures/DRAW Screenshots` ) |

> 💡 QB64-PE's own screen grab only works on Windows, which is why Linux and macOS use a capture tool. If none is installed, DRAW names the tools to install in the status bar. Wayland works through XWayland (verified on KDE Plasma 6).

### A global capture shortcut

`DRAW --capture` hands the capture to a running DRAW (no second window) or starts DRAW if it is not running. Started that way, DRAW is **one-shot**:

- No splash screen; it goes straight to the grab.
- `Ctrl+Enter` saves, copies and closes DRAW.
- `Esc` in the region picker closes DRAW too.

So a shortcut gives you capture → mark up → `Ctrl+Enter` , and you're back where you were. Bind it to a key:

- **KDE Plasma:** System Settings → Keyboard → Shortcuts → **Add New** → **Command or Script**. Command: `/path/to/DRAW.run --capture` . Then press the shortcut you want, for example `Meta+Shift+S` .
- **GNOME:** Settings → Keyboard → View and Customize Shortcuts → Custom Shortcuts → **+**. Command: `/path/to/DRAW.run --capture` .
- **Windows:** create a shortcut to `DRAW.exe` , add `--capture` to *Target*, and set *Shortcut key* in its Properties.
- **macOS:** Shortcuts app → new shortcut → **Run Shell Script** `/path/to/DRAW --capture` . Assign a keyboard shortcut in its details.

---

## Troubleshooting

| Problem | Fix |
|---|---|

| | |
|---|---|
| A workspace hid the menu bar and I'm stuck | `Ctrl+Shift+W` → Default, or `F11` to show everything |
| My panels came back wrong after leaving a workspace | Leaving restores the state from when you *entered*; toggles made inside it are dropped by design |
| Capture says "no screen capture tool found" | Install the tool for your desktop (see the table above) or set a Custom Command |
| The capture includes DRAW's window | Raise the Capture Delay (the window may need longer to fade on your desktop) |
| A workspace key does nothing | It only works in that workspace, not while typing, and not while a mouse button is held |
| How do I leave Annotate after a capture? | `Ctrl+Enter` (Done). Or `Ctrl+Shift+W` → Default to keep the screenshot open. `Esc` belongs to the tools. |

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